

A company uses an on-premises network-attached storage (NAS) system to provide file shares to its high performance computing (HPC) workloads. The company wants to migrate its latency-sensitive HPC workloads and its storage to the AWS Cloud. The company must be able to provide NFS and SMB multi-protocol access from the file system.

Which solution will meet these requirements with the LEAST latency? (Choose two.)

A. Deploy compute optimized EC2 instances into a cluster placement group. **Most Voted**

B. Deploy compute optimized EC2 instances into a partition placement group.

C. Attach the EC2 instances to an Amazon FSx for Lustre file system.

D. Attach the EC2 instances to an Amazon FSx for OpenZFS file system.

E. Attach the EC2 instances to an Amazon FSx for NetApp ONTAP file system. **Most Voted**

Correct Answer: AE

Community vote distribution

AE (88%)

8%

Comments

lucasbg **Highly Voted** 2 years, 6 months ago

Selected Answer: AE

You talked about smb and nfs, you talked fsx netapp ontap

C is wrong because lustre is a POSIX fs

upvoted 14 times

ZZZ_Sleep **Highly Voted** 2 years, 5 months ago

Selected Answer: AE

LEAST latency = cluster placement group

Amazon FSx for Lustre = SMB

Amazon FSx for OpenZFS = NFS

Amazon FSx for NetApp ONTAP = NFS, SMB, iSCSI

So, answer are A and E

upvoted 7 times

MatAlves **Most Recent** 1 year, 8 months ago

Selected Answer: AE

I got baited into quickly going for Lustre after reading, but forgot it doesn't not support NFS/SMB.

upvoted 5 times

tsdsmth 2 years, 4 months ago

Amazon FSx for Lustre does not support SMB. So it's A,E

upvoted 6 times

awsgeek75 2 years, 5 months ago

Selected Answer: AE

A Because HPC equivalent in AWS is EC2. Cluster placement for low-latency: <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html>

E: ONTAP gives NFS and SMB which is required

AE is correct

B does not solve low latency requirements

C No support for NFS and SMB

D OpenZFS is not required

upvoted 7 times

awsgeek75 2 years, 5 months ago

<https://aws.amazon.com/fsx/netapp-ontap/features/>

Amazon FSx for NetApp ONTAP provides access to shared file storage over all versions of the Network File System (NFS) and Server Message Block (SMB) protocols, and also supports multi-protocol access (i.e. concurrent NFS and SMB access) to the same data. As a result, you can access Amazon FSx for NetApp ONTAP from virtually any Linux, Windows, or macOS client.

upvoted 2 times

1rob 2 years, 5 months ago

Selected Answer: AD

A because cluster placement group means low latency, and D because OpenZFS has less latency compared to FSx for NetApp ONTAP. See <https://aws.amazon.com/fsx/when-to-choose-fsx/>

FSx for OpenZFS can handle SMB and NFS.

Despite that for onprem NAS appliances the aws Recommended Amazon FSx file system would be FSx for NetApp ONTAP, I still choose FSx for OpenZFS for the lower latency.

upvoted 2 times

Sumith4112 2 years, 6 months ago

Selected Answer: AE

A because cluster placement group means low latency.

E

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: AE

HPC, NFS, SMB = FSx for NetApp ONTAP file system

HPC, latency-sensitive = cluster placement group

upvoted 4 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

AE

<https://aws.amazon.com/fsx/when-to-choose-fsx/>

upvoted 2 times

achechen 2 years, 6 months ago

Selected Answer: AE

I don't think FSx for Lustre supports SMB. At least I could not find anything in the documentation. However, FSx for ONTAP delivers NFS and SMB support.

upvoted 4 times

chikuwan 2 years, 6 months ago

Selected Answer: AE

<https://aws.amazon.com/jp/fsx/lustre/features/>

upvoted 4 times

reika1914 2 years, 6 months ago

Selected Answer: AC

To meet the requirements of migrating latency-sensitive HPC workloads with multi-protocol access (NFS and SMB) to AWS with minimal latency, the following solutions would be the most appropriate:

A. Deploy compute optimized EC2 instances into a cluster placement group.

C. Attach the EC2 instances to an Amazon FSx for Lustre file system.

upvoted 2 times

pentium75 2 years, 5 months ago

FSx for Lustre provides Lustre, not SMB and not NFS

upvoted 2 times

Chiquitabandita 2 years, 6 months ago

Selected Answer: AE

[https://aws.amazon.com/fsx/netapp-ontap/features/#:~:text=Amazon%20FSx%20for%20NetApp%20ONTAP%20provides%20access%20to%20shared%20file,access\)%20to%20the%20same%20data.](https://aws.amazon.com/fsx/netapp-ontap/features/#:~:text=Amazon%20FSx%20for%20NetApp%20ONTAP%20provides%20access%20to%20shared%20file,access)%20to%20the%20same%20data.) "Amazon FSx for NetApp ONTAP provides access to shared file storage over all versions of the Network File System (NFS) and Server Message Block (SMB) protocols, and also supports multi-protocol access (i.e. concurrent NFS and SMB access) to the same data."

upvoted 5 times

LemonGremlin 2 years, 6 months ago

Selected Answer: AC

Option A: A cluster placement group provides low-latency and high-bandwidth connectivity between instances. This is particularly beneficial for high performance computing workloads that are latency-sensitive. Instances within a cluster placement group are placed in close proximity to each other within the same Availability Zone.

Option C: Amazon FSx for Lustre is a high-performance file system optimized for fast access to data. It is well-suited for high-performance computing workloads. It provides low-latency access to data and supports the NFS protocol.

upvoted 3 times

t0nx 2 years, 6 months ago

Thank you

upvoted 1 times

1rob 2 years, 5 months ago

FSx for Lustre is not about NFS or SMB. You will need a linux instance. First install the open-source Lustre client on that instance. Once it's installed, you can mount your file system using standard Linux commands. So C is not correct here because NFS and SMB support is required.

upvoted 3 times

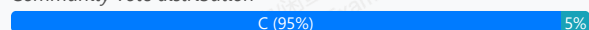
A company is relocating its data center and wants to securely transfer 50 TB of data to AWS within 2 weeks. The existing data center has a Site-to-Site VPN connection to AWS that is 90% utilized.

Which AWS service should a solutions architect use to meet these requirements?

- A. AWS DataSync with a VPC endpoint
- B. AWS Direct Connect
- C. AWS Snowball Edge Storage Optimized **Most Voted**
- D. AWS Storage Gateway

Correct Answer: C

Community vote distribution



Comments

Cyberkayu **Highly Voted** 2 years, 5 months ago

Selected Answer: C

90% utilization of the bandwidth = they discouraged the use of internet bandwidth for uploading, go seek for offline data seeding to AWS method
upvoted 9 times

NSA_Poker 2 years ago

data centers don't "discouraged the use of internet bandwidth."

upvoted 1 times

TariqKipkemei **Highly Voted** 2 years, 6 months ago

Selected Answer: C

50 TB of data to AWS within 2 weeks = Snowball Edge Storage Optimized

upvoted 5 times

KennethNg923 **Most Recent** 1 year, 11 months ago

Selected Answer: C

50 TB of data to AWS within 2 weeks + 90% utilization already be used -> snowball

upvoted 2 times

NSA_Poker 2 years ago

Selected Answer: B

(B) is correct.

Direct Connect is "A dedicated connection [is] made through a 1-Gbps, 10-Gbps, or 100-Gbps Ethernet port dedicated to a single customer." DX use: 802.1Q VLANs providing a dedicated private network connection to AWS. At 1-Gbps, transfer takes less than 5 days; at 100-Gbps it takes less than 67 minutes. Since it's a data center & not an oil rig in the middle of the Gulf of Mexico, data center should be able to get this service.

(C) is incorrect

"Consider Snowball Edge if you need to run computing in rugged, austere, mobile, or disconnected (or intermittently connected) environments. Also consider it for large-scale data transfers and migrations when bandwidth is not available for use of a high-speed online transfer service, such as AWS DataSync."

upvoted 1 times

JA2018 1 year, 6 months ago

hi bro, setting a new DX will usually takes longer than 2 weeks..... and we have not even started any Data transfer over the wire yet.

upvoted 1 times

NSA_Poker 2 years ago

B > A > C > D

upvoted 1 times

xBUGx 2 years, 3 months ago

Assuming vpn is 1Gbps, it can still transfer 50TB with in 5days with only 10% bandwidth available

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

A DataSync is for data

B Direct connect takes longer than 2 weeks

D StorageGateway is useless without more contex

C is only remaining choice.

upvoted 3 times

NSA_Poker 2 years ago

(A) The problem is not that DataSync deals with data (we have to 'transfer 50 TB of data'). The issue is with the VPC endpoint; it doesn't increase our bandwidth. DataSync task is capable fully utilizing 10-Gbps over a network link between your on-premises environment and AWS.

(B) works between 67-minutes & 5 days.

(C) Works but is NOT what I would recommend to a data center.

(D) over VPN connection that is 90% utilized, takes more than 6 weeks.

upvoted 1 times

ftaws 2 years, 5 months ago

Not memtioned network bandwidth. How we know that?

upvoted 2 times

allanm 10 months ago

Doesn't matter, 90% of it is utilised.

upvoted 1 times

A company hosts an application on Amazon EC2 On-Demand Instances in an Auto Scaling group. Application peak hours occur at the same time each day. Application users report slow application performance at the start of peak hours. The application performs normally 2-3 hours after peak hours begin. The company wants to ensure that the application works properly at the start of peak hours.

Which solution will meet these requirements?

- A. Configure an Application Load Balancer to distribute traffic properly to the instances.
- B. Configure a dynamic scaling policy for the Auto Scaling group to launch new instances based on memory utilization.
- C. Configure a dynamic scaling policy for the Auto Scaling group to launch new instances based on CPU utilization.
- D. Configure a scheduled scaling policy for the Auto Scaling group to launch new instances before peak hours. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

ZZZ_Sleep **Highly Voted** 1 year, 11 months ago

Selected Answer: D

occur at the same time each day = predictable

So, scheduled scaling policy, Answer is D.

Dynamic scaling policy work for unpredictable

upvoted 8 times

Arnaud92 **Highly Voted** 2 years ago

D. The application performs normally 2-3 hours after peak hours begin is a key! (schedule policy)

upvoted 5 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: D

ABC won't solve the performance issues at the start of peak hours.

D ensure that application is ready for use during the peak hours by scheduling an early launch

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: D

Technically both dynamic and scheduled scaling would work but there is strict requirement for the application to work properly at the start of peak hours and no mention of cost.

So scheduled scaling policy it is.

upvoted 4 times

TOR_0511 2 years ago

Selected Answer: D

Application users report slow application performance at the start of peak hours. The company wants to ensure that the application works properly at the start of peak hours

upvoted 1 times

A company runs applications on AWS that connect to the company's Amazon RDS database. The applications scale on weekends and at peak times of the year. The company wants to scale the database more effectively for its applications that connect to the database.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon DynamoDB with connection pooling with a target group configuration for the database. Change the applications to use the DynamoDB endpoint.

B. Use Amazon RDS Proxy with a target group for the database. Change the applications to use the RDS Proxy endpoint.

Most Voted

C. Use a custom proxy that runs on Amazon EC2 as an intermediary to the database. Change the applications to use the custom proxy endpoint.

D. Use an AWS Lambda function to provide connection pooling with a target group configuration for the database. Change the applications to use the Lambda function.

Correct Answer: B

Community vote distribution

B (100%)

Comments

TariqKipkemei **Highly Voted** 2 years ago

Selected Answer: B

Amazon RDS Proxy is a fully managed, highly available database proxy for Amazon Relational Database Service (RDS) that makes applications more resilient to database failures. Many applications, including those built on modern serverless architectures, can have a large number of open connections to the database server and may open and close database connections at a high rate, exhausting database memory and compute resources. Amazon RDS Proxy allows applications to pool and share connections established with the database, improving database efficiency and application scalability. With RDS Proxy, failover times for Aurora and RDS databases are reduced by up to 66%

upvoted 7 times

TOR_0511 **Highly Voted** 2 years ago

Selected Answer: B

A out because DynamoDB is a NoSQL DB

B As the question is referring about DB connections so this option has the LEAST operational overhead

upvoted 5 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: B

A: DynamoDB != RDS

C: Total nonsense

D: Lambda for providing connection pooling sound impractical if not impossible. Would be fun to watch someone do this though...

B RDS Proxy is specifically made for connection pooling.

upvoted 4 times

A company uses AWS Cost Explorer to monitor its AWS costs. The company notices that Amazon Elastic Block Store (Amazon EBS) storage and snapshot costs increase every month. However, the company does not purchase additional EBS storage every month. The company wants to optimize monthly costs for its current storage usage.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use logs in Amazon CloudWatch Logs to monitor the storage utilization of Amazon EBS. Use Amazon EBS Elastic Volumes to reduce the size of the EBS volumes.
- B. Use a custom script to monitor space usage. Use Amazon EBS Elastic Volumes to reduce the size of the EBS volumes.
- C. Delete all expired and unused snapshots to reduce snapshot costs.
- D. Delete all nonessential snapshots. Use Amazon Data Lifecycle Manager to create and manage the snapshots according to the company's snapshot policy requirements. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

t0nx **Highly Voted** 2 years, 6 months ago

Selected Answer: D

This option involves managing snapshots efficiently to optimize costs with minimal operational overhead.

Delete all nonessential snapshots: This reduces costs by eliminating unnecessary snapshot storage.

Use Amazon Data Lifecycle Manager (DLM): DLM can automate the creation and deletion of snapshots based on defined policies. This reduces operational overhead by automating snapshot management according to the company's snapshot policy requirements.

upvoted 8 times

JA2018 **Most Recent** 1 year, 6 months ago

Selected Answer: D

Allows the user to manage snapshots efficiently for cost optimization with the LEAST operational overhead.

#1: Delete all nonessential snapshots: This reduces costs by eliminating unnecessary storage used by such snapshots.

#2: Amazon Data Lifecycle Manager (DLM): This AWS managed service can automate the creation and deletion process of snapshots based on defined policies. Automating snapshot management according to snapshot policy requirements helps to cut down operational overhead.

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

Least operational overhead for your snapshot management is <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/snapshot-lifecycle.html>

C will just do it once but assuming they want an ongoing solution.

A: It will help with EBS size but won't fix the snapshot problems

B: Same as A, nothing to do with snapshots

upvoted 4 times

xBUGx 2 years, 2 months ago

Q says The company wants to optimize monthly costs for its current storage usage. I think they only want to do it once?

upvoted 1 times

KennethNg923 1 year, 11 months ago

I think "optimize monthly costs" include future monthly cost as well, so D is better than C

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: D

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/snapshot-lifecycle.html>
upvoted 4 times

A company is developing a new application on AWS. The application consists of an Amazon Elastic Container Service (Amazon ECS) cluster, an Amazon S3 bucket that contains assets for the application, and an Amazon RDS for MySQL database that contains the dataset for the application. The dataset contains sensitive information. The company wants to ensure that only the ECS cluster can access the data in the RDS for MySQL database and the data in the S3 bucket.

Which solution will meet these requirements?

A. Create a new AWS Key Management Service (AWS KMS) customer managed key to encrypt both the S3 bucket and the RDS for MySQL database. Ensure that the KMS key policy includes encrypt and decrypt permissions for the ECS task execution role.

Most Voted

B. Create an AWS Key Management Service (AWS KMS) AWS managed key to encrypt both the S3 bucket and the RDS for MySQL database. Ensure that the S3 bucket policy specifies the ECS task execution role as a user.

C. Create an S3 bucket policy that restricts bucket access to the ECS task execution role. Create a VPC endpoint for Amazon RDS for MySQL. Update the RDS for MySQL security group to allow access from only the subnets that the ECS cluster will generate tasks in.

D. Create a VPC endpoint for Amazon RDS for MySQL. Update the RDS for MySQL security group to allow access from only the subnets that the ECS cluster will generate tasks in. Create a VPC endpoint for Amazon S3. Update the S3 bucket policy to allow access from only the S3 VPC endpoint.

Correct Answer: A

Community vote distribution

A (44%)

D (41%)

C (15%)

Comments

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: A

We're asked to restrict access to both, RDS and S3, to "the ECS cluster" (not to a subnet or endpoint).

Not B: Does not restrict RDS at all. Wording about S3 is unusual.

Not C: Would work for S3, but would allow RDS access from whole subnet which may contain other resources besides the ECS cluster

Not D: Would allow RDS access from whole subnet which may contain other resources besides the ECS cluster. Would allow S3 access from VPC endpoint which might be accessed by other resources besides the ECS cluster.

upvoted 14 times

deacon967 1 year, 5 months ago

A doesn't restrict access. It only encrypts at rest. D restricts access. Encryption and Access Control are 2 different things

upvoted 2 times

GOTJ 1 year, 3 months ago

You are right with your option "D" concerns about letting other resources running on the same subnet that your ECS cluster accessing the database. But, likewise, the "task execution role" mentioned in your selected answer "A" is applied to a specific ECS task, not to the whole cluster (requirement)

upvoted 1 times

t0nx **Highly Voted** 2 years, 6 months ago

Selected Answer: D

Option D is the most comprehensive solution as it leverages VPC endpoints for both Amazon RDS and Amazon S3, along with proper network-level controls to restrict access to only the necessary resources from the ECS cluster.

upvoted 11 times

awsgeek75 2 years, 5 months ago

D only secures access to RDS and S3, it does not secure the sensitive data inside the RDS and S3.

upvoted 3 times

KarlosRC Most Recent 3 months ago

Selected Answer: D

Encryption ≠ Access Control

upvoted 1 times

Pucmaster 3 months, 2 weeks ago

Selected Answer: C

i think C

upvoted 1 times

b7261f6 7 months ago

Selected Answer: A

Sensitive Data in S3 & RDS - KMS Key (customer or AWS both will work)

Only access to ECAS Task - Role & KMS Key policy to give permission to access

upvoted 1 times

dungva1505 11 months ago

Selected Answer: D

I go for D.

Because the question requires a network access control.

A. It's not a network access control

B. Can't access to RDS from ECS

C. Have to access to S3 via internet

D. Only ECS can access to S3 via AWS private network. It's more secure

Someone consider about the other services can access to S3 and RDS if they lies on same subnet with ECS, but the question doesn't mention have other services. So I assumed we don't have any other services here.

upvoted 2 times

marcosnamekata 1 year, 1 month ago

Selected Answer: D

D- Shure

upvoted 2 times

diazed 1 year, 1 month ago

Selected Answer: C

I think this question is not written correctly.

upvoted 2 times

tch 1 year, 2 months ago

Selected Answer: A

But security groups allow you to control inbound and outbound traffic at the instance level, while network ACLs offer similar capabilities at the VPC subnet level.

upvoted 1 times

tch 1 year, 2 months ago

look portion of answer D: Update the RDS for MySQL security group to allow access from only the subnets that the ECS cluster will generate tasks in.

"allow access from only the subnets " .. is this correct for security group?

upvoted 1 times

Dantecito 1 year, 3 months ago

Selected Answer: C

I think the word "restricts" is confusing. But I interpret it as allow the ECS task execution role to access the S3 bucket and deny everything else.

upvoted 2 times

dragosky 1 year, 6 months ago

Selected Answer: D

The dataset contains sensitive information, nothing to be done here, just stating that it contains sensitive info.

upvoted 1 times

JA2018 1 year, 6 months ago

Selected Answer: C

Why not C?

Key point:

The question specifically asks to ensure that only the ECS cluster can access the data, which means restricting access to the S3 bucket and RDS database solely to the ECS task execution role.

Breakdown of option C:

S3 bucket policy: By creating a policy that only allows the ECS task execution role to access the S3 bucket, you effectively limit access to the data stored there.

VPC endpoint for RDS: A VPC endpoint creates a private connection to the RDS database, preventing any external access and ensuring that only resources within the VPC can connect.

RDS security group update: Further restricting access by configuring the RDS security group to only allow connections from the subnets where the ECS cluster will launch tasks.

upvoted 2 times

JA2018 1 year, 6 months ago

Why the other options are not as suitable:

Option A and B:

While using a KMS key to encrypt data is a good practice, it doesn't directly control access to the data. Attaching the KMS key to the ECS task execution role in the policy would still allow any entity with the key to decrypt the data, not just the ECS cluster.

upvoted 1 times

JA2018 1 year, 6 months ago

Option D:

This option only addresses the RDS access by creating a VPC endpoint, but it doesn't restrict access to the S3 bucket effectively, which is also mentioned in the question

upvoted 2 times

XXXXXINN 1 year, 8 months ago

I cannot believe how many people vote A.

the questions is asking only allow ECS cluster access RDS and access to S3.

2 keys here: 1. security group is usually used to security access between RDS and ECS cluster 2. access data in S3 securely, immediately, we should think about S3 VPC Gateway endpoints because this secures the traffic only travel via private network.

Answer A is just talking about encrypt data at rest, and that is not what the question is asking about

upvoted 4 times

MandAsh 1 year, 11 months ago

After reading comments changed to A. D will not protect data at rest it will only give n/w level security

upvoted 1 times

bujuman 2 years, 1 month ago

Selected Answer: A

According to me "The dataset contains sensitive information" is the main information that motivate the real requirement which is "The company wants to ensure that only the ECS cluster can access the data in the RDS for MySQL database and the data in the S3 bucket". So we have to take these two assertions into consideration.

And knowing that, as S3 default encryption capabilities, RDS Mysql DB Instance encryption is not active by default (check this link for details <https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Overview.Encryption.html>), option A is the best option to meet the requirements of accessing the datasets and the assets only from ECS cluster tasks and preserve, at the same time, data confidentiality and integrity. In other words, option A is the best one to ensure the data protection at REST for S3 and RDS and only accessed by ECS cluster.

upvoted 3 times

Hung23 2 years, 1 month ago

Selected Answer: C

Try to chat GPT Please

upvoted 1 times

seetpt 2 years, 3 months ago

Selected Answer: A

A seems right

upvoted 2 times

A company has a web application that runs on premises. The application experiences latency issues during peak hours. The latency issues occur twice each month. At the start of a latency issue, the application's CPU utilization immediately increases to 10 times its normal amount.

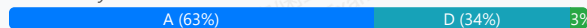
The company wants to migrate the application to AWS to improve latency. The company also wants to scale the application automatically when application demand increases. The company will use AWS Elastic Beanstalk for application deployment.

Which solution will meet these requirements?

- A. Configure an Elastic Beanstalk environment to use burstable performance instances in unlimited mode. Configure the environment to scale based on requests. **Most Voted**
- B. Configure an Elastic Beanstalk environment to use compute optimized instances. Configure the environment to scale based on requests.
- C. Configure an Elastic Beanstalk environment to use compute optimized instances. Configure the environment to scale on a schedule.
- D. Configure an Elastic Beanstalk environment to use burstable performance instances in unlimited mode. Configure the environment to scale on predictive metrics.

Correct Answer: A

Community vote distribution



Comments

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: A

"Scale on predictive metrics" does not sound like something that Beanstalk can do. In EC2 you can create a "predictive scaling policy", but apparently this is not supported by Beanstalk. That would rule out D.

We have no indication that the application is CPU-intensive in general. If CPU utilization "increases to 10 times its normal amount" then the "normal amount" cannot be higher than 10 %. This would rule out B and C.

upvoted 11 times

LemonGremlin **Highly Voted** 2 years, 6 months ago

Selected Answer: D

Burstable Performance Instances (T3 or T3a): These instances are designed for burstable workloads and provide a baseline level of CPU performance with the ability to burst above that baseline when needed. Bursting is particularly beneficial for handling sudden spikes in CPU utilization, such as those described in the scenario.

Unlimited Mode: Enabling "unlimited" mode allows instances to burst beyond their baseline performance without accumulating CPU credits. This is important for handling sudden and sustained increases in CPU utilization during peak hours.

Scale on Predictive Metrics: Configuring the environment to scale on predictive metrics allows AWS Elastic Beanstalk to proactively adjust the number of instances based on anticipated demand. This can help ensure that the environment is scaled up before the latency issues occur, addressing them in advance.

upvoted 10 times

ftaws 2 years, 5 months ago

Traffic is "immediately increases". We can't predict and can not use Predictive Metrics.

And requirement need auto scaling

upvoted 2 times

e10c971 **Most Recent** 6 months ago

Selected Answer: B

From AI Why B is correct

Compute optimized instances (C family) are designed for CPU-intensive workloads, which is exactly the problem here (CPU spikes 10x during peak). Scaling based on requests (or CPU utilization) ensures the environment automatically adds instances when demand increases, eliminating latency during peak hours.

Elastic Beanstalk supports Auto Scaling policies based on metrics like CPU or request count.

upvoted 1 times

Salilgen 1 year, 5 months ago

Selected Answer: A

IMO answer is A.

"The Auto Scaling group uses two Amazon CloudWatch alarms to trigger scaling operations. The default triggers scale when the average outbound network traffic from each instance is higher than 6 MiB or lower than 2 MiB over a period of five minutes. To use Auto Scaling effectively, configure triggers that are appropriate for your application, instance type, and service requirements. You can scale based on several statistics including latency, disk I/O, CPU utilization, and request count."

This is dynamic scaling.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/burstable-performance-instances.html>

<https://docs.aws.amazon.com/elasticbeanstalk/latest/dg/using-features-managing-env-types.html>

<https://docs.aws.amazon.com/elasticbeanstalk/latest/dg/using-features-managing-as.html>

upvoted 1 times

3c6417b 2 years ago

Selected Answer: B

Explain to me why it's not B?

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

B is incorrect because CPU usage percentage rising does not necessarily have anything to do with computing methods. Especially during peak hours, the jobs are coming in hot, but it's quite possible that each job can be done within seconds. It's just too many jobs, and that's not something compute optimized instances can tackle.

upvoted 1 times

Gape4 1 year, 11 months ago

I have the same question.

upvoted 1 times

sandordini 2 years, 1 month ago

Selected Answer: A

D - No such service as Elastic Beanstalk Predictive Scaling, And even if there was, no historical data in AWS for an application we are just about to migrate to AWS. Therefore: A

upvoted 6 times

lenotc 2 years, 2 months ago

Selected Answer: A

D is incorrect Predictive scaling not fit

upvoted 3 times

awsgeek75 2 years, 4 months ago

For those voting D, predictive scaling analyses historic data to predict the scaling needs. This scenario is a migration scenario so there won't be any historic data which is why D won't work. A (burst) is the only option after migration.

upvoted 5 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

BC are compute optimised instances which don't solve 10x CPU issues at start of the latency.

AD are burstable performance which will help with 10x increase CPU usage

D is not an available feature of Elastic Beanstalk (yet) or I cannot find it in config/docs. Happy to be corrected

A makes sense due to burst performance. Scale based on requests is possible and I'm assuming that latency is related to requests.

upvoted 6 times

1rob 2 years, 5 months ago

Selected Answer: A

Following <https://docs.aws.amazon.com/elasticbeanstalk/latest/dg/using-features-managing-as.html> I see: " You can scale based on several statistics including latency, disk I/O, CPU utilization, and request count. " So no 'scale on predictive metrics, so D is not okay.

Also, the company also wants to scale the application automatically when application demand increases, so scale on a schedule is not appropriate here. So C is not okay.

Burstable performance instances in unlimited mode can sustain high CPU utilization for any period of time whenever required, so an immediate demand of CPU resources is 'covered'. So I go for A.

upvoted 4 times

Min_93 2 years, 5 months ago

Selected Answer: D

Option A, which suggests using burstable performance instances in unlimited mode, is appropriate. However, option D is more specific to the requirement of scaling based on predictive metrics, which is crucial for handling the latency issues that occur at specific times each month.

Options B and C suggest using compute optimized instances and scaling based on requests or on a schedule. While these options might work for general scalability, they may not address the immediate and intense spikes in CPU utilization that are mentioned in the scenario.

Therefore, option D is the most appropriate solution for improving latency and automatically scaling the application based on predictive metrics using AWS Elastic Beanstalk.

upvoted 4 times

evelynsun 2 years, 5 months ago

Selected Answer: A

This solution meets the requirements because it allows the company to automatically scale the application's CPU capacity based on the number of requests it receives. The burstable performance instances provide high CPU performance when needed, which can help to reduce latency during peak hours.

not D: this solution has some drawbacks. First, it can be expensive to use burstable performance instances in unlimited mode, as the instances are charged per hour. Second, it can be difficult to predict the exact CPU requirements of the application, which can lead to over- or under-provisioning of CPU resources.

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

The company also wants to scale the application automatically when application demand increases = Scale based on requests

upvoted 3 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

B

Question is asking scale based on demand so better scale based on requests. Predictive metrics not defined and may be interpreted differently by many users.

upvoted 3 times

reika1914 2 years, 6 months ago

Selected Answer: D

Given the scenario described, the best solution among the provided options to meet the requirements of migrating the application to AWS, improving latency, and scaling the application automatically during increased demand would be:

D. Configure an Elastic Beanstalk environment to use burstable performance instances in unlimited mode. Configure the environment to scale on predictive metrics.

upvoted 2 times

t0nx 2 years, 6 months ago

Selected Answer: D

In this scenario, the application experiences latency issues during peak hours with a sudden increase in CPU utilization. Using burstable performance instances in unlimited mode allows the application to burst beyond the baseline performance when needed. Configuring the environment to scale on predictive metrics enables proactive scaling based on anticipated increases in demand.

upvoted 4 times

A company has customers located across the world. The company wants to use automation to secure its systems and network infrastructure. The company's security team must be able to track and audit all incremental changes to the infrastructure.

Which solution will meet these requirements?

- A. Use AWS Organizations to set up the infrastructure. Use AWS Config to track changes.
- B. Use AWS CloudFormation to set up the infrastructure. Use AWS Config to track changes. **Most Voted**
- C. Use AWS Organizations to set up the infrastructure. Use AWS Service Catalog to track changes.
- D. Use AWS CloudFormation to set up the infrastructure. Use AWS Service Catalog to track changes.

Correct Answer: B

Community vote distribution

B (100%)

Comments

TariqKipkemei **Highly Voted** 2 years, 6 months ago

Selected Answer: B

use automation to secure its systems and network infrastructure = AWS CloudFormation
track and audit all incremental changes to the infrastructure = AWS Config
upvoted 10 times

Min_93 **Highly Voted** 2 years, 5 months ago

Selected Answer: B

Option B is the most suitable because it combines the benefits of infrastructure as code (CloudFormation) with tracking and auditing capabilities (AWS Config). With CloudFormation, the company can define and deploy its infrastructure in a repeatable and automated way, ensuring consistency and adherence to security standards. AWS Config then complements this by providing visibility into changes and configuration details.
upvoted 5 times

Gape4 **Most Recent** 1 year, 11 months ago

Selected Answer: B

I will go for B
upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

Organisations is not really related to this
AWS Service Catalog is like a IaC source control so D is a close option. However B looks more logical.
upvoted 3 times

awsgeek75 2 years, 4 months ago

The difference is in wording: "The company's security team must be able to track and audit all incremental changes to the infrastructure"

If this has to be done BEFORE the deployment then D is the option
If this is AFTER the deployment then B is the option

Hopefully exam will have better language. Good luck!
upvoted 3 times

A startup company is hosting a website for its customers on an Amazon EC2 instance. The website consists of a stateless Python application and a MySQL database. The website serves only a small amount of traffic. The company is concerned about the reliability of the instance and needs to migrate to a highly available architecture. The company cannot modify the application code.

Which combination of actions should a solutions architect take to achieve high availability for the website? (Choose two.)

- A. Provision an internet gateway in each Availability Zone in use.
- B. Migrate the database to an Amazon RDS for MySQL Multi-AZ DB instance. **Most Voted**
- C. Migrate the database to Amazon DynamoDB, and enable DynamoDB auto scaling.
- D. Use AWS DataSync to synchronize the database data across multiple EC2 instances.
- E. Create an Application Load Balancer to distribute traffic to an Auto Scaling group of EC2 instances that are distributed across two Availability Zones. **Most Voted**

Correct Answer: BE

Community vote distribution

BE (100%)

Comments

TariqKipkemei **Highly Voted** 2 years ago

Selected Answer: BE

To achieve high availability for the website, Migrate the database to an Amazon RDS for MySQL Multi-AZ DB instance and Create an Application Load Balancer to distribute traffic to an Auto Scaling group of EC2 instances that are distributed across two Availability Zones.

upvoted 9 times

Sergiuss95 **Highly Voted** 1 year, 7 months ago

Selected Answer: BE

I sold my soul to the devil to pass the exam

upvoted 6 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: BE

B: RDS HA

E: Application HA

C: Company cannot change code so this won't work

A: Does not make sense with other options

D: Makes no sense with other options

upvoted 3 times

Cyberkayu 1 year, 11 months ago

A. no failed over mechanism

C. DynamoDB is no SQL, cannot use with MySQL

D. Not HA, just sync/replication tools.

Answer BE.

upvoted 3 times

A company is moving its data and applications to AWS during a multiyear migration project. The company wants to securely access data on Amazon S3 from the company's AWS Region and from the company's on-premises location. The data must not traverse the internet. The company has established an AWS Direct Connect connection between its Region and its on-premises location.

Which solution will meet these requirements?

- A. Create gateway endpoints for Amazon S3. Use the gateway endpoints to securely access the data from the Region and the on-premises location.
- B. Create a gateway in AWS Transit Gateway to access Amazon S3 securely from the Region and the on-premises location.
- C. Create interface endpoints for Amazon S3. Use the interface endpoints to securely access the data from the Region and the on-premises location. **Most Voted**
- D. Use an AWS Key Management Service (AWS KMS) key to access the data securely from the Region and the on-premises location.

Correct Answer: C

Community vote distribution

C (82%)

Other

Comments

Ernestokoro **Highly Voted** 2 years, 6 months ago

Ans is C: >> You can access Amazon S3 from your VPC using gateway VPC endpoints. After you create the gateway endpoint, you can add it as a target in your route table for traffic destined from your VPC to Amazon S3.

There is no additional charge for using gateway endpoints. Amazon S3 supports both gateway endpoints and interface endpoints. With a gateway endpoint, you can access Amazon S3 from your VPC, without requiring an internet gateway or NAT device for your VPC, and with no additional cost. However, gateway endpoints do not allow access from on-premises networks, from peered VPCs in other AWS Regions, or through a transit gateway. For those scenarios, you must use an interface endpoint, which is available for an additional cost. For more information, see Types of VPC endpoints for Amazon S3 in the Amazon S3 User Guide.

<https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html>

upvoted 12 times

MatAlves **Highly Voted** 1 year, 8 months ago

Gateway Endpoint -> only within same VPC

Interface Endpoint -> On-premises (VPN or Direct Connect), or different Region over VPC peering.

upvoted 11 times

bora4motion **Most Recent** 1 year ago

Selected Answer: C

to access it from on prem you will need an interface endpoint

upvoted 1 times

chest_jd 1 year, 1 month ago

Selected Answer: A

VPC has access to S3 through gateway endpoint or interface endpoint. Gateway is better. If we have Direct connect to VPC from on-premises network we get access to S3.

upvoted 1 times

Gape4 1 year, 11 months ago

Selected Answer: C

Please C

upvoted 2 times

1Alpha1 2 years, 4 months ago

Selected Answer: C

Gateway endpoints do not allow access from on-premises networks, from peered VPCs in other AWS Regions, or through a transit gateway. For those scenarios, you must use an interface endpoint, which is available for an additional cost.

<https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html>

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/privatelink-interface-endpoints.html>

With AWS PrivateLink for Amazon S3, you can provision interface VPC endpoints (interface endpoints) in your virtual private cloud (VPC). These endpoints are directly accessible from applications that are on premises over VPN and AWS Direct Connect, or in a different AWS Region over VPC peering.

upvoted 4 times

pentium75 2 years, 5 months ago

Selected Answer: C

Not A, Gateway endpoint can be accessed only from inside the VPC it's in

Not B, Transit Gateway alone won't help

Not D, KMS has nothing to do with this

upvoted 4 times

fea9bdf 2 years, 5 months ago

Answer seems to be C

gateway endpoints do not allow access from on-premises networks, from peered VPCs in other AWS Regions, or through a transit gateway. For those scenarios, you must use an interface endpoint, which is available for an additional cost. For more information, see Types of VPC endpoints for Amazon S3 in the Amazon S3 User Guide.

upvoted 4 times

ale_brd_111 2 years, 5 months ago

Selected Answer: C

gateway endpoint uses public ip address even if traffic does not directly route thru the internet, also they are no meant to be used from on-premises Answer is C

upvoted 3 times

Min_93 2 years, 5 months ago

Selected Answer: C

Options A, B, and D are not the most suitable for the following reasons:

A. Create gateway endpoints for Amazon S3:

Gateway endpoints are used for accessing S3 from within a VPC, but they do not extend connectivity to on-premises locations.

B. Create a gateway in AWS Transit Gateway:

AWS Transit Gateway is designed for routing traffic between VPCs and on-premises networks but is not used as a direct gateway for S3 access.

D. Use an AWS Key Management Service (AWS KMS) key:

AWS KMS is a key management service and does not provide direct access to S3. It's used for managing encryption keys.

Therefore, option C, creating interface endpoints for Amazon S3, is the most appropriate solution for securely accessing S3 from both the AWS Region and the on-premises location.

upvoted 2 times

Min_93 2 years, 5 months ago

Gateway endpoints for Amazon S3

Interface endpoints for Amazon S3

In both cases, your network traffic remains on the AWS network.

Use Amazon S3 public IP addresses

Use private IP addresses from your VPC to access Amazon S3

Use the same Amazon S3 DNS names

Require endpoint-specific Amazon S3 DNS names

Do not allow access from on premises

Allow access from on premises

Do not allow access from another AWS Region

Allow access from a VPC in another AWS Region by using VPC peering or AWS Transit Gateway

Not billed

Billed

upvoted 2 times

ftaws 2 years, 5 months ago

Selected Answer: B

Transit Gateway support inter region.

interface gateway not use in S3

upvoted 1 times

Min_93 2 years, 5 months ago

com.amazonaws.ap-southeast-1.s3 amazon Interface

Interface is now available for S3

upvoted 1 times

Beshowasfy 2 years, 6 months ago

Selected Answer: A

GW Endpoint is only for S3 and DynamoDB, interface endpoint for other services so C is wrong

upvoted 4 times

ale_brd_111 2 years, 5 months ago

you can't access gateway endpoint from on-premises

upvoted 3 times

XXXXXINN 1 year, 8 months ago

but you can via direct connection

upvoted 2 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

S3 gateway endpoints do not currently support access from resources in a different Region, different VPC, or from an on-premises (non-AWS) environment.

<https://aws.amazon.com/blogs/architecture/choosing-your-vpc-endpoint-strategy-for-amazon-s3/#:~:text=associated.%20S3%20gateway-,endpoints,-do%20not%20currently>

upvoted 2 times

SHAAHIBHUSHANAWS 2 years, 6 months ago

C

. S3 gateway endpoints do not currently support access from resources in a different Region, different VPC, or from an on-premises (non-AWS) environment. However, if you're willing to manage a complex custom architecture, you can use proxies. In all those scenarios, where access is from resources external to VPC, S3 interface endpoints access S3 in a secure way.

<https://aws.amazon.com/blogs/architecture/choosing-your-vpc-endpoint-strategy-for-amazon-s3/>

upvoted 4 times

VladanO 2 years, 6 months ago

Selected Answer: A

<https://docs.aws.amazon.com/vpc/latest/privatelink/gateway-endpoints.html>

Gateway VPC endpoints provide reliable connectivity to Amazon S3 and DynamoDB without requiring an internet gateway or a NAT device for your VPC.

There is no additional charge for using gateway endpoints.

upvoted 2 times

pentium75 2 years, 5 months ago

You can't use GW endpoint from on-premises

upvoted 2 times

Prahacker69 1 year, 5 months ago

You can with direct connect

upvoted 1 times

t0nx 2 years, 6 months ago

Selected Answer: C

CCCCC

upvoted 2 times

A company created a new organization in AWS Organizations. The organization has multiple accounts for the company's development teams. The development team members use AWS IAM Identity Center (AWS Single Sign-On) to access the accounts. For each of the company's applications, the development teams must use a predefined application name to tag resources that are created.

A solutions architect needs to design a solution that gives the development team the ability to create resources only if the application name tag has an approved value.

Which solution will meet these requirements?

- A. Create an IAM group that has a conditional Allow policy that requires the application name tag to be specified for resources to be created.
- B. Create a cross-account role that has a Deny policy for any resource that has the application name tag.
- C. Create a resource group in AWS Resource Groups to validate that the tags are applied to all resources in all accounts.
- D. Create a tag policy in Organizations that has a list of allowed application names. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

awsgeek75 **Highly Voted** 1 year, 11 months ago

Selected Answer: D

A: Don't think this is possible.

B: Cross account role with deny policy? Never seen anything like this

C: Resource groups have nothing to do with allowed tags

D: Correct https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_tag-policies.html

upvoted 7 times

pentium75 **Most Recent** 1 year, 11 months ago

Selected Answer: D

Other options don't make sense

upvoted 4 times

m_y_s 1 year, 12 months ago

Selected Answer: D

A tag policy can also specify that noncompliant tagging operations on specified resource types are enforced. In other words, noncompliant tagging requests on specified resource types are prevented from completing.

upvoted 2 times

Beshowasfy 2 years ago

Selected Answer: D

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_tag-policies.html

upvoted 3 times

SHAAHIBHUSHANAWS 2 years ago

D

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_tag-policies.html

upvoted 2 times

rcptryk 2 years ago

Selected Answer: D

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_tag-policies.html

upvoted 3 times

A company runs its databases on Amazon RDS for PostgreSQL. The company wants a secure solution to manage the master user password by rotating the password every 30 days.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use Amazon EventBridge to schedule a custom AWS Lambda function to rotate the password every 30 days.
- B. Use the modify-db-instance command in the AWS CLI to change the password.
- C. Integrate AWS Secrets Manager with Amazon RDS for PostgreSQL to automate password rotation. **Most Voted**
- D. Integrate AWS Systems Manager Parameter Store with Amazon RDS for PostgreSQL to automate password rotation.

Correct Answer: C

Community vote distribution

C (100%)

Comments

TariqKipkemei **Highly Voted** 2 years ago

Selected Answer: C

password rotation = AWS Secrets Manager
upvoted 10 times

rcptryk **Highly Voted** 2 years ago

Selected Answer: C

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/rds-secrets-manager.html>
upvoted 6 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: C

"Least operational overhead"
A: Lambda overhead so not correct
B: CLI = overhead
D: Yes, it can be done but requires more work for integration.

C: This is correct way of doing it.
<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/rds-secrets-manager.html#rds-secrets-manager-overview>
upvoted 3 times

pentium75 1 year, 11 months ago

Selected Answer: C

Secrets Manager allows that, least overhead
upvoted 4 times

A company performs tests on an application that uses an Amazon DynamoDB table. The tests run for 4 hours once a week. The company knows how many read and write operations the application performs to the table each second during the tests. The company does not currently use DynamoDB for any other use case. A solutions architect needs to optimize the costs for the table.

Which solution will meet these requirements?

- A. Choose on-demand mode. Update the read and write capacity units appropriately.
- B. Choose provisioned mode. Update the read and write capacity units appropriately. **Most Voted**
- C. Purchase DynamoDB reserved capacity for a 1-year term.
- D. Purchase DynamoDB reserved capacity for a 3-year term.

Correct Answer: B

Community vote distribution

B (61%)

A (39%)

Comments

OSHOAIB **Highly Voted** 1 year, 11 months ago

Selected Answer: A

On-demand mode Option A: On-demand mode is suitable for workloads that are unpredictable or that do not have significant or consistent database traffic. It automatically scales to accommodate workload demands and charges for the read and write throughput that the application consumes. For infrequent testing, this could be cost-effective because you only pay for what you use during the testing period and don't incur costs when the table is not being accessed.

Whereas for the Option B, if you only run tests once a week for 4 hours, you might pay for unused capacity for the rest of the week unless you manually scale down the capacity after tests are completed, which adds operational overhead.

upvoted 7 times

allanm 10 months ago

You're missing one key information - "The company knows how many read and write operations the application performs to the table each second during the tests." If you know the capacity, then on-demand doesn't make sense. You'll need to go with provisioned resources.

upvoted 1 times

1Alpha1 **Highly Voted** 1 year, 10 months ago

Selected Answer: B

With provisioned capacity mode, you specify the number of reads and writes per second that you expect your application to require, and you are billed based on that. Furthermore if you can forecast your capacity requirements you can also reserve a portion of DynamoDB provisioned capacity and optimize your costs even further.

<https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/capacity.html>

upvoted 6 times

Pucmaster **Most Recent** 3 months, 2 weeks ago

Selected Answer: A

A is correct

upvoted 1 times

SnaxAttax 6 months, 2 weeks ago

Selected Answer: A

On demand means you are only paying for the read/write you use, which is very little here. B, C, D would all have you wasting money given how little this DB is actually used.

upvoted 1 times

jacobtriestech 8 months, 2 weeks ago

Selected Answer: A

On-demand mode automatically scales to demand, and you only pay per request.

upvoted 1 times

111380c 10 months ago

Selected Answer: A

provisioned mode - charge by hours, even not used

upvoted 1 times

ln2718 10 months, 3 weeks ago

Selected Answer: A

This is either a poorly written or trick question. On one hand the performance is known, which would tend towards answer B and provisioned, however only 4 hours a week eliminates the gains from such.

ChatGPT - running your DynamoDB tables in provisioned mode for only four hours a week can lead to suboptimal cost efficiency if not managed carefully. Recommendations for managing costs with intermittent workloads consider using on-demand capacity mode: For workloads that are truly unpredictable or only run for short periods, on-demand mode can simplify cost management and reduce expenses.

upvoted 1 times

Dz1990 1 year ago

Selected Answer: B

The exam writers are being EXTRA sneaky here! While on-demand seems logical for 4 hours/week usage, the key trap is this phrase: "The company knows how many read and write operations the application performs to the table each second."

This is AWS exam code for "predictable workload - use provisioned mode."

Known/Predictable Workload + Cost Optimization = Provisioned Mode

Unknown/Variable Workload + Cost Optimization = On-Demand Mode

When they give you specific performance data, they expect you to USE it (provisioned), not ignore it (on-demand). AWS exams reward recognizing keyword signals rather than doing obvious cost calculations.

Answer B because the exam tests if you catch the "predictable workload pattern" over pure math. Classic AWS exam trickery!

upvoted 1 times

yangbo 1 year, 2 months ago

Selected Answer: A

if after 4hours change the RCU/WCU ,select B

if after 4hours do not change the RCU/WCU ,select A

upvoted 1 times

mestule 1 year, 10 months ago

Selected Answer: B

DynamoDB On-Demand pricing is about 6.94x the cost of provisioned capacity. If your applications have predictable traffic patterns and you don't mind spending the time to understand those patterns, using DynamoDB's provisioned throughput capacity can save you money.

Also can't set any capacity units for on-demand mode, so A is false in it's premise.

<https://www.serverless.com/blog/dynamodb-on-demand-serverless>

upvoted 5 times

anikolov 1 year, 10 months ago

Selected Answer: A

A: is most cost effective (which is a question/requirement) - 4h per week for Tests purpose

upvoted 2 times

awsgeek75 1 year, 11 months ago

Selected Answer: B

CD are expensive as reserved capacity even with discounts would spend most time in idle mode (over paid, under utilized)

A: On demand is good if you have unpredictable usage,

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.ReadWriteCapacityMode.html#HowItWorks.OnDemand>

B: Provisioned is good if you the usage: <https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/ProvisionedThroughput.html>

"The company knows how many read and write operations the application performs to the table each second during the tests." so ideally they can set this as max

upvoted 4 times

theonlyhero 1 year, 11 months ago

I initially thought it would be A, but when they mentioned "Update the read and write capacity units appropriately." which are automatically set in "on-demand" switched to B

upvoted 2 times

skynetjay 1 year, 11 months ago

Selected Answer: B

Provisioned Mode should be the answer seeing that the workloads are predictable and DynamoDB isn't used for any other thing.
upvoted 2 times

pentium75 1 year, 11 months ago

Selected Answer: B

Agree with B, on-demand mode might not scale fast enough after the DB has been idle for 164 hours. As they know exactly the number of reads and writes per second, should use provisioned mode, and set capacity to 1 RCU and 1 WCU while the DB is not in use.
upvoted 3 times

meenkaza 1 year, 11 months ago

Selected Answer: B

Provisioned Mode (Option B): Provisioned mode allows you to specify the desired read and write capacity units. Since the workload occurs once a week for 4 hours, you can provision the read and write capacity units accordingly to handle the expected load during that time. This can be a more cost-effective option than on-demand pricing for predictable workloads.
upvoted 2 times

A company runs its applications on Amazon EC2 instances. The company performs periodic financial assessments of its AWS costs. The company recently identified unusual spending.

The company needs a solution to prevent unusual spending. The solution must monitor costs and notify responsible stakeholders in the event of unusual spending.

Which solution will meet these requirements?

- A. Use an AWS Budgets template to create a zero spend budget.
- B. Create an AWS Cost Anomaly Detection monitor in the AWS Billing and Cost Management console. **Most Voted**
- C. Create AWS Pricing Calculator estimates for the current running workload pricing details.
- D. Use Amazon CloudWatch to monitor costs and to identify unusual spending.

Correct Answer: B

Community vote distribution

B (100%)

Comments

meenkaza **Highly Voted** 1 year, 11 months ago

Selected Answer: B

AWS Cost Anomaly Detection (Option B): AWS Cost Anomaly Detection is designed to automatically detect unusual spending patterns based on machine learning algorithms. It can identify anomalies and send notifications when it detects unexpected changes in spending. This aligns well with the requirement to prevent unusual spending and notify stakeholders.

upvoted 10 times

awsgeek75 **Most Recent** 1 year, 11 months ago

Selected Answer: B

Unusual spending = Cost anomaly hence B

upvoted 4 times

pentium75 1 year, 11 months ago

Selected Answer: B

<https://aws.amazon.com/aws-cost-management/aws-cost-anomaly-detection/>

upvoted 4 times

A marketing company receives a large amount of new clickstream data in Amazon S3 from a marketing campaign. The company needs to analyze the clickstream data in Amazon S3 quickly. Then the company needs to determine whether to process the data further in the data pipeline.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create external tables in a Spark catalog. Configure jobs in AWS Glue to query the data.
- B. Configure an AWS Glue crawler to crawl the data. Configure Amazon Athena to query the data. **Most Voted**
- C. Create external tables in a Hive metastore. Configure Spark jobs in Amazon EMR to query the data.
- D. Configure an AWS Glue crawler to crawl the data. Configure Amazon Kinesis Data Analytics to use SQL to query the data.

Correct Answer: B

Community vote distribution

B (90%)

10%

Comments

meenkaza **Highly Voted** 2 years, 5 months ago

Selected Answer: B

AWS Glue with Athena (Option B): AWS Glue is a fully managed extract, transform, and load (ETL) service, and Athena is a serverless query service that allows you to analyze data directly in Amazon S3 using SQL queries. By configuring an AWS Glue crawler to crawl the data, you can create a schema for the data, and then use Athena to query the data directly without the need to load it into a separate database. This minimizes operational overhead.

upvoted 7 times

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: B

Neither Glue nor EMR nor Kinesis are used "to query the data"

upvoted 5 times

Johnoppong101 **Most Recent** 1 year, 10 months ago

Selected Answer: B

You've come a loooong way...keep going...

Kinesis Data Analytics applications continuously read and process streaming data in real time.

Data is already at rest in S3. So Athena.

<https://docs.aws.amazon.com/kinesisanalytics/latest/dev/how-it-works.html>

upvoted 3 times

bogdannb 1 year, 11 months ago

Selected Answer: D

It says to quickly analyze the data, Athena can't do it so it's D

upvoted 2 times

JA2018 1 year, 6 months ago

But Athena can directly access S3

upvoted 1 times

JA2018 1 year, 6 months ago

requirement: LEAST operational overhead

upvoted 1 times

OSHOAIB 2 years, 5 months ago

Selected Answer: B

Option B - leverages serverless services that minimise management tasks and allows the company to focus on querying and analysing the data with the LEAST operational overhead.

upvoted 3 times

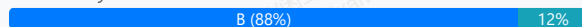
A company runs an SMB file server in its data center. The file server stores large files that the company frequently accesses for up to 7 days after the file creation date. After 7 days, the company needs to be able to access the files with a maximum retrieval time of 24 hours.

Which solution will meet these requirements?

- A. Use AWS DataSync to copy data that is older than 7 days from the SMB file server to AWS.
- B. Create an Amazon S3 File Gateway to increase the company's storage space. Create an S3 Lifecycle policy to transition the data to S3 Glacier Deep Archive after 7 days. **Most Voted**
- C. Create an Amazon FSx File Gateway to increase the company's storage space. Create an Amazon S3 Lifecycle policy to transition the data after 7 days.
- D. Configure access to Amazon S3 for each user. Create an S3 Lifecycle policy to transition the data to S3 Glacier Flexible Retrieval after 7 days.

Correct Answer: B

Community vote distribution



Comments

awsgeek75 **Highly Voted** 1 year, 11 months ago

Selected Answer: B

A: DataSync is not used for this

C: FSx File Gateway requires NFS on both sides so won't work with S3

D: Doesn't say how to transfer data to S3

B: S3 File Gateway will connect SMB to S3. Lifecycle policy will move objects to S3 Glacier Deep Archive which support 12 hours retrieval
<https://aws.amazon.com/blogs/aws/new-amazon-s3-storage-class-glacier-deep-archive/>

upvoted 8 times

PegasusForever **Highly Voted** 1 year, 11 months ago

Answer is B, Amazon S3 File Gateway supports SMB and NFS, Amazon FSx File Gateway SMB for windows workloads.

upvoted 6 times

NayeraB **Most Recent** 1 year, 9 months ago

It feels like C is there just to mess with everyone

upvoted 5 times

pentium75 1 year, 11 months ago

Selected Answer: B

Not C because FSx File Gateway saves files in FSx for Windows file server, not S3.

Not D because users should access the files via SMB

upvoted 4 times

chickenmf 1 year, 9 months ago

"FSx File Gateway saves files in FSx for Windows File Server, not S3"

-- me spreading misinformation on the Internet >:)

upvoted 1 times

chickenmf 1 year, 9 months ago

While it is optimized for compatibility with Windows environments, the files stored in Amazon S3 through the FSx File Gateway are not limited to Windows-only access.

upvoted 2 times

cciesam 1 year, 11 months ago

Selected Answer: B

S3 file gateway supports SMB and S3 Glacier Deep Archive can retrieve data within 12 hours.
<https://aws.amazon.com/storagegateway/file/s3/>

<https://docs.aws.amazon.com/prescriptive-guidance/latest/backup-recovery/amazon-s3-glacier.html>
upvoted 5 times

Roger_Liu 1 year, 11 months ago

Selected Answer: B

I prefer to choose Amazon S3 File Gateway.
<https://docs.aws.amazon.com/filegateway/latest/files3/file-gateway-concepts.html>
upvoted 5 times

meenkaza 1 year, 11 months ago

Selected Answer: C

Amazon FSx File Gateway with S3 Lifecycle policy (Option C): Amazon FSx is a fully managed file storage service, and with a File Gateway, it allows seamless integration between on-premises file servers and AWS storage. By creating an Amazon FSx File Gateway and implementing an S3 Lifecycle policy to transition data to S3 after 7 days, you can achieve the desired storage and retrieval characteristics.
upvoted 3 times

pentium75 1 year, 11 months ago

Wrong. An "FSx File Gateway" stores the files on AWS side in FSx for Windows file server, NOT in S3. Thus you can't apply the "S3 Lifecycle Policy".
upvoted 5 times

A company runs a web application on Amazon EC2 instances in an Auto Scaling group. The application uses a database that runs on an Amazon RDS for PostgreSQL DB instance. The application performs slowly when traffic increases. The database experiences a heavy read load during periods of high traffic.

Which actions should a solutions architect take to resolve these performance issues? (Choose two.)

- A. Turn on auto scaling for the DB instance.
- B. Create a read replica for the DB instance. Configure the application to send read traffic to the read replica. **Most Voted**
- C. Convert the DB instance to a Multi-AZ DB instance deployment. Configure the application to send read traffic to the standby DB instance.
- D. Create an Amazon ElastiCache cluster. Configure the application to cache query results in the ElastiCache cluster. **Most Voted**
- E. Configure the Auto Scaling group subnets to ensure that the EC2 instances are provisioned in the same Availability Zone as the DB instance.

Correct Answer: BD

Community vote distribution

BD (90%)

10%

Comments

awsgeek75 **Highly Voted** 1 year, 11 months ago

Selected Answer: BD

A: RDS DB instance Autoscaling is not a thing
C: You cannot read from standby even if this was done.
E: Does not solve any problem

Correct answer

B: Read replicas distribute load and help improving performance
D: Caching of any kind will help with performance

Remember: "The database experiences a heavy read load during periods of high traffic."
upvoted 10 times

meenkaza **Highly Voted** 1 year, 11 months ago

Selected Answer: BD

B. Create a read replica for the DB instance. Configure the application to send read traffic to the read replica.

By creating a read replica, you offload read traffic from the primary DB instance to the replica, distributing the load and improving overall performance during periods of heavy read traffic.

D. Create an Amazon ElastiCache cluster. Configure the application to cache query results in the ElastiCache cluster.

Amazon ElastiCache can be used to cache frequently accessed data, reducing the load on the database. This is particularly effective for read-heavy workloads, as it allows the application to retrieve data from the cache rather than making repeated database queries.

upvoted 5 times

pentium75 1 year, 11 months ago

ElastiCache requires application changes, "the solutions architect" cannot simply "configure the application to cache query results".
upvoted 3 times

pentium75 1 year, 11 months ago

On second thought, this might still be correct.

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/creating-elasticache-cluster-with-RDS-settings.html>

upvoted 2 times

xBUGx **Most Recent** 1 year, 8 months ago

Selected Answer: BD

RDS auto scaling helps capacity issue, not heavy read workload issue.
upvoted 3 times

06042022 1 year, 11 months ago

Selected Answer: BD

By creating a read replica, you offload read traffic from the primary DB instance to the replica, distributing the load and improving overall performance during periods of heavy read traffic.

Amazon ElastiCache can be used to cache frequently accessed data, reducing the load on the database. This is particularly effective for read-heavy workloads, as it allows the application to retrieve data from the cache rather than making repeated database queries.

upvoted 5 times

Tekk97 1 year, 11 months ago

i think we need Multi az DB, with ElastiCache

upvoted 1 times

pentium75 1 year, 11 months ago

Selected Answer: BD

Not A - There is no such thing as "auto scaling for a DB instance". There is automatic storage scaling, but storage is not the issue here.

B - Yes, read replica will help with "heavy read load"

Not C - "send read traffic to the standby DB instance" does not work

D - "Configure the application ..." might be a bit simplified, but ElastiCache helps with read load

Not E - That might have impact on latency, but not on database load; and all instances in same AZ would be against WAF

upvoted 5 times

OSHOAIB 1 year, 11 months ago

Amazon RDS does support Storage Auto Scaling :)

<https://aws.amazon.com/about-aws/whats-new/2019/06/rds-storage-auto-scaling/>

upvoted 1 times

awsgeek75 1 year, 11 months ago

Storage auto scaling is not same as instance autoscaling. Storage is not a problem here.

upvoted 3 times

Riajul 1 year, 11 months ago

Selected Answer: AB

A and B should be most correct ans

upvoted 3 times

awsgeek75 1 year, 10 months ago

A is autoscaling for DB, it won't fix read problem.

upvoted 2 times

Riajul 1 year, 11 months ago

Should be A and B

upvoted 1 times

A company uses Amazon EC2 instances and Amazon Elastic Block Store (Amazon EBS) volumes to run an application. The company creates one snapshot of each EBS volume every day to meet compliance requirements. The company wants to implement an architecture that prevents the accidental deletion of EBS volume snapshots. The solution must not change the administrative rights of the storage administrator user.

Which solution will meet these requirements with the LEAST administrative effort?

- A. Create an IAM role that has permission to delete snapshots. Attach the role to a new EC2 instance. Use the AWS CLI from the new EC2 instance to delete snapshots.
- B. Create an IAM policy that denies snapshot deletion. Attach the policy to the storage administrator user.
- C. Add tags to the snapshots. Create retention rules in Recycle Bin for EBS snapshots that have the tags.
- D. Lock the EBS snapshots to prevent deletion. **Most Voted**

Correct Answer: D

Community vote distribution

D (68%)

C (32%)

Comments

meenkaza **Highly Voted** 2 years, 5 months ago

Selected Answer: D

Locking EBS Snapshots (Option D): The "lock" feature in AWS allows you to prevent accidental deletion of resources, including EBS snapshots. This can be set at the snapshot level, providing a straightforward and effective way to meet the requirements without changing the administrative rights of the storage administrator user.

upvoted 9 times

KarlosRC **Most Recent** 3 months ago

Selected Answer: C

Lock the EBS snapshots – no such function

upvoted 1 times

In2718 9 months, 3 weeks ago

Selected Answer: C

Chat AI -

Add tags to snapshots. Create retention rules in Recycle Bin for EBS snapshots that have the tags. Recycle Bin allows you to recover snapshots for a retention period if they are deleted (accidentally or intentionally). This prevents permanent loss without removing admin permissions. Minimal administration — just tagging and one-time rule setup.

upvoted 1 times

111380c 10 months, 1 week ago

Selected Answer: C

"Locking" EBS snapshots is not a native feature in AWS (as of now). Snapshot locking is available for Amazon Data Lifecycle Manager (DLM), but it only applies to snapshots created by DLM, and still requires setup effort.

upvoted 1 times

ChhatwaniB 1 year, 1 month ago

Selected Answer: D

<https://docs.aws.amazon.com/ebs/latest/userguide/ebs-snapshot-lock.html>

upvoted 1 times

maoer78 1 year, 3 months ago

Selected Answer: C

Recycle Bin for EBS Snapshots

upvoted 2 times

zdi561 1 year, 4 months ago

Selected Answer: C

D is not right 'Enable the Snapshot Lock feature on important snapshots to prevent any deletion attempts even the administrator. To prevent accidental deletion of EBS volume snapshots, the most effective method is to enable the "Recycle Bin" feature in AWS, which temporarily stores deleted snapshots for a specified retention period, allowing you to recover them if needed

upvoted 2 times

aquarian_ngc 1 year, 11 months ago

correct option D

upvoted 1 times

Gape4 1 year, 11 months ago

Selected Answer: D

I will go for D

upvoted 1 times

awsgeek75 2 years, 5 months ago

D: Exactly what a locked EBS snapshot is used for

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-snapshot-lock.html>

upvoted 4 times

pentium75 2 years, 5 months ago

Selected Answer: D

Typical use case for object lock aka D

upvoted 4 times

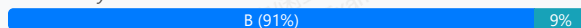
A company's application uses Network Load Balancers, Auto Scaling groups, Amazon EC2 instances, and databases that are deployed in an Amazon VPC. The company wants to capture information about traffic to and from the network interfaces in near real time in its Amazon VPC. The company wants to send the information to Amazon OpenSearch Service for analysis.

Which solution will meet these requirements?

- A. Create a log group in Amazon CloudWatch Logs. Configure VPC Flow Logs to send the log data to the log group. Use Amazon Kinesis Data Streams to stream the logs from the log group to OpenSearch Service.
- B. Create a log group in Amazon CloudWatch Logs. Configure VPC Flow Logs to send the log data to the log group. Use Amazon Kinesis Data Firehose to stream the logs from the log group to OpenSearch Service. **Most Voted**
- C. Create a trail in AWS CloudTrail. Configure VPC Flow Logs to send the log data to the trail. Use Amazon Kinesis Data Streams to stream the logs from the trail to OpenSearch Service.
- D. Create a trail in AWS CloudTrail. Configure VPC Flow Logs to send the log data to the trail. Use Amazon Kinesis Data Firehose to stream the logs from the trail to OpenSearch Service.

Correct Answer: B

Community vote distribution



Comments

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: B

CloudTrail is for logging administrative actions, we need CloudWatch. We want the data in another AWS service (OpenSearch), not Kinesis, thus we need Firehose, not Streams.

upvoted 10 times

meenkaza **Highly Voted** 2 years, 5 months ago

Selected Answer: B

Amazon CloudWatch Logs and VPC Flow Logs (Option B): VPC Flow Logs capture information about the IP traffic going to and from network interfaces in a VPC. By configuring VPC Flow Logs to send the log data to a log group in Amazon CloudWatch Logs, you can then use Amazon Kinesis Data Firehose to stream the logs from the log group to Amazon OpenSearch Service for analysis. This approach provides near real-time streaming of logs to the analytics service.

upvoted 5 times

andybernard411 **Most Recent** 1 year ago

Selected Answer: A

Currently, Firehose does not support the delivery of CloudWatch Logs to Amazon OpenSearch Service destination because Amazon CloudWatch combines multiple log events into one Firehose record and Amazon OpenSearch Service cannot accept multiple log events in one record. As an alternative, you can consider Using subscription filter for Amazon OpenSearch Service in CloudWatch Logs.

<https://docs.aws.amazon.com/firehose/latest/dev/writing-with-cloudwatch-logs.html>

upvoted 1 times

Jacky_S 1 year, 11 months ago

Selected Answer: A

base on the research, it should be Answer A, because question is asking for a "near real time" which Kinesis Data Stream is offering the data with less than 1 second latency. But Kinesis Data Firehose is offering the data with more than 1 second.

<https://docs.aws.amazon.com/opensearch-service/latest/developerguide/integrations.html#integrations-kinesis>

<https://stackoverflow.com/questions/44608274/is-there-any-difference-in-processing-times-between-aws-kinesis-firehose-and-str>

<https://docs.aws.amazon.com/streams/latest/dev/using-other-services-cw-logs.html>

upvoted 1 times

FlyingHawk 1 year, 5 months ago

While Amazon Kinesis Data Streams can stream data, it requires additional setup (e.g., Lambda functions) to process and send logs to OpenSearch Service. This adds complexity compared to Kinesis Data Firehose, which is purpose-built for this use case.

upvoted 2 times

Jacky_S 1 year, 11 months ago

base on the research, it should be Answer A, because question is asking for a "near real time" which Kinesis Data Stream is offering the data with less than 1 second latency. But Kinese Data Firehost is offering the data with more than 1 second.

<https://docs.aws.amazon.com/opensearch-service/latest/developerguide/integrations.html#integrations-kinesis>

<https://stackoverflow.com/questions/44608274/is-there-any-difference-in-processing-times-between-aws-kinesis-firehose-and-str>

upvoted 1 times

zinabu 2 years, 2 months ago

log analysis place= aws cloudwatch log

data capturing on the entire vpc=aws flow log

near real time data analysis and send to OpenSearch service= kinesis data fire hose

upvoted 2 times

1Alpha1 2 years, 4 months ago

Selected Answer: B

OpenSearch patterns for CloudWatch Logs:

1) "Near Real Time": CloudWatch logs --> Subscription Filter --> Kinesis Data Firehose --> Amazon OpenSearch (option *B*)

2) "Real Time": CloudWatch logs --> Subscription Filter --> Lambda --> Amazon OpenSearch

upvoted 5 times

A company is developing an application that will run on a production Amazon Elastic Kubernetes Service (Amazon EKS) cluster. The EKS cluster has managed node groups that are provisioned with On-Demand Instances.

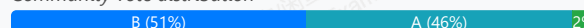
The company needs a dedicated EKS cluster for development work. The company will use the development cluster infrequently to test the resiliency of the application. The EKS cluster must manage all the nodes.

Which solution will meet these requirements MOST cost-effectively?

- A. Create a managed node group that contains only Spot Instances.
- B. Create two managed node groups. Provision one node group with On-Demand Instances. Provision the second node group with Spot Instances. **Most Voted**
- C. Create an Auto Scaling group that has a launch configuration that uses Spot Instances. Configure the user data to add the nodes to the EKS cluster.
- D. Create a managed node group that contains only On-Demand Instances.

Correct Answer: B

Community vote distribution



Comments

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: A

I think the question is easy to misunderstand, whether you should create the whole setup or just the development cluster. But from the wording ("The [production] EKS cluster has (!) managed node groups ... The company needs a dedicated EKS cluster for development work"), I conclude that we should only create the development cluster.

As this will be used "infrequently" for testing purposes only, and it must be "most cost-effective", I'd go with A - new cluster with "one managed node group that contains only Spot instances".

upvoted 10 times

Drew3000 1 year, 8 months ago

I hate this question.... I think I will go with B just because wording also. A company is developing an application that "WILL" run on a production Amazon Elastic Kubernetes Service

upvoted 1 times

awsgeek75 1 year, 11 months ago

The wording of question and options is so confusing. The last line is a throw off also "The EKS cluster must manage all the nodes" Which EKS cluster? A new one or the existing one.

Both A and B are correct depending on how you decipher the question.

I really hope the exam question uses better language!

upvoted 5 times

frmrkc **Highly Voted** 1 year, 10 months ago

Selected Answer: B

This question is convoluted and missing some details.

We need:

- control plane running on on-demand EC2s
- worker nodes running on spot instances

Read this to understand correct solution:

<https://aws.amazon.com/blogs/containers/amazon-eks-now-supports-provisioning-and-managing-ec2-spot-instances-in-managed-node-groups/>

upvoted 7 times

ln2718 **Most Recent** 10 months, 3 weeks ago

Selected Answer: B

The answer should be B, because you could lose the entire spot instances in development and then the developers complain to the CIO and then the CIO fires you for putting everything on spot instances trying to save money, but instead causing an outage that cost the company ten times more if you would have just done a decent job and thought through the real-life situation.

upvoted 1 times

Odd48f2 11 months, 1 week ago

Selected Answer: B

The balance of node groups instances is the best option, for example the cluster has one on-demand EKS managed node group for cluster management and operational tools. This is where customers run workloads like datastores, monitoring tools, and any applications that are not interruption tolerant and needs to remain available. The cluster also should have multiple EKS managed node groups with Spot Instances, each diversified with similarly sized instance types, in order to tap into multiple Spot capacity pools.

This solution, you make sure during the test of production application that the Spot Instances interruption will not affected the results.

<https://aws.amazon.com/blogs/containers/amazon-eks-now-supports-provisioning-and-managing-ec2-spot-instances-in-managed-node-groups/>

upvoted 1 times

bora4motion 1 year, 1 month ago

Selected Answer: A

this is one stupid question with a lot of fluff. The question is around the dev cluster, the answer is A.

upvoted 1 times

tch 1 year, 2 months ago

Selected Answer: B

in the exam.... answer A is too easy...

AWS want the candidate to apply the two workload... the PROD vs DEV...

upvoted 2 times

CTao 1 year, 3 months ago

Selected Answer: A

infrequently + cost effective -> Spot

upvoted 1 times

ieffiong 1 year, 3 months ago

Selected Answer: A

For a development Amazon Elastic Kubernetes Service (Amazon EKS) cluster that is used infrequently to test application resiliency, the most cost-effective solution is to create a managed node group consisting solely of Spot Instances (Option A). Spot Instances offer significant cost savings—up to 90% compared to On-Demand Instances—making them ideal for non-critical, intermittent workloads like development and testing.

upvoted 1 times

Dantecito 1 year, 3 months ago

Selected Answer: B

I'll go with B because no matter what production should not go in spot instances

upvoted 1 times

ieffiong 1 year, 3 months ago

This has nothing to do with production instances though. The question is referring to the development instance.

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: D

Clearly prod and dev are two different envs and two separate EKS cluster. In order to do meaningful resilience test, dev must matches the prod, so D is the answer.

upvoted 1 times

bujuman 1 year, 7 months ago

Selected Answer: B

If we look closer to the last requirement "The EKS cluster must manage all the nodes." Option B is the only feasible and cost-effective one.

upvoted 3 times

FlyingHawk 1 year, 5 months ago

I think this requirment mean to use manage node group to manage all worker nodes, the nodes for control panel will be managed by AWS, so A is correct

upvoted 1 times

1Alpha1 1 year, 10 months ago

Selected Answer: A

Based on the document [1], we can know that only self-managed node group can deploy the container on EC2 dedicated hosts . Which mean that customer need to manually create launch template, auto scaling group, and register it to the EKS cluster. The creation process should be same as general EC2 auto scaling creation. For now, EKS managed node group only supported on-demand and spot.

MOST cost-effectively: *Spot Instances*

<https://repost.aws/questions/QUugoX4f1gRHW0MGHRTHFFFA/how-to-create-eks-cluster-with-dedicated-host-node-group>
upvoted 3 times

anikolov 1 year, 10 months ago

Selected Answer: A

"The company will use the development cluster infrequently to test the resiliency of the application" = Spot instances = cost effective
upvoted 2 times

06042022 1 year, 11 months ago

Selected Answer: B

The keywords are infrequent and resiliency..

This solution allows you to have a mix of On-Demand Instances and Spot Instances within the same EKS cluster. You can use the On-Demand Instances for the development work where you need dedicated resources and then leverage Spot Instances for testing the resiliency of the application. Spot Instances are generally more cost-effective but can be terminated with short notice, so using a combination of On-Demand and Spot Instances provides a balance between cost savings and stability.

Option A (Create a managed node group that contains only Spot Instances) might be cost-effective, but it could introduce potential challenges for tasks that require dedicated resources and might not be the best fit for all scenarios.

upvoted 3 times

mr123dd 1 year, 11 months ago

Selected Answer: B

The GBT vote A, I know the spot instance is the cheapest, but the question says "dedicated EKS cluster for development", so I vote B
upvoted 2 times

FlyingHawk 1 year, 5 months ago

A dedicated EKS cluster means a separate EKS cluster that is isolated from the production cluster. This ensures that development work does not interfere with the production environment

upvoted 1 times

OSHOAIB 1 year, 11 months ago

Selected Answer: A

Option A leverages the cost savings of Spot Instances, which is ideal for a development environment where the application is tested infrequently, and there is flexibility in when the nodes can be interrupted. This aligns with the goal of cost-efficiency and takes advantage of EKS's ability to manage the nodes directly.

upvoted 1 times

cciesam 1 year, 11 months ago

Selected Answer: B

B is the best ans.

upvoted 1 times

pentium75 1 year, 11 months ago

Why do you think so?

upvoted 2 times

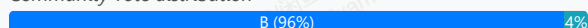
A company stores sensitive data in Amazon S3. A solutions architect needs to create an encryption solution. The company needs to fully control the ability of users to create, rotate, and disable encryption keys with minimal effort for any data that must be encrypted.

Which solution will meet these requirements?

- A. Use default server-side encryption with Amazon S3 managed encryption keys (SSE-S3) to store the sensitive data.
- B. Create a customer managed key by using AWS Key Management Service (AWS KMS). Use the new key to encrypt the S3 objects by using server-side encryption with AWS KMS keys (SSE-KMS). **Most Voted**
- C. Create an AWS managed key by using AWS Key Management Service (AWS KMS). Use the new key to encrypt the S3 objects by using server-side encryption with AWS KMS keys (SSE-KMS).
- D. Download S3 objects to an Amazon EC2 instance. Encrypt the objects by using customer managed keys. Upload the encrypted objects back into Amazon S3.

Correct Answer: B

Community vote distribution



Comments

meenkaza **Highly Voted** 2 years, 5 months ago

Selected Answer: B

SSE-KMS with Customer Managed Key (Option B): This option allows you to create a customer managed key using AWS KMS. With a customer managed key, you have full control over key lifecycle management, including the ability to create, rotate, and disable keys with minimal effort. SSE-KMS also integrates with AWS Identity and Access Management (IAM) for fine-grained access control.

upvoted 11 times

MatAlves **Highly Voted** 1 year, 8 months ago

Selected Answer: B

Having both awsgeek75 and pentium75 in the comment section makes me more confident about my own answers.

upvoted 6 times

rubiteb **Most Recent** 2 years, 3 months ago

Selected Answer: C

Customer needs to control the 'user's ability' and not the management of the keys. Option C will prevent users to have this ability.

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

Has to be customer manages to AC are not useful

D is just wrong way of doing this

B give full control to customer while using S3 server side encryption.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: B

A and C do not allow the company "to fully control the ability of users to create, rotate, and disable encryption keys". D is anything but "minimal effort".

upvoted 4 times

Riajul 2 years, 5 months ago

Selected Answer: B

Option B should be correct
upvoted 3 times

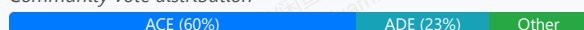
A company wants to back up its on-premises virtual machines (VMs) to AWS. The company's backup solution exports on-premises backups to an Amazon S3 bucket as objects. The S3 backups must be retained for 30 days and must be automatically deleted after 30 days.

Which combination of steps will meet these requirements? (Choose three.)

- A. Create an S3 bucket that has S3 Object Lock enabled. **Most Voted**
- B. Create an S3 bucket that has object versioning enabled.
- C. Configure a default retention period of 30 days for the objects. **Most Voted**
- D. Configure an S3 Lifecycle policy to protect the objects for 30 days.
- E. Configure an S3 Lifecycle policy to expire the objects after 30 days. **Most Voted**
- F. Configure the backup solution to tag the objects with a 30-day retention period

Correct Answer: ACE

Community vote distribution



Comments

te1973 **Highly Voted** 2 years ago

This is a good example for a completely non-sense AWS exam question. In order to delete the object like requested in the question you need (E). This is required in either versioned or non-versioned buckets. Basically the task is done here. But let's assume we want to make it extra secure and retain the files for 30 days. Then we need object lock (A). You cannot have object lock without versioning (B). You also need to set a retention period then (C). So you either have A,B,C,E or you have E. Choosing exactly 3 options is completely nonsense here. But what do I know.

upvoted 9 times

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: ACE

In theory, E alone would be enough because the objects are "retained for 30 days" without any configuration as long as no one deletes them. But let's assume that they want us to prevent deletion.

A: Yes, required to prevent deletion. Object Lock requires Versioning, so if we "create an S3 bucket that has S3 Object Lock enabled" that this also has object versioning enabled, otherwise we would not be able to create it.

B: No. We need versioning, but we cannot "create" the bucket twice. If we create it "with object lock enabled" then versioning is enabled too, but NOT the other way round (creating it with versioning enabled will not automatically enable object lock).

upvoted 5 times

pentium75 2 years, 5 months ago

C: Yes, "default retention period" specifies how long object lock will be applied to new objects by default, we need this to protect objects from deletion.

D: No, S3 Lifecycle Policy can "transition" or "expire" but not "protect".

E: Yes, this will delete the objects after 30 days (C just removes the object lock after 30 days but does not delete the objects).

F: No, 'tag with a retention period' is not common AWS wording, "tags" are something different in AWS context

upvoted 4 times

andrea10 **Most Recent** 10 months, 1 week ago

Selected Answer: BDE

A (S3 Object Lock) is used for legal/compliance needs to prevent deletion — not for automatic deletion. Since you want backups to be deleted after 30 days, Object Lock is not needed and would actually get in the way.

upvoted 1 times

ln2718 10 months, 2 weeks ago

Selected Answer: ABE

The correct answer is A, B, E - the answer from another question bank and ChatGPT, --- S3 Object Lock requires S3 Versioning to be enabled on the bucket.

upvoted 1 times

Dz1990 1 year ago

Selected Answer: ACE

After analyzing the community debate, I'm sticking with ACE.

AWS Logic: They wouldn't give you two "create bucket" options (A and B) if both were needed. When you "Create bucket with Object Lock enabled" (A), versioning is automatically included - you can't have Object Lock without it.

The sneaky part: AWS is testing if you understand Object Lock includes versioning, not if you'll pick redundant options. If they wanted both A and B, they'd phrase them as separate configuration steps, not both as "create bucket" actions.

Most experienced test-takers chose ACE recognizing this pattern. AWS tests concepts, not redundant configurations.

ACE is correct.

upvoted 2 times

tch 1 year, 2 months ago

Selected Answer: ACE

To manage the lifecycle of your objects, create an S3 Lifecycle configuration for your bucket. An S3 Lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. There are two types of actions:

- Transition actions – These actions define when objects transition to another storage class. For example, you might choose to transition objects to the S3 Standard-IA storage class 30 days after creating them, or archive objects to the S3 Glacier Flexible Retrieval storage class one year after creating them.
- Expiration actions – These actions define when objects expire. Amazon S3 deletes expired objects on your behalf. For example, you might choose to expire objects after they have been stored for a regulatory compliance period.

upvoted 1 times

tch 1 year, 2 months ago

To use S3 Object Lock with a bucket (or objects within a bucket), you must first enable versioning for the bucket, as you won't be able to turn versioning on later. Retention periods and legal holds apply to individual object versions

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: ABE

C is not right because a standard S3 object has no default retention period, meaning it can be deleted at any time unless you specifically configure a retention period using the "S3 Object Lock" feature, which allows you to set a custom retention period for individual objects or an entire bucket, with the minimum retention period being one day, B is right because you have to do it, it does not matter you have to do it to lock an object or do it yourself. D is a general description of action A.

upvoted 1 times

omega_coaching 1 year, 5 months ago

Selected Answer: ABE

For enabling S3 object lock we need to enable Bucket versioning.. SO and must be ABE

upvoted 2 times

MatAlves 1 year, 8 months ago

Selected Answer: ACE

"Object Lock works only in buckets that have S3 Versioning enabled"

However, we can't have 2 options (A and B) telling to create the bucket. So, A is only possible if versioning is already enabled.

We need retention period (C), since this is not a case for legal holds:

"Object Lock provides two ways to manage object retention: retention periods and legal holds."

E - obvious reasons.

Ref. <https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

upvoted 1 times

chwieobjom 1 year, 9 months ago

this is shit

upvoted 2 times

mohammadthainat 2 years, 2 months ago

Selected Answer: ACE

1- The S3 backups must be retained for 30 days -->

For that you must enable S3 Object Lock (versioning must be enabled) in Compliance Mode and set Retention Period to 30 days. Thus, to achieve this you need 3 options <A, B and C>

2- The S3 backups must be automatically deleted after 30 days. -->

For that you must Create Lifecycle Rule with action Expire current versions of objects (versioning must be enabled) and set Expiration Period to 30 days. Thus to achieve this you need 2 options <B and E>

 is a must here as both locking the objects and deleting them can't be achieved without it. But, when choosing "A.Create an S3 bucket that has S3 Object Lock enabled." this explicitly indicated that versioning is enabled in your bucket.

upvoted 4 times

awsgeek75 2 years, 4 months ago

Selected Answer: ADE

B: No versioning is required

D: Lifecycle is for transitioning or expiring. There is no protection lifecycle policy

F: No such tag

Enable object lock, retain for 30 days (<https://docs.aws.amazon.com/AmazonS3/latest/userguide/batch-ops-retention-date.html>) and expire after 30 days.

upvoted 5 times

MatAlves 1 year, 8 months ago

"Object Lock works only in buckets that have S3 Versioning enabled"

However, I still agree with ACE, since the bucket has already been created, so we can't have 2 answers telling to create the bucket.

And yes, for this case, we need retention period:

"Object Lock provides two ways to manage object retention: retention periods and legal holds."

Ref. <https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

upvoted 3 times

awsgeek75 2 years, 4 months ago

I meant ACE! not ADE!

upvoted 3 times

PegasusForever 2 years, 5 months ago

ABE -> <https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

A. Create an S3 bucket that has S3 Object Lock enabled. -> You set a Retention period of 30 days with this feature.

B. Create an S3 bucket that has object versioning enabled -> Object Lock works only in buckets that have S3 Versioning enabled

E. Configure an S3 Lifecycle policy to expire the objects after 30 days. -> It is valid using the lifecycle policy.

upvoted 2 times

PegasusForever 2 years, 5 months ago

After analyzing the question deeply and reading: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>, I keep A and B, change E per C.

A. Create an S3 bucket that has S3 Object Lock enabled.

B. Create an S3 bucket that has object versioning enabled.

Change E must be automatically deleted after 30 days (objects will be marked as expired not deleted). per C. Configure a default retention period of 30 days for the objects. It feature delete the object.

upvoted 2 times

PegasusForever 2 years, 5 months ago

Selected Answer: ACE

A. Create an S3 bucket that has S3 Object Lock enabled. Enable the S3 Object Lock feature on S3.

C. Configure a default retention period of 30 days for the objects. To lock the objects for 30 days.

E. Configure an S3 Lifecycle policy to expire the objects after 30 days. -> to delete the objects after 30 days.

upvoted 2 times

cciesam 2 years, 5 months ago

Selected Answer: ACE

ACE is the correct ans.

upvoted 5 times

Riajul 2 years, 5 months ago

Selected Answer: ADE

ADE should be correct

upvoted 1 times

pentium75 2 years, 5 months ago

Why?

S3 Lifecycle Policy can "transition" or "expire" but not "protect"

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/lifecycle-expire-general-considerations.html>

upvoted 2 times

Naijaboy99 2 years, 5 months ago

Correct Answer is A C E

upvoted 2 times

meenkaza 2 years, 5 months ago

Selected Answer: ADE

A. Create an S3 bucket that has S3 Object Lock enabled.

S3 Object Lock provides the ability to enforce retention periods on objects, preventing deletion or modification for a specified duration.

D. Configure an S3 Lifecycle policy to protect the objects for 30 days.

By configuring a lifecycle policy, you can define a transition action to move objects to the S3 Glacier storage class (or any other storage class) after 30 days.

E. Configure an S3 Lifecycle policy to expire the objects after 30 days.

upvoted 1 times

pentium75 2 years, 5 months ago

S3 Lifecycle Policy can "transition" or "expire" but not "protect"

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/lifecycle-expire-general-considerations.html>

upvoted 3 times

tch 1 year, 2 months ago

To manage the lifecycle of your objects, create an S3 Lifecycle configuration for your bucket. An S3 Lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. There are two types of actions:

- Transition actions – These actions define when objects transition to another storage class. For example, you might choose to transition objects to the S3 Standard-IA storage class 30 days after creating them, or archive objects to the S3 Glacier Flexible Retrieval storage class one year after creating them.

-

- Expiration actions – These actions define when objects expire. Amazon S3 deletes expired objects on your behalf. For example, you might choose to expire objects after they have been stored for a regulatory compliance period.

upvoted 1 times

A solutions architect needs to copy files from an Amazon S3 bucket to an Amazon Elastic File System (Amazon EFS) file system and another S3 bucket. The files must be copied continuously. New files are added to the original S3 bucket consistently. The copied files should be overwritten only if the source file changes.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create an AWS DataSync location for both the destination S3 bucket and the EFS file system. Create a task for the destination S3 bucket and the EFS file system. Set the transfer mode to transfer only data that has changed. **Most Voted**
- B. Create an AWS Lambda function. Mount the file system to the function. Set up an S3 event notification to invoke the function when files are created and changed in Amazon S3. Configure the function to copy files to the file system and the destination S3 bucket.
- C. Create an AWS DataSync location for both the destination S3 bucket and the EFS file system. Create a task for the destination S3 bucket and the EFS file system. Set the transfer mode to transfer all data.
- D. Launch an Amazon EC2 instance in the same VPC as the file system. Mount the file system. Create a script to routinely synchronize all objects that changed in the origin S3 bucket to the destination S3 bucket and the mounted file system.

Correct Answer: A

Community vote distribution

A (100%)

Comments

meenkaza **Highly Voted** 1 year, 11 months ago

AWS DataSync (Option A): AWS DataSync is designed for efficient and reliable copying of data between different storage solutions. By setting up an AWS DataSync task with the transfer mode set to transfer only data that has changed, you ensure that only the new or modified files are copied. This minimizes data transfer and operational overhead.

upvoted 5 times

pentium75 1 year, 11 months ago

Actually this is not fully correct:

"By setting up an AWS DataSync task with the transfer mode set to transfer only data that has changed, you ensure that only the new or modified files are copied."

"Transfer only data that has changed ... copies only the data and metadata that differs between the source and destination location."

So, if we have a source with existing items and an empty destination (like in this example), "transfer only data that has changed" will transfer all the existing items though in the true sense of the word they have not "changed".

upvoted 4 times

mohammadthainat **Most Recent** 1 year, 8 months ago

Selected Answer: A

DataSync will do an Initial Scan of both S3 buckets. Identifying Differences. Then, Transferring Changes, so technically DataSync will transfer All the data at first run then it will only transfer newly added/modified objects subsequently.

upvoted 4 times

Kezuko 1 year, 8 months ago

Have always did this using B, guess now that I know A is less operational

upvoted 4 times

awsgeek75 1 year, 11 months ago

Selected Answer: A

BD are more operation overhead although B can work in principle

AC uses managed service to transfer data. A fulfils the requirement of "copied files should be overwritten only if the source file changes" so A is

correct. B will just copy regardless of the change
upvoted 2 times

awsgeek75 1 year, 11 months ago

Meant C will transfer everything and copy data without comparing for change
upvoted 2 times

pentium75 1 year, 11 months ago

Selected Answer: A

Transfer only data that has changed – DataSync copies only the data and metadata that differs between the source and destination location.

Transfer all data – DataSync copies everything in the source to the destination without comparing differences between the locations.

<https://docs.aws.amazon.com/datasync/latest/userguide/configure-metadata.html>

(B would work too but is more "operational overhead.")

upvoted 4 times

cciesam 1 year, 11 months ago

Selected Answer: A

ans: A

upvoted 3 times

A company uses Amazon EC2 instances and stores data on Amazon Elastic Block Store (Amazon EBS) volumes. The company must ensure that all data is encrypted at rest by using AWS Key Management Service (AWS KMS). The company must be able to control rotation of the encryption keys.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create a customer managed key. Use the key to encrypt the EBS volumes. **Most Voted**
- B. Use an AWS managed key to encrypt the EBS volumes. Use the key to configure automatic key rotation.
- C. Create an external KMS key with imported key material. Use the key to encrypt the EBS volumes.
- D. Use an AWS owned key to encrypt the EBS volumes.

Correct Answer: A

Community vote distribution

A (88%)

12%

Comments

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: A

"Able to control rotation of the encryption keys" = customer managed key (created by AWS but managed by the customer in KMS)
upvoted 6 times

meenkaza **Highly Voted** 1 year, 11 months ago

Selected Answer: A

option A (Create a customer managed key. Use the key to encrypt the EBS volumes) is the most suitable option with the least operational overhead for the given requirements.
upvoted 5 times

Chen77 **Most Recent** 1 year, 3 months ago

Selected Answer: B

The company must ensure encryption at rest using AWS KMS and must control key rotation with minimal operational overhead.

AWS Managed Keys are automatically managed by AWS, including automatic key rotation every year.
This approach eliminates manual key management, reducing administrative effort.
upvoted 1 times

AAbirdy 1 year, 11 months ago

Selected Answer: A

The company must be able to control rotation of the encryption keys = customer managed key
upvoted 5 times

awsgeek75 1 year, 11 months ago

Selected Answer: A

"The company must be able to control rotation of the encryption keys."
BD does not allow company owned keys
C is too much operational overhead
upvoted 4 times

dikshya1233 1 year, 11 months ago

Selected Answer: B

The solution that meets the requirements with the LEAST operational overhead is:

B. Use an AWS managed key to encrypt the EBS volumes. Use the key to configure automatic key rotation.

With AWS managed keys (AWS managed CMKs), AWS takes care of key management tasks, including key rotation. This reduces operational overhead as AWS automatically handles key rotation without requiring manual intervention. It is a convenient option for users who want to ensure encryption at rest with minimal effort in managing encryption keys.

upvoted 2 times

awsgeek75 1 year, 11 months ago

AWS Managed keys don't meet the requirements "The company must be able to control rotation of the encryption keys."

upvoted 3 times

Shobhit2021 1 year, 11 months ago

Selected Answer: A

A is correct option

upvoted 2 times

fea9bdf 1 year, 11 months ago

Answer is C

Details are on this link below:

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/default-bucket-encryption.html>

Amazon S3 buckets have bucket encryption enabled by default, and new objects are automatically encrypted by using server-side encryption with Amazon S3 managed keys (SSE-S3). This encryption applies to all new objects in your Amazon S3 buckets, and comes at no cost to you.

If you need more control over your encryption keys, such as managing key rotation and access policy grants, you can elect to use server-side encryption with AWS Key Management Service (AWS KMS) keys (SSE-KMS), or dual-layer server-side encryption with AWS KMS keys (DSSE-KMS). For more information about SSE-KMS, see [Specifying server-side encryption with AWS KMS \(SSE-KMS\)](#). For more information about DSSE-KMS, see [Using dual-layer server-side encryption with AWS KMS keys \(DSSE-KMS\)](#).

upvoted 1 times

pentium75 1 year, 11 months ago

How does this relate to answer C? With "imported key material" you cannot "control rotation of the encryption keys" (except by importing new keys). SSE-KMS (as mentioned in your explanation = customer managed key = A

upvoted 2 times

Riajul 1 year, 11 months ago

Should be option A

upvoted 2 times

Naijaboy99 1 year, 11 months ago

option B is the correct answer with least operational overhead on admins

upvoted 1 times

Naijaboy99 1 year, 11 months ago

@meenkaza was right the answer is A

upvoted 3 times

OSHOAIB 1 year, 11 months ago

AWS managed keys do allow for automatic rotation, but the company does NOT have control over the rotation - AWS manages this automatically without company intervention.

upvoted 2 times

A company needs a solution to enforce data encryption at rest on Amazon EC2 instances. The solution must automatically identify noncompliant resources and enforce compliance policies on findings.

Which solution will meet these requirements with the LEAST administrative overhead?

- A. Use an IAM policy that allows users to create only encrypted Amazon Elastic Block Store (Amazon EBS) volumes. Use AWS Config and AWS Systems Manager to automate the detection and remediation of unencrypted EBS volumes. **Most Voted**
- B. Use AWS Key Management Service (AWS KMS) to manage access to encrypted Amazon Elastic Block Store (Amazon EBS) volumes. Use AWS Lambda and Amazon EventBridge to automate the detection and remediation of unencrypted EBS volumes.
- C. Use Amazon Macie to detect unencrypted Amazon Elastic Block Store (Amazon EBS) volumes. Use AWS Systems Manager Automation rules to automatically encrypt existing and new EBS volumes.
- D. Use Amazon Inspector to detect unencrypted Amazon Elastic Block Store (Amazon EBS) volumes. Use AWS Systems Manager Automation rules to automatically encrypt existing and new EBS volumes.

Correct Answer: A

Community vote distribution

A (96%)

4%

Comments

meenkaza **Highly Voted** 1 year, 11 months ago

Selected Answer: A

IAM Policy and AWS Config (Option A): By creating an IAM policy that allows users to create only encrypted EBS volumes, you proactively prevent the creation of unencrypted volumes. Using AWS Config, you can set up rules to detect noncompliant resources, and AWS Systems Manager Automation can be used for automated remediation. This approach provides a proactive and automated solution.

upvoted 13 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: A

Not B - because when you want to detect unencrypted EBS volumes using Lambda and EventBridge, the codes you'll need to compose is so much more than you think. Just leave that whole shebang to AWS Config and then you'll choose A accordingly.

upvoted 1 times

88f8032 1 year, 7 months ago

Selected Answer: B

Isn't B simpler?

upvoted 1 times

awsgeek75 1 year, 11 months ago

Selected Answer: A

B: Too much work

C: Macie is for PII and sensitive data not for encrypted volumes

D: Inspector for OS patching and vulnerability detections

upvoted 2 times

f2e2419 1 year, 11 months ago

why not B?

upvoted 1 times

OSHOAIB 1 year, 11 months ago

Selected Answer: A

Option A - enforces the creation of encrypted volumes via IAM policies and uses AWS Config for detection and AWS Systems Manager for remediation with the LEAST administrative overhead.

upvoted 3 times

pentium75 1 year, 11 months ago

Selected Answer: A

A as exactly described here: <https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/automatically-encrypt-existing-and-new-amazon-ebs-volumes.html>

Not B, that could in theory work but would be massive operational overhead

Not C, Macie detects PII data, not unencrypted volumes

Not D, Inspector detects vulnerabilities, not unencrypted volumes

upvoted 4 times

A company is migrating its multi-tier on-premises application to AWS. The application consists of a single-node MySQL database and a multi-node web tier. The company must minimize changes to the application during the migration. The company wants to improve application resiliency after the migration.

Which combination of steps will meet these requirements? (Choose two.)

- A. Migrate the web tier to Amazon EC2 instances in an Auto Scaling group behind an Application Load Balancer. **Most Voted**
- B. Migrate the database to Amazon EC2 instances in an Auto Scaling group behind a Network Load Balancer.
- C. Migrate the database to an Amazon RDS Multi-AZ deployment. **Most Voted**
- D. Migrate the web tier to an AWS Lambda function.
- E. Migrate the database to an Amazon DynamoDB table.

Correct Answer: AC

Community vote distribution

AC (100%)

Comments

meenkaza **Highly Voted** 1 year, 11 months ago

Selected Answer: AC

Web Tier Migration (Option A): Migrating the web tier to Amazon EC2 instances in an Auto Scaling group behind an Application Load Balancer (ALB) provides horizontal scalability, automatic scaling, and improved resiliency. Auto Scaling helps in managing and maintaining the desired number of EC2 instances based on demand, and the ALB distributes incoming traffic across multiple instances.

Database Migration to Amazon RDS Multi-AZ (Option C): Migrating the database to Amazon RDS in a Multi-AZ deployment provides high availability and automatic failover. In a Multi-AZ deployment, Amazon RDS maintains a standby replica in a different Availability Zone, and in the event of a failure, it automatically promotes the replica to the primary instance. This enhances the resiliency of the database.

upvoted 11 times

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: AC

- A - ALB is ideal for web application
- B - NLB would work too but ALB is better
- C - same functionality as on-premises just with 'improved resiliency'
- D - would require significant "changes to the application"
- E - would require significant "changes to the application"

upvoted 5 times

fea9bdf **Most Recent** 1 year, 11 months ago

Also Dynamo DB is noSQL, that can not be an option here

upvoted 3 times

Naijaboy99 1 year, 11 months ago

option A C

upvoted 1 times

A company wants to migrate its web applications from on premises to AWS. The company is located close to the eu-central-1 Region. Because of regulations, the company cannot launch some of its applications in eu-central-1. The company wants to achieve single-digit millisecond latency.

Which solution will meet these requirements?

- A. Deploy the applications in eu-central-1. Extend the company's VPC from eu-central-1 to an edge location in Amazon CloudFront.
- B. Deploy the applications in AWS Local Zones by extending the company's VPC from eu-central-1 to the chosen Local Zone. **Most Voted**
- C. Deploy the applications in eu-central-1. Extend the company's VPC from eu-central-1 to the regional edge caches in Amazon CloudFront.
- D. Deploy the applications in AWS Wavelength Zones by extending the company's VPC from eu-central-1 to the chosen Wavelength Zone.

Correct Answer: B

Community vote distribution

B (78%)

D (22%)

Comments

pentium75 **Highly Voted** 1 year, 11 months ago

Selected Answer: B

"AWS Local Zones are a type of AWS infrastructure deployment that place compute, storage, database, and other select services closer to large population, industry, and IT centers, enabling you to deliver applications that require single-digit millisecond latency to end-users."

A and C tell us to "deploy the applications in eu-central-1" which is exactly what we're not supposed to do.

AWS Wavelength zones are AWS deployments in CSP's networks, has nothing to do with this question.

https://aws.amazon.com/about-aws/global-infrastructure/localzones/features/?nc1=h_ls

upvoted 10 times

awsgeek75 **Highly Voted** 1 year, 11 months ago

Selected Answer: B

AC is not right "Because of regulations, the company cannot launch some of its applications in eu-central-1"

D: AWS Wavelength is for mobile network

B: Local Zones can be used to launch apps close to a region but not in a region like EUC1 so this works

upvoted 7 times

bodakrishna **Most Recent** 1 year, 9 months ago

Correct B:

AWS Local Zones are an extension of AWS infrastructure and bring AWS services closer to end-users, providing ultra-low latency for applications that require single-digit millisecond latencies. By deploying the applications in AWS Local Zones, the company can meet the latency requirements while also complying with regulations that prevent certain applications from being hosted in the eu-central-1 Region.

upvoted 6 times

OSHOAIB 1 year, 11 months ago

Selected Answer: B

Option B - AWS Local Zones place AWS compute, storage, database, and other select services closer to end-users. This would allow the company to deploy applications within geographic proximity to eu-central-1 without being directly in the region, potentially meeting regulatory requirements and achieving low latency.

Whereas Option D - AWS Wavelength Zones are designed to provide developers the ability to build applications that deliver single-digit millisecond latencies to MOBILE and connected devices. And it's more focused on 5G Apps and may not be directly relevant to Web Apps hosting.

upvoted 2 times

pdragon1981 1 year, 11 months ago

Selected Answer: B

I would go also for B, was in doubt from B or D but I agree with pentium75 the wavelength zones are not designed for this use case however AWS local zones can provide single-digit millisecond latency as described in the link

<https://aws.amazon.com/about-aws/global-infrastructure/localzones/>

upvoted 2 times

Naijaboy99 1 year, 11 months ago

option B

upvoted 4 times

meenkaza 1 year, 11 months ago

Selected Answer: D

AWS Wavelength (Option D): AWS Wavelength Zones bring AWS services to the edge of the 5G network, providing ultra-low latency for applications that require single-digit millisecond latencies. Deploying applications in Wavelength Zones allows the company to extend its VPC from the eu-central-1 Region to the chosen Wavelength Zone, providing the required low-latency access.

upvoted 6 times

pentium75 1 year, 11 months ago

"Wavelength Zones are AWS infrastructure deployments that embed AWS compute and storage services within communications service providers' (CSP) 5G networks". They reduce latency for mobile users in the CSP's network, but this is not asked here. Local Zones provide "single-digit millisecond latency".

upvoted 3 times

Roger_Liu 1 year, 11 months ago

It looks like D is correct from diagram in the following url.

<https://docs.aws.amazon.com/wavelength/latest/developerguide/how-wavelengths-work.html>

upvoted 1 times

A company's ecommerce website has unpredictable traffic and uses AWS Lambda functions to directly access a private Amazon RDS for PostgreSQL DB instance. The company wants to maintain predictable database performance and ensure that the Lambda invocations do not overload the database with too many connections.

What should a solutions architect do to meet these requirements?

- A. Point the client driver at an RDS custom endpoint. Deploy the Lambda functions inside a VPC.
- B. Point the client driver at an RDS proxy endpoint. Deploy the Lambda functions inside a VPC. **Most Voted**
- C. Point the client driver at an RDS custom endpoint. Deploy the Lambda functions outside a VPC.
- D. Point the client driver at an RDS proxy endpoint. Deploy the Lambda functions outside a VPC.

Correct Answer: B

Community vote distribution

B (100%)

Comments

ogerber **Highly Voted** 1 year, 10 months ago

Selected Answer: B

Option B.

Reduce number of connection to RDS -> RDS Proxy.

"A Lambda function that's outside of a VPC can't access an RDS instance that's inside a VPC."

<https://repost.aws/knowledge-center/connect-lambda-to-an-rds-instance>

upvoted 10 times

Kezuko **Highly Voted** 1 year, 8 months ago

Selected Answer: B

Have to be inside VPC in order to access the RDS instance for Lambda

upvoted 5 times

ogerber **Most Recent** 1 year, 10 months ago

Option B.

Reduce number of connection to RDS -> RDS Proxy.

"A Lambda function that's outside of a VPC can't access an RDS instance that's inside a VPC."

<https://repost.aws/knowledge-center/connect-lambda-to-an-rds-instance>

upvoted 4 times

Moon239 1 year, 10 months ago

Same as question 802 in SAA-C02

upvoted 4 times

A company is creating an application. The company stores data from tests of the application in multiple on-premises locations.

The company needs to connect the on-premises locations to VPCs in an AWS Region in the AWS Cloud. The number of accounts and VPCs will increase during the next year. The network architecture must simplify the administration of new connections and must provide the ability to scale.

Which solution will meet these requirements with the LEAST administrative overhead?

- A. Create a peering connection between the VPCs. Create a VPN connection between the VPCs and the on-premises locations.
- B. Launch an Amazon EC2 instance. On the instance, include VPN software that uses a VPN connection to connect all VPCs and on-premises locations.
- C. Create a transit gateway. Create VPC attachments for the VPC connections. Create VPN attachments for the on-premises connections. **Most Voted**
- D. Create an AWS Direct Connect connection between the on-premises locations and a central VPC. Connect the central VPC to other VPCs by using peering connections.

Correct Answer: C

Community vote distribution

C (100%)

Comments

1Alpha1 **Highly Voted** 2 years, 4 months ago

Selected Answer: C

multiple on-premises locations + increasing number of accounts and VPCs --> connections using *transit gateway*
upvoted 5 times

Andy_09 **Highly Voted** 2 years, 4 months ago

Think C would be the correct answer here.
upvoted 5 times

Tendu **Most Recent** 1 year, 3 months ago

Selected Answer: C

Transit Gateway is a way to go in this situation.
upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

A - OK but for every new VPC or account, you need to manually create new peering connections and VPNs. Too much administrative overhead!

B - Launching EC2s are never an easy job. You not only need to do everything manually mentioned in A, but now you also have to manage the EC2 instance and VPN software. Plus, if the EC2 instance goes down, all connectivity is lost, which is called "single-point of failure".

C - Everything seems cool, it's just uhm lightly higher initial setup cost, but it saves time and money in the long run.

D - Direct Connect provides fast & reliable connections, but just like A & B, as the number of VPCs grows, you'll go crazy due to all those unmanageable nodes.

upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: C

Anytime I see "the number of VPCs will increase", I immediately look for "transit gateway" as the least administrative overhead.

upvoted 2 times

MatAlves 1 year, 8 months ago

"AWS Transit Gateway provides a hub and spoke design for connecting VPCs and on-premises networks as a fully managed service without requiring you to provision third-party virtual appliances. No VPN overlay is required, and AWS manages high availability and scalability."

<https://docs.aws.amazon.com/whitepapers/latest/building-scalable-secure-multi-vpc-network-infrastructure/transit-gateway.html#:~:text=AWS%20Transit%20Gateway%20provides%20a,manages%20high%20availability%20and%20scalability.>

upvoted 2 times

MatAlves 1 year, 8 months ago

Transit Gateway enables customers to connect thousands of VPCs. You can attach all your hybrid connectivity (VPN and Direct Connect connections) to a single gateway, consolidating and controlling your organization's entire AWS routing configuration in one place (refer to the following figure)

upvoted 2 times

ogerber 2 years, 3 months ago

Selected Answer: C

high number of accounts and VPC to connect to on prem -> exactly the transit gateway use case

upvoted 3 times

KZ06 2 years, 4 months ago

Hi,

Seems like after question 684, the discussions are quite less and seem recent comments. Are these new sets of questions updated?

Anyone having any idea around this?

upvoted 2 times

MatAlves 1 year, 8 months ago

Anytime I see "the number of VPCs will increase", I immediately look for "transit gateway" as the least administrative overhead.

upvoted 2 times

Cali182 2 years, 4 months ago

Selected Answer: C

vote for C

upvoted 3 times

EZforeverman 2 years, 4 months ago

I think it's C. LEAST administrative overhead. D can work but AWS direct connection and VPC peering configuration require too much administrative overhead

upvoted 3 times

A company that uses AWS needs a solution to predict the resources needed for manufacturing processes each month. The solution must use historical values that are currently stored in an Amazon S3 bucket. The company has no machine learning (ML) experience and wants to use a managed service for the training and predictions.

Which combination of steps will meet these requirements? (Choose two.)

- A. Deploy an Amazon SageMaker model. Create a SageMaker endpoint for inference.
- B. Use Amazon SageMaker to train a model by using the historical data in the S3 bucket.
- C. Configure an AWS Lambda function with a function URL that uses Amazon SageMaker endpoints to create predictions based on the inputs.
- D. Configure an AWS Lambda function with a function URL that uses an Amazon Forecast predictor to create a prediction based on the inputs. **Most Voted**
- E. Train an Amazon Forecast predictor by using the historical data in the S3 bucket. **Most Voted**

Correct Answer: DE

Community vote distribution



Comments

MatAlves **Highly Voted** 1 year, 8 months ago

Selected Answer: AB

"Amazon Forecast is no longer available to new customers. Existing customers of Amazon Forecast can continue to use the service as normal"

"After careful consideration, we have made the decision to close new customer access to Amazon Forecast, effective July 29, 2024."

This question will either be removed or reformulated to exclude Forecast as the service is no longer available to new customers.

upvoted 11 times

Salilgen 1 year, 5 months ago

I agree: this is probably an old questions, but I think answer was D,E
Otherwise you need three options: A,B and C (to create predictions)

upvoted 1 times

betttty **Highly Voted** 2 years, 4 months ago

Explanation:

Training the Model with SageMaker (Option B):

Use Amazon SageMaker to train a machine learning model based on historical data. SageMaker simplifies the process of training, deploying, and managing machine learning models.

Creating Predictions with Amazon Forecast (Option D):

Use Amazon Forecast to create a predictor based on historical data. Forecast is designed for time-series forecasting, making it suitable for predicting resources needed for manufacturing processes each month.

Combining SageMaker for training and Amazon Forecast for predictions provides a comprehensive solution, and AWS Lambda can be used to integrate these services into your workflow.

upvoted 5 times

JackyCCK 2 years, 2 months ago

combination of steps so it cannot be B,D.
B D is two different solution

upvoted 1 times

Sato_2777 Most Recent 1 month, 1 week ago

Selected Answer: DE

Predict + No ML experience → Amazon Forecast
upvoted 1 times

In2718 9 months, 3 weeks ago

Selected Answer: DE

Chat AI -
D. Configure an AWS Lambda function with a function URL that uses an Amazon Forecast predictor... Lambda can expose an endpoint that queries Forecast for predictions.

E. Train an Amazon Forecast predictor by using the historical data in the S3 bucket.

Amazon Forecast is a managed service for time series forecasting (sales, resource usage, etc.), designed for users with little/no ML expertise.

A and B require ML knowledge

upvoted 1 times

In2718 10 months, 2 weeks ago

Selected Answer: AB

Status update - reading on the latest changes to Sagemaker, there is an option as well for no-code solution, so your guess is as good as mine on this question.

upvoted 1 times

In2718 10 months, 2 weeks ago

Selected Answer: DE

Because the company has no ML experience, the following... An Amazon Forecast predictor is a trained machine learning model that uses time series data to generate forecasts. It leverages various algorithms to learn from your data and produce accurate predictions for future values. Sagemaker - need ML experience, Forecaster, no ML experience

upvoted 1 times

lalaraj 10 months, 3 weeks ago

Selected Answer: DE

D and E

Why:

Amazon Forecast is a fully managed time-series forecasting service that requires no ML expertise.

E. Train an Amazon Forecast predictor with the historical data in S3.

D. Invoke that predictor (e.g., behind a Lambda with a function URL or API Gateway) to generate monthly predictions.

Why not SageMaker (A/B/C): SageMaker is powerful but assumes more ML familiarity and management overhead than Forecast, which is purpose-built for this use case.

upvoted 1 times

hilker1983 1 year, 5 months ago

Selected Answer: DE

D. Configure an AWS Lambda function with a function URL that uses an Amazon Forecast predictor to create a prediction based on the inputs.

Amazon Forecast is a managed service specifically designed for time series forecasting and can be used without extensive ML knowledge. This option allows for easy integration with Lambda for making predictions.

E. Train an Amazon Forecast predictor by using the historical data in the S3 bucket.

This option is essential for creating the forecasting model using the historical data stored in S3. Amazon Forecast is designed to handle this type of task and is user-friendly for those without ML experience.

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: DE

Amazon Forecast is no longer available to new customers as of July 29, 2024. However, for the AWS Certified Solutions Architect – Associate (SAA-C03) exam, Amazon Forecast may still appear as a valid service since the exam content often lags behind recent announcements.

Since the company has no ML expertise, A B C are all out, because they require prior SageMaker knowledge. That would make the answers: D & E.

But IRL (in real life, not a web service), you must consider SageMaker Canvas, because it "includes the key capabilities found in Forecast, including the ability to train an ensemble of forecasting models using both statistical and neural network algorithms. It creates the best model for your dataset by generating base models for each algorithm, evaluating their performance, and then combining the top-performing models into an ensemble".

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

<https://aws.amazon.com/blogs/machine-learning/transition-your-amazon-forecast-usage-to-amazon-sagemaker-canvas/>
upvoted 2 times

FlyingHawk 1 year, 4 months ago

We should reword the DE as below:

Train a time series forecasting model using Amazon SageMaker Canvas with the historical data in the S3 bucket.

SageMaker Canvas allows you to import data directly from Amazon S3, prepare the dataset, and train time series forecasting models through its intuitive interface.

Configure an AWS Lambda function with a function URL that uses the SageMaker Canvas model to create predictions based on the inputs.

After training the model, you can deploy it and use an AWS Lambda function to invoke the model endpoint for generating predictions programmatically.

upvoted 1 times

XXXXXINN 1 year, 8 months ago

so what about AB since the Forecast is no longer available to new customers?

upvoted 3 times

JoeTromundo 1 year, 8 months ago

Selected Answer: AB

Amazon Forecast is no longer available to new customers.

<https://aws.amazon.com/blogs/machine-learning/transition-your-amazon-forecast-usage-to-amazon-sagemaker-canvas/>

upvoted 5 times

TwinSpark 2 years ago

Selected Answer: DE

Amazon forecast can be trained by using data from S3:

<https://docs.aws.amazon.com/forecast/latest/dg/getting-started.html>

upvoted 3 times

bujuman 2 years, 1 month ago

Selected Answer: DE

Because of these assertions

- The company has no machine learning (ML) experience
- The company wants to use a managed service

We could be tempted to go for SageMaker that is the core AWS managed service for ML purposes.

But, but, if we consider this valuable information:

- A company that uses AWS needs a solution to predict the resources needed for manufacturing processes.

With a bit of research, we will find out that AWS also holds time-series forecasting service based on machine learning (ML).

https://aws.amazon.com/forecast/?nc1=h_ls

So I understand options DE are the best answers even though this service is not mentioned anywhere in the current SAA-C03 course version

upvoted 5 times

Hung23 2 years, 2 months ago

Selected Answer: BE

BE from CHATGPT

upvoted 1 times

lenotc 2 years, 2 months ago

Selected Answer: BE

SageMaker and Forecast can directly utilize data within an S3

B) E)

<https://aws.amazon.com/blogs/compute/build-workflows-for-amazon-forecast-with-aws-step-functions/>

<https://docs.aws.amazon.com/sagemaker/latest/dg/train-model.html>

upvoted 1 times

TheLaPlanta 2 years, 2 months ago

Selected Answer: AB

A + B dude

upvoted 2 times

Ravan 2 years, 3 months ago

Selected Answer: AB

Yes, exactly. Steps B and A together constitute a comprehensive solution:

- Step B involves using Amazon SageMaker to train a machine learning model using historical data stored in the S3 bucket.
- Step A involves deploying the trained model as a SageMaker endpoint, allowing for real-time inference on new data.

This combination leverages Amazon SageMaker's managed services for both training and inference, meeting the company's requirements efficiently.
upvoted 4 times

A company manages AWS accounts in AWS Organizations. AWS IAM Identity Center (AWS Single Sign-On) and AWS Control Tower are configured for the accounts. The company wants to manage multiple user permissions across all the accounts.

The permissions will be used by multiple IAM users and must be split between the developer and administrator teams. Each team requires different permissions. The company wants a solution that includes new users that are hired on both teams.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create individual users in IAM Identity Center for each account. Create separate developer and administrator groups in IAM Identity Center. Assign the users to the appropriate groups. Create a custom IAM policy for each group to set fine-grained permissions.

B. Create individual users in IAM Identity Center for each account. Create separate developer and administrator groups in IAM Identity Center. Assign the users to the appropriate groups. Attach AWS managed IAM policies to each user as needed for fine-grained permissions.

C. Create individual users in IAM Identity Center. Create new developer and administrator groups in IAM Identity Center. Create new permission sets that include the appropriate IAM policies for each group. Assign the new groups to the appropriate accounts. Assign the new permission sets to the new groups. When new users are hired, add them to the appropriate group. **Most Voted**

D. Create individual users in IAM Identity Center. Create new permission sets that include the appropriate IAM policies for each user. Assign the users to the appropriate accounts. Grant additional IAM permissions to the users from within specific accounts. When new users are hired, add them to IAM Identity Center and assign them to the accounts.

Correct Answer: C

Community vote distribution

C (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

I would:

1. Use AWS IAM Identity Center (SSO) to centrally manage user access across all accounts.
2. Create Permission Sets in Identity Center:
 - One for the Developer Team, granting access to resources with developer-specific permissions (AmazonEC2FullAccess or custom policies or something).
 - Another one for the Administrator Team, granting administrator-specific permissions (AdministratorAccess or something).
3. Sync users and groups from the corporate identity provider (or manage groups directly in IAM Identity Center) to automatically apply these permissions.
4. Assign these Permission Sets to groups of users in Identity Center rather than individuals, ensuring new users automatically inherit the appropriate permissions based on their group.

That's why you choose C, with the least operational overhead.

upvoted 1 times

xBUGx 1 year, 9 months ago

Selected Answer: C

C is least overhead

upvoted 1 times

1Alpha1 1 year, 10 months ago

Selected Answer: C

Check out this one. https://www.youtube.com/watch?v=y_n9xN5mg1g

upvoted 1 times

Moon239 1 year, 10 months ago

Selected Answer: C

<https://docs.aws.amazon.com/controltower/latest/userguide/sso.html>

upvoted 2 times

Cali182 1 year, 10 months ago

Selected Answer: C

Correct is C

upvoted 2 times

Andy_09 1 year, 10 months ago

The correct answer should be C

upvoted 3 times

A company wants to standardize its Amazon Elastic Block Store (Amazon EBS) volume encryption strategy. The company also wants to minimize the cost and configuration effort required to operate the volume encryption check.

Which solution will meet these requirements?

- A. Write API calls to describe the EBS volumes and to confirm the EBS volumes are encrypted. Use Amazon EventBridge to schedule an AWS Lambda function to run the API calls.
- B. Write API calls to describe the EBS volumes and to confirm the EBS volumes are encrypted. Run the API calls on an AWS Fargate task.
- C. Create an AWS Identity and Access Management (IAM) policy that requires the use of tags on EBS volumes. Use AWS Cost Explorer to display resources that are not properly tagged. Encrypt the untagged resources manually.
- D. Create an AWS Config rule for Amazon EBS to evaluate if a volume is encrypted and to flag the volume if it is not encrypted.

Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

MatAlves 1 year, 8 months ago

D: no-brainer.

upvoted 1 times

asdfcdsxdxc 2 years, 3 months ago

Selected Answer: D

D looks right

upvoted 3 times

bodakrishna 2 years, 3 months ago

AWS Config allows you to define rules to automatically check the configuration of AWS resources against desired configurations. By creating a custom AWS Config rule specifically for Amazon EBS volumes to evaluate if they are encrypted, you can ensure consistent encryption across all volumes. If a volume is found to be unencrypted, it can be flagged for further action. This solution automates the process of encryption checking, minimizing manual effort and ensuring standardization across the environment. Additionally, AWS Config provides a cost-effective solution compared to continuously running scripts or tasks.

upvoted 3 times

mestule 2 years, 4 months ago

Selected Answer: D

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. It can check whether your resource comply with certain conditions (such as being encrypted), and it can flag or take action on resources that do not comply.

upvoted 4 times

betttty 2 years, 4 months ago

D :

you could use a managed rule to quickly start assessing whether your Amazon Elastic Block Store (Amazon EBS) volumes are encrypted or whether specific tags are applied to your resources.

https://docs.aws.amazon.com/config/latest/developerguide/evaluate-config_use-managed-rules.html

upvoted 4 times

Andy_09 2 years, 4 months ago

Correct answer is D

upvoted 3 times

A company regularly uploads GB-sized files to Amazon S3. After the company uploads the files, the company uses a fleet of Amazon EC2 Spot Instances to transcode the file format. The company needs to scale throughput when the company uploads data from the on-premises data center to Amazon S3 and when the company downloads data from Amazon S3 to the EC2 instances.

Which solutions will meet these requirements? (Choose two.)

- A. Use the S3 bucket access point instead of accessing the S3 bucket directly.
- B. Upload the files into multiple S3 buckets.
- C. Use S3 multipart uploads. **Most Voted**
- D. Fetch multiple byte-ranges of an object in parallel. **Most Voted**
- E. Add a random prefix to each object when uploading the files.

Correct Answer: CD

Community vote distribution

CD (100%)

Comments

betttty **Highly Voted** 2 years, 4 months ago

CD

C: Increase the file upload throughput

D: increase the file download throughput

upvoted 10 times

sandordini **Highly Voted** 2 years, 1 month ago

Selected Answer: CD

C: Upload: Multipart clear,

D: Download: You can fetch a byte-range from an object, transferring only the specified portion. You can use concurrent connections to Amazon S3 to fetch different byte ranges from within the same object. This helps you achieve higher aggregate throughput versus a single whole-object request

A: S3 Access Points can be easily scaled, but are typically used to simplify data access for any AWS service or customer application that stores data in S3.

E: Prefixes: You can increase your read or write performance by using parallelization. For example, if you create 10 prefixes in an Amazon S3 bucket to parallelize reads, you could scale your read performance to 55,000 read requests per second.

But wording in this answer is strange...

upvoted 5 times

Salahdin **Most Recent** 1 year, 6 months ago

Selected Answer: CD

CD

C: Increase the file upload throughput

D: increase the file download throughpu

upvoted 2 times

dds69 2 years, 2 months ago

Selected Answer: CD

C&D are correct

upvoted 3 times

Bazzix 2 years, 2 months ago

Selected Answer: CD

Cd are correct

upvoted 3 times

bodakrishna 2 years, 3 months ago

C & D Correct

upvoted 3 times

Darshan07 2 years, 4 months ago

Selected Answer: CD

CD are the correct options

upvoted 3 times

Cali182 2 years, 4 months ago

Selected Answer: CD

CD is the correct for me

upvoted 2 times

Andy_09 2 years, 4 months ago

Correct answer is CD

upvoted 2 times

A solutions architect is designing a shared storage solution for a web application that is deployed across multiple Availability Zones. The web application runs on Amazon EC2 instances that are in an Auto Scaling group. The company plans to make frequent changes to the content. The solution must have strong consistency in returning the new content as soon as the changes occur.

Which solutions meet these requirements? (Choose two.)

A. Use AWS Storage Gateway Volume Gateway Internet Small Computer Systems Interface (iSCSI) block storage that is mounted to the individual EC2 instances.

B. Create an Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system on the individual EC2 instances.

Most Voted

C. Create a shared Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume on the individual EC2 instances.

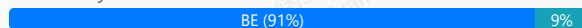
D. Use AWS DataSync to perform continuous synchronization of data between EC2 hosts in the Auto Scaling group.

E. Create an Amazon S3 bucket to store the web content. Set the metadata for the Cache-Control header to no-cache. Use Amazon CloudFront to deliver the content.

Most Voted

Correct Answer: BE

Community vote distribution



Comments

Andy_09 **Highly Voted** 2 years, 4 months ago

Correct answer BE

upvoted 8 times

Cali182 **Highly Voted** 2 years, 4 months ago

Selected Answer: BE

B & E seems to be the most logic

upvoted 5 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: BE

A - Volume Gateway is typically on-prem and optimized for hybrid setups, not for fully cloud-based solutions.

B - EFS provides a fully managed, shared file storage system that works across multiple AZs.

C - Even with Multi-Attach, sharing EBS volumes across AZs is pretty complex and error-prone.

D - DataSync is designed for periodic synchronization and not real-time, frequent updates.

E - The "no-cache" header ensures the content is always fetched fresh, and CloudFront delivers it efficiently with reduced latency. Although S3 is less efficient for frequent updates compared to a shared file system like EFS, but not a problem here.

upvoted 3 times

Penjerla 1 year, 6 months ago

Selected Answer: BC

Option A is out as storage gateway is suited for hybrid cloud use cases where on-premises workloads need access to AWS cloud storage

upvoted 1 times

Salilgen 1 year, 5 months ago

C is wrong: you cannot share EBS in multiple AZ

upvoted 1 times

sandordini 2 years, 1 month ago

Regarding storage, I'd go for EFS, although it never mentions the requirement for file storage.

Datasync can copy data between several storage types, including EFS, agents can be installed on EC2, but you cannot perform continuous synchronization of EC2 instances. Only storage.

Cloudfront can publish both passive (s3) and active content (EC2+EFS) but wording doesn't tell a thing about such a share. And if it's a passive site why do we even have 2 storage types...

I'd say, for me, the least bad solution seems to be B + E.

upvoted 2 times

Hung23 2 years, 2 months ago

Selected Answer: BE

I choose BE

upvoted 2 times

alawada 2 years, 2 months ago

BD looks most logical to me - continuous changes required an update via DataSync

upvoted 2 times

A company is deploying an application in three AWS Regions using an Application Load Balancer. Amazon Route 53 will be used to distribute traffic between these Regions.

Which Route 53 configuration should a solutions architect use to provide the MOST high-performing experience?

- A. Create an A record with a latency policy. **Most Voted**
- B. Create an A record with a geolocation policy.
- C. Create a CNAME record with a failover policy.
- D. Create a CNAME record with a geoproximity policy.

Correct Answer: A

Community vote distribution

A (88%)

9%

Comments

Kezuko **Highly Voted** 1 year, 8 months ago

Selected Answer: A

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html>
upvoted 7 times

h0ng97_spare_002 **Highly Voted** 1 year, 8 months ago

Selected Answer: A

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html>
upvoted 6 times

In2718 **Most Recent** 10 months, 2 weeks ago

Selected Answer: A

three AWS Regions using an Application Load Balancer?? I think they meant to say three AZs. ALBs do not span regions.
upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: A

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html>

To provide the MOST high-performing experience, the Latency-based routing policy is the most appropriate. It ensures requests are routed to the AWS Region that provides the fastest response times based on real-world network conditions.
upvoted 1 times

sandordini 1 year, 7 months ago

Selected Answer: A

1. Given the chance >always use Alias over a Cname<
2. Latency-based routing is for user experience. (low latency)

Failover is for DR, Geolocation for local restrictions/rights/language/currency, and geo-proximity is a more complex, biased location-based routing, not part of the SA Associate exam.
upvoted 4 times

NSA_Poker 1 year, 6 months ago

Alias?! "A record" is NOT an Alias record; it's an ADDRESS RECORD. There's a difference in between an address record (A record) & an Alias.

An Address record will map to 1 or more IP addresses.
An Alias record will map to another name like a CNAME does.

We eliminate C&D bc we need an IP address that will give us the best performance; we distribute traffic to a certain IP address based on policy. geolocation policy is defined by where the request comes from. latency policy is defined by how fast (performance) we can reply.

upvoted 3 times

mohammadthainat 1 year, 8 months ago

Selected Answer: A

Geoproximity Policy routing users to resources based on their geographic location, routing based on geographic location may not always be the absolute lowest latency.

latency-based routing prioritizes user experience.

upvoted 4 times

TruthWS 1 year, 8 months ago

A is true

upvoted 3 times

cedser8 1 year, 9 months ago

Selected Answer: D

The correct is D, the question says "using an Application Load Balancer" the ALB has a DNS name assigned not an IP. A type A record will only allow you to point to an IPv4. If I'm wrong, happy to be corrected.

upvoted 3 times

dkw2342 1 year, 8 months ago

Answer A is correct.

Route53 uses an internal record type called ALIAS, but from a DNS point of view it's still an A record.

Just try it yourself, create an ALB and create a DNS record in Route53. While you can technically use a CNAME (for subdomains, see below), the wizard will guide you to use an A ALIAS record, which also makes the most sense.

The problem with CNAME records is that it's not possible to create them at the root level of the domain. Let's say your domain is somedomain.com - you can't create a CNAME for the apex of the domain (mydomain.com), only for subdomains (subdomain.mydomain.com).

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/resource-record-sets-choosing-alias-non-alias.html>

upvoted 5 times

bodakrishna 1 year, 9 months ago

ChatGPT:

The most high-performing experience in this scenario would be achieved by using:

D. Create a CNAME record with a geoproximity policy.

Geoproximity routing allows you to route traffic based on the geographic location of your users and your resources. This would distribute traffic to the AWS Region that is closest to the user, optimizing performance by reducing latency. It's particularly useful when deploying applications across multiple regions to ensure users are directed to the closest region, minimizing network latency and providing the best user experience.

upvoted 1 times

sandordini 1 year, 7 months ago

And, exactly, this is the reason why you should not rely on a LANGUAGE MODEL when you need a solution architect's advice .

upvoted 5 times

osmk 1 year, 9 months ago

A

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-latency.html>

upvoted 2 times

haci 1 year, 9 months ago

Selected Answer: A

Based on previous questions, I believe A is correct. Because; the closest geolocated server doesn't necessarily provide the best performance. Geolocated load balancing is mostly used for serving location-specific content.

upvoted 3 times

1Alpha1 1 year, 10 months ago

Selected Answer: A

Q. What is Amazon Route 53's Latency Based Routing (LBR) feature?

LBR (Latency Based Routing) is a new feature for Amazon Route 53 that helps you improve your application's performance for a global audience. You can run applications in multiple AWS regions and Amazon Route 53, using dozens of edge locations worldwide, will route end users to the AWS region that provides the lowest latency.

<https://aws.amazon.com/route53/faqs/>

upvoted 3 times

Cali182 1 year, 10 months ago

Selected Answer: B

Why would you use a CNAME record?? Most suitable seems to be option B

upvoted 1 times

Typewriter101 1 year, 9 months ago

Not really sure but ALBs do not have a static ip address they have domains assigned to them and also an Elastic ip can't be attached to an ALB. So mainly a cname would be preferred here.

upvoted 2 times

Typewriter101 1 year, 9 months ago

But generally speaking it's not a bad idea. But yes A record alias name can point to it. and i don't think it's B cause even if it's geolocation it may not always result in a high-performing exp.

upvoted 2 times

osmk 1 year, 9 months ago

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-latency.html>

upvoted 2 times

Andy_09 1 year, 10 months ago

Sorry changing to B.

upvoted 1 times

Andy_09 1 year, 10 months ago

D looks correct.

upvoted 2 times

A company has a web application that includes an embedded NoSQL database. The application runs on Amazon EC2 instances behind an Application Load Balancer (ALB). The instances run in an Amazon EC2 Auto Scaling group in a single Availability Zone.

A recent increase in traffic requires the application to be highly available and for the database to be eventually consistent.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Replace the ALB with a Network Load Balancer. Maintain the embedded NoSQL database with its replication service on the EC2 instances.
- B. Replace the ALB with a Network Load Balancer. Migrate the embedded NoSQL database to Amazon DynamoDB by using AWS Database Migration Service (AWS DMS).
- C. Modify the Auto Scaling group to use EC2 instances across three Availability Zones. Maintain the embedded NoSQL database with its replication service on the EC2 instances.
- D. Modify the Auto Scaling group to use EC2 instances across three Availability Zones. Migrate the embedded NoSQL database to Amazon DynamoDB by using AWS Database Migration Service (AWS DMS). **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

LeonSauveterre 1 year, 5 months ago

Selected Answer: D

Application Load Balancer (ALB) is better suited for HTTP/HTTPS traffic. A & B are out.

Maintaining replication and failover for the embedded database across AZs is frustratingly complex and operationally intensive. Migration to DynamoDB requires initial effort but reduces long-term operational overhead. So D.

upvoted 2 times

freedafeng 1 year, 10 months ago

I honestly don't think you can use db Migration service to migrate an embedding db.

upvoted 1 times

HTHK 2 years, 1 month ago

DDDDDDDDDDDDDDDDDDDD

upvoted 1 times

TruthWS 2 years, 2 months ago

Option D: let focus on HA + Scaling

upvoted 1 times

Kezuko 2 years, 2 months ago

Selected Answer: D

ASG for application HA + DynamoDB Scale for HA

upvoted 4 times

rubiteb 2 years, 3 months ago

B as it's highly available and has less operational overhead than D.

upvoted 1 times

dkw2342 2 years, 2 months ago

ALB -> NLB makes no sense and solution lacks HA for the app layer.

upvoted 1 times

NayeraB 2 years, 3 months ago

But wouldn't migrating an embedded database to a new one introduce operational overhead now and in the future?

upvoted 1 times

MatAlves 1 year, 8 months ago

No, the very opposite:

"Amazon DynamoDB is a fully managed, serverless, key-value NoSQL database that runs high-performance applications at any scale."

upvoted 1 times

1Alpha1 2 years, 4 months ago

Selected Answer: D

DynamoDB + Modifying the Auto Scaling group

upvoted 2 times

Cali182 2 years, 4 months ago

Selected Answer: D

Dynamo DB presents more advantages, because it would need less administrative effort

upvoted 2 times

Andy_09 2 years, 4 months ago

The correct option should be D

upvoted 4 times

A company is building a shopping application on AWS. The application offers a catalog that changes once each month and needs to scale with traffic volume. The company wants the lowest possible latency from the application. Data from each user's shopping cart needs to be highly available. User session data must be available even if the user is disconnected and reconnects.

What should a solutions architect do to ensure that the shopping cart data is preserved at all times?

- A. Configure an Application Load Balancer to enable the sticky sessions feature (session affinity) for access to the catalog in Amazon Aurora.
- B. Configure Amazon ElastiCache for Redis to cache catalog data from Amazon DynamoDB and shopping cart data from the user's session. **Most Voted**
- C. Configure Amazon OpenSearch Service to cache catalog data from Amazon DynamoDB and shopping cart data from the user's session.
- D. Configure an Amazon EC2 instance with Amazon Elastic Block Store (Amazon EBS) storage for the catalog and shopping cart. Configure automated snapshots.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Tanidanindo **Highly Voted** 2 years, 2 months ago

Selected Answer: B

- ElastiCache is a managed in-memory data store service that is well-suited for managing session data in a distributed architecture.
- upvoted 5 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: B

User session data must be available even if the user is disconnected and reconnects. Choose ElastiCache for Redis. That remains true for so many similar questions.

upvoted 1 times

RC6 1 year, 12 months ago

Selected Answer: B

B looks correct

upvoted 2 times

asdfcdsxdcf 2 years, 3 months ago

why not A?

upvoted 3 times

24b2e9e 1 year, 11 months ago

because data to store(cache) will be bigger in size

upvoted 2 times

knben 2 years, 3 months ago

Selected Answer: B

session data must be available even if the user is disconnected and reconnects -> ElastiCache for Redis

upvoted 2 times

1Alpha1 2 years, 4 months ago

Selected Answer: B

B: ELB <--> ASG <--> ElastiCache <--> DynamoDB

upvoted 3 times

Andy_09 2 years, 4 months ago

B looks correct

upvoted 4 times

A company is building a microservices-based application that will be deployed on Amazon Elastic Kubernetes Service (Amazon EKS). The microservices will interact with each other. The company wants to ensure that the application is observable to identify performance issues in the future.

Which solution will meet these requirements?

- A. Configure the application to use Amazon ElastiCache to reduce the number of requests that are sent to the microservices.
- B. Configure Amazon CloudWatch Container Insights to collect metrics from the EKS clusters. Configure AWS X-Ray to trace the requests between the microservices. **Most Voted**
- C. Configure AWS CloudTrail to review the API calls. Build an Amazon QuickSight dashboard to observe the microservice interactions.
- D. Use AWS Trusted Advisor to understand the performance of the application.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Cali182 **Highly Voted** 1 year, 10 months ago

Selected Answer: B

Option B

Amazon CloudWatch Container Insights: This service provides monitoring and troubleshooting capabilities for containerized applications. It collects and aggregates metrics, logs, and events from Amazon EKS clusters and containers. This helps in monitoring the performance and health of microservices.

upvoted 9 times

Andy_09 **Highly Voted** 1 year, 10 months ago

Correct answer is B

upvoted 6 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: B

A - ElastiCache can reduce latency but does not directly provide observability.

B - CloudWatch Container Insights provides detailed metrics and logs for containers. AWS X-Ray traces requests across microservices

C - CloudTrail is designed for security auditing and API call tracking, not detailed observability or performance monitoring.

D - Trusted Advisor focuses on high-level recommendations and does not provide real-time observability for application interactions or performance issues.

upvoted 1 times

Linuslin 1 year, 6 months ago

Selected Answer: B

B is the correct answer.

Use CloudWatch Container Insights to collect, aggregate, and summarize metrics and logs from your containerized applications and microservices.

https://docs.aws.amazon.com/zh_tw/AmazonCloudWatch/latest/monitoring/ContainerInsights.html

AWS X-Ray collects data about requests that your application serves, and it helps you view, filter, and gain insights into that data to identify issues and opportunities for optimization.

https://docs.aws.amazon.com/zh_tw/prescriptive-guidance/latest/logging-monitoring-for-application-owners/x-ray.html

upvoted 5 times

A company needs to provide customers with secure access to its data. The company processes customer data and stores the results in an Amazon S3 bucket.

All the data is subject to strong regulations and security requirements. The data must be encrypted at rest. Each customer must be able to access only their data from their AWS account. Company employees must not be able to access the data.

Which solution will meet these requirements?

- A. Provision an AWS Certificate Manager (ACM) certificate for each customer. Encrypt the data client-side. In the private certificate policy, deny access to the certificate for all principals except an IAM role that the customer provides.
- B. Provision a separate AWS Key Management Service (AWS KMS) key for each customer. Encrypt the data server-side. In the S3 bucket policy, deny decryption of data for all principals except an IAM role that the customer provides.
- C. Provision a separate AWS Key Management Service (AWS KMS) key for each customer. Encrypt the data server-side. In each KMS key policy, deny decryption of data for all principals except an IAM role that the customer provides. **Most Voted**
- D. Provision an AWS Certificate Manager (ACM) certificate for each customer. Encrypt the data client-side. In the public certificate policy, deny access to the certificate for all principals except an IAM role that the customer provides.

Correct Answer: C

Community vote distribution

C (71%)

B (29%)

Comments

tch 1 year, 3 months ago

Selected Answer: C

A key policy is a resource policy for an AWS KMS key. Key policies are the primary way to control access to KMS keys. Every KMS key must have exactly one key policy. The statements in the key policy determine who has permission to use the KMS key and how they can use it.

upvoted 1 times

tch 1 year, 3 months ago

You can also use IAM policies and grants to control access to the KMS key, but every KMS key must have a key policy.
<https://docs.aws.amazon.com/kms/latest/developerguide/key-policies.html>

upvoted 1 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

ACM certificates are for securing communication (e.g., HTTPS/SSL), not for encrypting data at rest in S3. A and D are out.

Between B and C, they both create a separate key for each customer, and both work. So which is more cumbersome, managing S3 policies or managing KMS key policies?

Of course the latter one. In B, S3 bucket policy centralizes access rules for all keys, while in C, policies are distributed across individual keys. That's why I decide to choose B.

upvoted 3 times

RC6 1 year, 12 months ago

Selected Answer: C

C looks correct

upvoted 2 times

BBR01 2 years, 1 month ago

Selected Answer: C

Actually I think neither B or C is correctly worded. If talking about key policy, should be "Modify the key's policy to grant the IAM user permissions for the kms:GenerateDataKey and kms:Decrypt actions at minimum."

If talking about bucket policy, should be "Deny GetObjects of particular customer without condition kms key equals 1234abcd...."

upvoted 4 times

mohammadthainat 2 years, 2 months ago

Selected Answer: C

Encryption at rest --> KMS

Each customer must be able to access only their data --> KMS Key Policies

upvoted 4 times

LeonSauveterre 1 year, 5 months ago

Each customer must be able to access only their data --> Deny decryption of data for all principals except an IAM role that the customer provides. This doesn't necessarily have to be KMS key policies. B & C both work in this scenario.

upvoted 3 times

Neung983 2 years, 3 months ago

Selected Answer: B

B.

Here's why this option is the best fit:

Server-Side Encryption: Encrypting data server-side with KMS ensures encryption happens transparently within AWS, eliminating the need for complex client-side management and potential security risks associated with user-managed keys.

Customer-Specific Keys: Utilizing separate KMS keys for each customer provides granular access control and encryption isolation. Each customer can only decrypt their data using their specific KMS key.

S3 Bucket Policy: By denying decryption permissions for all principals except the dedicated customer IAM role in the S3 bucket policy, unauthorized access, even from company employees, is prevented. This aligns with the requirement of customer-specific data access.

upvoted 3 times

Cali182 2 years, 4 months ago

Selected Answer: C

Option C

From Chapt

Option A is incorrect because using ACM certificates is typically for establishing secure communication over HTTPS and doesn't directly relate to encrypting data at rest in S3.

Option B is incorrect because while it suggests using AWS KMS keys for encryption, it mentions using S3 bucket policies for access control, which would not be appropriate for controlling decryption permissions.

Option D is incorrect because it suggests using ACM certificates for client-side encryption, which is not typically used for encrypting data at rest in S3, and the approach described would not effectively control access to the encrypted data.

upvoted 4 times

Andy_09 2 years, 4 months ago

Correct answer should be C

upvoted 3 times

A solutions architect creates a VPC that includes two public subnets and two private subnets. A corporate security mandate requires the solutions architect to launch all Amazon EC2 instances in a private subnet. However, when the solutions architect launches an EC2 instance that runs a web server on ports 80 and 443 in a private subnet, no external internet traffic can connect to the server.

What should the solutions architect do to resolve this issue?

- A. Attach the EC2 instance to an Auto Scaling group in a private subnet. Ensure that the DNS record for the website resolves to the Auto Scaling group identifier.
- B. Provision an internet-facing Application Load Balancer (ALB) in a public subnet. Add the EC2 instance to the target group that is associated with the ALB. Ensure that the DNS record for the website resolves to the ALB. **Most Voted**
- C. Launch a NAT gateway in a private subnet. Update the route table for the private subnets to add a default route to the NAT gateway. Attach a public Elastic IP address to the NAT gateway.
- D. Ensure that the security group that is attached to the EC2 instance allows HTTP traffic on port 80 and HTTPS traffic on port 443. Ensure that the DNS record for the website resolves to the public IP address of the EC2 instance.

Correct Answer: B

Community vote distribution

B (84%)

Other

Comments

sandordini **Highly Voted** 2 years, 1 month ago

Selected Answer: B

Not A - Autoscaling Irrelevant

B - ALB, route tables for the public subnet with a route to Priv subnet

C - "NAT gateway" is "to allow [outbound] internet traffic", but this is about inbound traffic

D - Instances are in the private subnet, therefore it won't work from the public.

upvoted 6 times

anikolov **Highly Voted** 2 years, 4 months ago

Selected Answer: B

B: Provision an internet-facing Application Load Balancer (ALB) in a public subnet makes more sense

upvoted 5 times

LeonSauveterre **Most Recent** 1 year, 5 months ago

Selected Answer: B

A - Auto Scaling groups are for managing instance scaling.

B - The internet-facing ALB is deployed in a public subnet and acts as a bridge between the internet and the private EC2 instance. Then, The ALB forwards HTTP/HTTPS traffic to the EC2 instance in the private subnet.

C - A NAT gateway provides outbound internet access for private instances, but it does not solve inbound traffic issues.

D - Private EC2 instances do not have public IP addresses at all.

upvoted 1 times

8621a7c 1 year, 7 months ago

Selected Answer: C

Is the question ask to solve the external connection?

upvoted 1 times

waldirlsantos 2 years, 1 month ago

Why not "D"?

upvoted 1 times

boluwatito 2 years, 2 months ago

Selected Answer: D

nsure that the security group attached to the EC2 instance allows inbound traffic on ports 80 and 443 from the desired sources (e.g., any IP or specific IP ranges). This allows external internet traffic to reach the web server running on the EC2 instance

upvoted 1 times

GOTJ 1 year, 5 months ago

But the EC2s must be located in a private subnet, which is not direct internet accesible by definition

upvoted 1 times

TruthWS 2 years, 2 months ago

B - because ALB do it better NAT

upvoted 2 times

Cali182 2 years, 4 months ago

Selected Answer: C

Option C from Chatgt

upvoted 1 times

jaswantn 2 years, 4 months ago

NAT Gateway stays in public subnet, not in private subnet. So, C can't be.

upvoted 6 times

lenotc 2 years, 2 months ago

NAT Gateway outbound connections

upvoted 2 times

mestule 2 years, 4 months ago

Selected Answer: B

B makes most sense

upvoted 4 times

Andy_09 2 years, 4 months ago

Changing to option D

upvoted 1 times

Andy_09 2 years, 4 months ago

C should be the correct answer

upvoted 1 times

A company is deploying a new application to Amazon Elastic Kubernetes Service (Amazon EKS) with an AWS Fargate cluster. The application needs a storage solution for data persistence. The solution must be highly available and fault tolerant. The solution also must be shared between multiple application containers.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create Amazon Elastic Block Store (Amazon EBS) volumes in the same Availability Zones where EKS worker nodes are placed. Register the volumes in a StorageClass object on an EKS cluster. Use EBS Multi-Attach to share the data between containers.
- B. Create an Amazon Elastic File System (Amazon EFS) file system. Register the file system in a StorageClass object on an EKS cluster. Use the same file system for all containers. **Most Voted**
- C. Create an Amazon Elastic Block Store (Amazon EBS) volume. Register the volume in a StorageClass object on an EKS cluster. Use the same volume for all containers.
- D. Create Amazon Elastic File System (Amazon EFS) file systems in the same Availability Zones where EKS worker nodes are placed. Register the file systems in a StorageClass object on an EKS cluster. Create an AWS Lambda function to synchronize the data between file systems.

Correct Answer: B

Community vote distribution

B (100%)

Comments

boluwatito **Highly Voted** 1 year, 8 months ago

Selected Answer: B

Overall, Amazon EFS provides a highly available, fault-tolerant, and shared storage solution with minimal operational overhead, making it the ideal choice for persisting data in an Amazon EKS Fargate cluster.

upvoted 5 times

TruthWS **Most Recent** 1 year, 8 months ago

B bcs EBS only attach one EC2

upvoted 3 times

asdfcdsxdcf 1 year, 9 months ago

Selected Answer: B

B looks correct

upvoted 3 times

Naveena_Devanga 1 year, 9 months ago

B, The solution also must be shared between multiple application containers so attaching to each container is not a practical solution.

upvoted 4 times

Marunio 1 year, 10 months ago

Selected Answer: B

B is correct answer because it is high available - EBS isn't HA for that so A isn't dealing with request.

upvoted 3 times

jaswantn 1 year, 10 months ago

Option A... EBS with multi attach does not provide HA so option B is more appropriate.

upvoted 2 times

dkw2342 1 year, 8 months ago

It's just plain wrong. Not getting HA with EBS multi attach is really the least of your problems. Mounting a regular FS in read/write mode on more than one machine will cause data corruption. You'd need a clustered filesystem.

upvoted 2 times

Andy_09 1 year, 10 months ago

Correct answer is B

upvoted 4 times

A company has an application that uses Docker containers in its local data center. The application runs on a container host that stores persistent data in a volume on the host. The container instances use the stored persistent data.

The company wants to move the application to a fully managed service because the company does not want to manage any servers or storage infrastructure.

Which solution will meet these requirements?

- A. Use Amazon Elastic Kubernetes Service (Amazon EKS) with self-managed nodes. Create an Amazon Elastic Block Store (Amazon EBS) volume attached to an Amazon EC2 instance. Use the EBS volume as a persistent volume mounted in the containers.
- B. Use Amazon Elastic Container Service (Amazon ECS) with an AWS Fargate launch type. Create an Amazon Elastic File System (Amazon EFS) volume. Add the EFS volume as a persistent storage volume mounted in the containers. **Most Voted**
- C. Use Amazon Elastic Container Service (Amazon ECS) with an AWS Fargate launch type. Create an Amazon S3 bucket. Map the S3 bucket as a persistent storage volume mounted in the containers.
- D. Use Amazon Elastic Container Service (Amazon ECS) with an Amazon EC2 launch type. Create an Amazon Elastic File System (Amazon EFS) volume. Add the EFS volume as a persistent storage volume mounted in the containers.

Correct Answer: B

Community vote distribution

B (92%)

8%

Comments

Marunio **Highly Voted** 2 years, 3 months ago

Selected Answer: B

Mounting S3 in Fargate is not supported commonly. You'd have to make it manually. EFS is very well supported with Fargate. <https://stackoverflow.com/questions/66391791/how-to-mount-s3-bucket-to-ecs-fargate-container>

<https://docs.aws.amazon.com/AmazonECS/latest/bestpracticesguide/storage.html>
upvoted 8 times

Andy_09 **Highly Voted** 2 years, 4 months ago

B looks correct
upvoted 6 times

waldirlsantos **Most Recent** 2 years, 1 month ago

Selected Answer: B

EFS is listed like a best practice for this cases
"ou can use Amazon ECS to run stateful containerized applications at scale by using AWS storage services, such as Amazon EFS, Amazon EBS, or FSx for Windows File Server, that provide data persistence to inherently ephemeral containers. The term data persistence means that the data itself outlasts the process that created it."
upvoted 2 times

MattBJ 2 years, 2 months ago

Selected Answer: B

B is correct
upvoted 2 times

ogerber 2 years, 3 months ago

Selected Answer: C

The company does not want to manage any servers or storage infrastructure.
I would go with C

upvoted 1 times

MatAlves 1 year, 8 months ago

Both B and C work with AWS launch type. So you have to decide between EFS vs S3.

How are you going to MOUNT s3 buckets in the containers?!

upvoted 2 times

MatAlves 1 year, 8 months ago

It's simply not good practice, unless you're confusing S3 with S3 storage gateway.

upvoted 2 times

A gaming company wants to launch a new internet-facing application in multiple AWS Regions. The application will use the TCP and UDP protocols for communication. The company needs to provide high availability and minimum latency for global users.

Which combination of actions should a solutions architect take to meet these requirements? (Choose two.)

- A. Create internal Network Load Balancers in front of the application in each Region. **Most Voted**
- B. Create external Application Load Balancers in front of the application in each Region.
- C. Create an AWS Global Accelerator accelerator to route traffic to the load balancers in each Region. **Most Voted**
- D. Configure Amazon Route 53 to use a geolocation routing policy to distribute the traffic.
- E. Configure Amazon CloudFront to handle the traffic and route requests to the application in each Region

Correct Answer: AC

Community vote distribution

AC (84%)

CD (16%)

Comments

Andy_09 **Highly Voted** 1 year, 10 months ago

Correct answer should be AC

upvoted 10 times

mestule 1 year, 10 months ago

Agreed.

When you add an internal Load Balancer or an Amazon EC2 instance endpoint in AWS Global Accelerator, you enable internet traffic to flow directly to and from the endpoint in Virtual Private Clouds (VPCs) by targeting it in a private subnet. The VPC that contains the load balancer or EC2 instance must have an internet gateway attached to it, to indicate that the VPC accepts internet traffic. However, you don't need public IP addresses on the load balancer or EC2 instance. You also don't need an associated internet gateway route for the subnet.

upvoted 8 times

ogerber **Highly Voted** 1 year, 10 months ago

Selected Answer: AC

Gaming + TCP / UDP => always think NLB and global accelerator

upvoted 5 times

yangbo **Most Recent** 1 year, 3 months ago

Selected Answer: CD

A is wrong . AWS Global Accelerator cannot directly connect to Internal Network Load Balancers (NLBs). It can only connect to Internet-facing NLBs or Application Load Balancers (ALBs), and cannot connect to Internal NLBs because Internal NLBs are designed to serve traffic within the VPC and are not exposed to the internet.

upvoted 3 times

yangbo 1 year, 3 months ago

client - dns - AWS Global Accelerator - NLB -EC2

upvoted 1 times

sandordini 1 year, 7 months ago

Selected Answer: AC

Gaming, TCP&UDP, HA, Low latency >> NLB + AWS Global Accelerator

upvoted 2 times

waldirlsantos 1 year, 8 months ago

Selected Answer: AC

Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP. NLB + GA for UDP, TCP

upvoted 2 times

Kezuko 1 year, 8 months ago

Selected Answer: AC

UDP -> NLB and Global Accelerator

upvoted 4 times

1Alpha1 1 year, 10 months ago

Selected Answer: AC

AC - the app is using TCP & UDP

upvoted 3 times

jaswantn 1 year, 10 months ago

For global user where TCP and UDP protocols are used and HA with minimum latency is needed.... Global Accelerator with NLB is the solution combination .

upvoted 3 times

A city has deployed a web application running on Amazon EC2 instances behind an Application Load Balancer (ALB). The application's users have reported sporadic performance, which appears to be related to DDoS attacks originating from random IP addresses. The city needs a solution that requires minimal configuration changes and provides an audit trail for the DDoS sources.

Which solution meets these requirements?

- A. Enable an AWS WAF web ACL on the ALB, and configure rules to block traffic from unknown sources.
- B. Subscribe to Amazon Inspector. Engage the AWS DDoS Response Team (DRT) to integrate mitigating controls into the service.
- C. Subscribe to AWS Shield Advanced. Engage the AWS DDoS Response Team (DRT) to integrate mitigating controls into the service. **Most Voted**
- D. Create an Amazon CloudFront distribution for the application, and set the ALB as the origin. Enable an AWS WAF web ACL on the distribution, and configure rules to block traffic from unknown sources

Correct Answer: C

Community vote distribution

C (100%)

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago

C is the correct answer

upvoted 7 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: C

Answer C

upvoted 1 times

NSA_Poker 2 years ago

Selected Answer: C

(A & D) are incorrect.

AWS WAF Web ACL - contain WAF rules that define how to inspect web requests and what to do when a web request matches the inspection criteria. We don't have the inspection criteria necessary to use WAF Web ACL effectively bc DDoS attacks are originating from random IP addresses. The AWS DDoS Response Team can respond to the randomness.

(B) is incorrect.

Amazon Inspector - a service that analyzes your EC2 instances to identify potential security and configuration issues.

Inspector is not good at dealing with an actual DDOS attack like AWS Shield Advanced.

upvoted 1 times

sandordini 2 years, 1 month ago

Selected Answer: C

DDoS = AWS Shield

upvoted 2 times

Mikado211 2 years, 2 months ago

Selected Answer: C

C is the correct answer, AWS Shield Advanced.

upvoted 1 times

asdfcdsxdcf 2 years, 3 months ago

Selected Answer: C

C looks correct
upvoted 1 times

Naveena_Devanga 2 years, 3 months ago

C is the correct answer.

Amazon Inspector is an automated vulnerability management service whereas AWS Shield Advanced is a managed service that helps you protect your application against external threats, like DDoS attacks, volumetric bots, and vulnerability exploitation attempts. For higher levels of protection against attacks.

upvoted 2 times

Darshan07 2 years, 4 months ago

Selected Answer: C

C is the correct answer

upvoted 1 times

A company copies 200 TB of data from a recent ocean survey onto AWS Snowball Edge Storage Optimized devices. The company has a high performance computing (HPC) cluster that is hosted on AWS to look for oil and gas deposits. A solutions architect must provide the cluster with consistent sub-millisecond latency and high-throughput access to the data on the Snowball Edge Storage Optimized devices. The company is sending the devices back to AWS.

Which solution will meet these requirements?

- A. Create an Amazon S3 bucket. Import the data into the S3 bucket. Configure an AWS Storage Gateway file gateway to use the S3 bucket. Access the file gateway from the HPC cluster instances.
- B. Create an Amazon S3 bucket. Import the data into the S3 bucket. Configure an Amazon FSx for Lustre file system, and integrate it with the S3 bucket. Access the FSx for Lustre file system from the HPC cluster instances. **Most Voted**
- C. Create an Amazon S3 bucket and an Amazon Elastic File System (Amazon EFS) file system. Import the data into the S3 bucket. Copy the data from the S3 bucket to the EFS file system. Access the EFS file system from the HPC cluster instances.
- D. Create an Amazon FSx for Lustre file system. Import the data directly into the FSx for Lustre file system. Access the FSx for Lustre file system from the HPC cluster instances.

Correct Answer: B

Community vote distribution

B (65%)

D (35%)

Comments

Linuslin **Highly Voted** 2 years ago

Selected Answer: B

No direct integration between Snowball and Fsx for Lustre. It must be via S3.
Snowball Edge (Storage Optimized) --> S3 --integrate--> FSx for Lustre

<https://docs.aws.amazon.com/fsx/latest/LustreGuide/create-dra-linked-data-repo.html>

<https://aws.amazon.com/tw/blogs/aws/enhanced-amazon-s3-integration-for-amazon-fsx-for-lustre/>
upvoted 16 times

Cali182 **Highly Voted** 2 years, 4 months ago

Selected Answer: D

Option D
Option A, B, and C involve using Amazon S3 or Amazon EFS as an intermediary storage layer, which may introduce additional latency and overhead, not meeting the requirement of consistent sub-millisecond latency. Therefore, Option D is the most suitable solution for this scenario.
upvoted 13 times

domper20232023 2 years, 3 months ago

The format on the Snowball device would be s3 compatible only. The FSx for Lustre file system can be created and then linked to the S3 bucket. The Lustre file system can then be mounted on the HPC workloads that need sub-millisecond latency to store data. Option B would be the correct option, assuming only S3 support on snowball.
upvoted 9 times

46a8566 **Most Recent** 4 months, 3 weeks ago

Selected Answer: B

FSx for Lustre can lazily load data from S3 on first access and automatically sync changes back to S3 — perfect for large datasets like 200 TB.
upvoted 1 times

a8a1e0e 1 year, 4 months ago

Selected Answer: B

Integration from Snowall to Fsc only with S3

upvoted 1 times

FlyingHawk 1 year, 5 months ago

Selected Answer: B

it is not possible to import data directly from Snowball Edge Storage Optimized devices to Amazon FSx for Lustre, FSx for Lustre is designed to work seamlessly with data stored in Amazon S3, providing integration for high-performance access and lazy loading. Snowball Edge devices are designed to securely transfer data to AWS services, particularly Amazon S3.
<https://aws.amazon.com/blogs/storage/automatically-import-objects-from-amazon-s3-into-amazon-fsx-for-lustre/>

upvoted 1 times

Abdullah2004 1 year, 9 months ago

Selected Answer: D

The correct answer is D for sure

upvoted 1 times

MatAlves 1 year, 8 months ago

You cannot import data from snowball in any other destination other than S3. So no, D is INCORRECT. Don't get tricked just because they mentioned HPC.

upvoted 2 times

n999 1 year, 10 months ago

Selected Answer: B

B for sure

upvoted 4 times

DZRomero 1 year, 11 months ago

Selected Answer: D

Import data to AWS services: When AWS receives the device, the data is automatically imported into the designated AWS service or Amazon S3 bucket based on your configuration. For example, if you need to access the data from an HPC cluster running on AWS, you would import the data into an Amazon FSx for Lustre file system or Amazon S3, and then access it from your HPC cluster instances.

No need S3 bucket

upvoted 1 times

trinh_le 2 years, 1 month ago

Selected Answer: B

You cannot access the FSx for Lustre file system from the HPC cluster instances and this is only possible via S3

upvoted 5 times

sandordini 2 years, 1 month ago

Selected Answer: D

HPC = Lustre

upvoted 3 times

sandordini 2 years, 1 month ago

Extension: HPC= Lustre, but Snowball = S3, therefore: B
Sync from Snowball to S3 -> Link/integrate with Lustre

Correct answer: C

upvoted 2 times

sandordini 2 years, 1 month ago

Which is of course not C but B... :D Sorry...

So correct answer: :D

upvoted 2 times

sukjubae 2 years, 2 months ago

B is right

upvoted 3 times

alawada 2 years, 2 months ago

Selected Answer: D is right answer because it mentions sub-millisecond latency and high-throughput access

upvoted 1 times

mgrimandi 2 years, 2 months ago

B

<https://medium.com/@abylead/amazon-fsx-for-migration-and-certification-f3cb7b4dd843>

upvoted 2 times

MattBJ 2 years, 2 months ago

Selected Answer: B

B is correct

upvoted 3 times

iczcezar 2 years, 3 months ago

Selected Answer: D

Option D, creating an Amazon FSx for Lustre file system and importing the data directly into it, is indeed the most suitable solution for this scenario. By bypassing an intermediary storage layer and directly importing the data into FSx for Lustre, the solution ensures optimal performance with consistent sub-millisecond latency and high throughput, meeting the requirements of the HPC cluster. Thank you for pointing out the clarity.

upvoted 1 times

FZA24 2 years, 3 months ago

Selected Answer: B

It should be B.

No direct integration between Snowball and Fsx for Lustre

upvoted 4 times

FZA24 2 years, 3 months ago

It must be via S3

upvoted 2 times

67a3f49 2 years, 3 months ago

Cali182 you cannot directly copy from Snowball Edge to FSx for luster

upvoted 1 times

A company has NFS servers in an on-premises data center that need to periodically back up small amounts of data to Amazon S3.

Which solution meets these requirements and is MOST cost-effective?

- A. Set up AWS Glue to copy the data from the on-premises servers to Amazon S3.
- B. Set up an AWS DataSync agent on the on-premises servers, and sync the data to Amazon S3. **Most Voted**
- C. Set up an SFTP sync using AWS Transfer for SFTP to sync data from on premises to Amazon S3.
- D. Set up an AWS Direct Connect connection between the on-premises data center and a VPC, and copy the data to Amazon S3.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago

B is the correct option

upvoted 9 times

BillaRanga **Highly Voted** 2 years, 3 months ago

Selected Answer: B

A -> Used for ETL not copying

B -> Works

C -> Works, but overkill for the described scenario of periodic small backups, high cost

D -> Works but it may not be necessary for transferring small amounts of data periodically. High setup cost

upvoted 9 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: B

AnswerB

Should be sufficient

upvoted 2 times

Darshan07 2 years, 4 months ago

Selected Answer: B

B is the correct option

upvoted 2 times

An online video game company must maintain ultra-low latency for its game servers. The game servers run on Amazon EC2 instances. The company needs a solution that can handle millions of UDP internet traffic requests each second.

Which solution will meet these requirements MOST cost-effectively?

- A. Configure an Application Load Balancer with the required protocol and ports for the internet traffic. Specify the EC2 instances as the targets.
- B. Configure a Gateway Load Balancer for the internet traffic. Specify the EC2 instances as the targets.
- C. Configure a Network Load Balancer with the required protocol and ports for the internet traffic. Specify the EC2 instances as the targets. **Most Voted**
- D. Launch an identical set of game servers on EC2 instances in separate AWS Regions. Route internet traffic to both sets of EC2 instances.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago
UDP needs NLB
upvoted 9 times

echonesis **Most Recent** 1 year, 6 months ago
UDP -> L4 Protocol -> NLB
upvoted 1 times

MatAlves 1 year, 8 months ago
Selected Answer: C
UDP > NLB.
upvoted 2 times

zinabu 2 years, 2 months ago
Ans : C
OfCourse we can use both NLB and GLB balancers for UDP traffic but NLB is more cost effective than GLB that is why we choice C.
upvoted 1 times

asdfcdsxdhc 2 years, 3 months ago
Selected Answer: C
TCP/UDP = NLB
upvoted 3 times

osmk 2 years, 3 months ago
C -> <https://docs.aws.amazon.com/elasticloadbalancing/latest/network/introduction.html>
upvoted 4 times

Marunio 2 years, 3 months ago
Selected Answer: C
UDP -> NLB.

ALB is for HTTP/HTTPS.

Gateway Load Balancer is for 3rd party virtual appliances like Firewalls etc not the traffic distribution.

<https://aws.amazon.com/compare/the-difference-between-the-difference-between-application-network-and-gateway-load-balancing/#:~:text=An%20NLB%20operates%20on%20layer,level%20along%20with%20gateway%20functionality.>

upvoted 3 times

Gagg 2 years, 3 months ago

Selected Answer: C

UDP, should use network load balancer

upvoted 2 times

nj1999 2 years, 4 months ago

C, NLB

upvoted 4 times

A company runs a three-tier application in a VPC. The database tier uses an Amazon RDS for MySQL DB instance.

The company plans to migrate the RDS for MySQL DB instance to an Amazon Aurora PostgreSQL DB cluster. The company needs a solution that replicates the data changes that happen during the migration to the new database.

Which combination of steps will meet these requirements? (Choose two.)

- A. Use AWS Database Migration Service (AWS DMS) Schema Conversion to transform the database objects. **Most Voted**
- B. Use AWS Database Migration Service (AWS DMS) Schema Conversion to create an Aurora PostgreSQL read replica on the RDS for MySQL DB instance.
- C. Configure an Aurora MySQL read replica for the RDS for MySQL DB instance.
- D. Define an AWS Database Migration Service (AWS DMS) task with change data capture (CDC) to migrate the data. **Most Voted**
- E. Promote the Aurora PostgreSQL read replica to a standalone Aurora PostgreSQL DB cluster when the replica lag is zero.

Correct Answer: AD

Community vote distribution

AD (100%)

Comments

h0ng97_spare_002 **Highly Voted** 2 years, 2 months ago

Selected Answer: AD

A: Correct. because need convert from MySQL to PostgreSQL

B: Wrong. Schema Conversion does not create an Aurora read replica

C: Wrong. Company wants to migrate to Aurora PostgreSQL, not Aurora MySQL

D: Correct. CDC task helps to capture ongoing change from source data store

E: Wrong. Although using Aurora Read Replica is an option for DB migration within the same Region, this question is asking for "combination of steps", which this option does not have another compatible option to pair with

Therefore, answer is "AD"

upvoted 8 times

mestule **Highly Voted** 2 years, 4 months ago

AD makes sense to me, but I am not sure if that's the best answer.

upvoted 6 times

Andy_09 2 years, 4 months ago

Agreed. AD makes more sense !!

upvoted 4 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: AD

AnswerAD

upvoted 2 times

xBUGx 2 years, 2 months ago

Lag many never be zero, then it will ne er be promoted to primary

upvoted 2 times

haci 2 years, 3 months ago

Selected Answer: AD

It's quite similar with Q.235, based on that discussion A-D makes more sense.
upvoted 4 times

1e22522 1 year, 10 months ago
of course, sin mas
upvoted 1 times

Andy_09 2 years, 4 months ago
Correct answer BE
upvoted 4 times

A company hosts a database that runs on an Amazon RDS instance that is deployed to multiple Availability Zones. The company periodically runs a script against the database to report new entries that are added to the database. The script that runs against the database negatively affects the performance of a critical application. The company needs to improve application performance with minimal costs.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Add functionality to the script to identify the instance that has the fewest active connections. Configure the script to read from that instance to report the total new entries.
- B. Create a read replica of the database. Configure the script to query only the read replica to report the total new entries.**
Most Voted
- C. Instruct the development team to manually export the new entries for the day in the database at the end of each day.
- D. Use Amazon ElastiCache to cache the common queries that the script runs against the database.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Moon239 **Highly Voted** 2 years, 4 months ago

Selected Answer: B

Read replica

upvoted 5 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: B

Answer B

upvoted 2 times

giovanna_mag 2 years, 3 months ago

Selected Answer: B

B, read replica

upvoted 3 times

mestule 2 years, 4 months ago

Selected Answer: B

B looks correct

upvoted 2 times

A company is using an Application Load Balancer (ALB) to present its application to the internet. The company finds abnormal traffic access patterns across the application. A solutions architect needs to improve visibility into the infrastructure to help the company understand these abnormalities better.

What is the MOST operationally efficient solution that meets these requirements?

- A. Create a table in Amazon Athena for AWS CloudTrail logs. Create a query for the relevant information.
- B. Enable ALB access logging to Amazon S3. Create a table in Amazon Athena, and query the logs. **Most Voted**
- C. Enable ALB access logging to Amazon S3. Open each file in a text editor, and search each line for the relevant information.
- D. Use Amazon EMR on a dedicated Amazon EC2 instance to directly query the ALB to acquire traffic access log information.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago

B is the correct answer
upvoted 11 times

Marunio **Highly Voted** 2 years, 3 months ago

Selected Answer: B

A - Cloudtrail is for API Calls and changes on AWS account.
B - Going for athena in S3. - Correct
C - Manual work
D - Distractor
upvoted 9 times

Saliigen **Most Recent** 1 year, 5 months ago

Selected Answer: B

<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-access-logs.html>
upvoted 1 times

Scheldon 1 year, 11 months ago

Selected Answer: B

Answer B
upvoted 2 times

Naveena_Devanga 2 years, 3 months ago

B -
Amazon Athena is an interactive query service that makes it easy to analyze data directly in Amazon Simple Storage Service (Amazon S3) using standard SQL.
upvoted 5 times

c48b4e2 2 years, 3 months ago

Why there is a "Correct answer" (the green bordered one) at all while most of the time the community thinks (correctly) otherwise?
upvoted 2 times

betttty 2 years, 4 months ago

why not A?
upvoted 3 times

Kezuko 2 years, 2 months ago

Access logs is an optional feature of Elastic Load Balancing that is disabled by default
upvoted 3 times

A company wants to use NAT gateways in its AWS environment. The company's Amazon EC2 instances in private subnets must be able to connect to the public internet through the NAT gateways.

Which solution will meet these requirements?

- A. Create public NAT gateways in the same private subnets as the EC2 instances.
- B. Create private NAT gateways in the same private subnets as the EC2 instances.
- C. Create public NAT gateways in public subnets in the same VPCs as the EC2 instances. **Most Voted**
- D. Create private NAT gateways in public subnets in the same VPCs as the EC2 instances.

Correct Answer: C

Community vote distribution

C (100%)

Comments

anikolov **Highly Voted** 1 year, 10 months ago

Selected Answer: C

Should be C: Public NAT GW in Public Subnet to have access to internet. Private NAT GW is used for VPC or on-prem
upvoted 16 times

mestule **Highly Voted** 1 year, 10 months ago

Selected Answer: C

I think the correct is C, because D would require more than just private NAT gateway.

Private – Instances in private subnets can connect to other VPCs or your on-premises network through a private NAT gateway. You can route traffic from the NAT gateway through a transit gateway or a virtual private gateway. You cannot associate an elastic IP address with a private NAT gateway. You can attach an internet gateway to a VPC with a private NAT gateway, but if you route traffic from the private NAT gateway to the internet gateway, the internet gateway drops the traffic.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html>

upvoted 9 times

Kezuko **Most Recent** 1 year, 8 months ago

Selected Answer: C

Public NAT Gateway in public subnets for the internet access
<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html>
upvoted 3 times

knben 1 year, 9 months ago

Selected Answer: C

Public NAT GW in Public Subnet to have access to internet
upvoted 2 times

Andy_09 1 year, 10 months ago

Looks correct
upvoted 2 times

A company has an organization in AWS Organizations. The company runs Amazon EC2 instances across four AWS accounts in the root organizational unit (OU). There are three nonproduction accounts and one production account. The company wants to prohibit users from launching EC2 instances of a certain size in the nonproduction accounts. The company has created a service control policy (SCP) to deny access to launch instances that use the prohibited types.

Which solutions to deploy the SCP will meet these requirements? (Choose two.)

A. Attach the SCP to the root OU for the organization.

B. Attach the SCP to the three nonproduction Organizations member accounts. **Most Voted**

C. Attach the SCP to the Organizations management account.

D. Create an OU for the production account. Attach the SCP to the OU. Move the production member account into the new OU.

E. Create an OU for the required accounts. Attach the SCP to the OU. Move the nonproduction member accounts into the new OU. **Most Voted**

Correct Answer: BE

Community vote distribution

BE (90%)

7%

Comments

anikolov **Highly Voted** 2 years, 4 months ago

Selected Answer: BE

My vote is for BE

upvoted 13 times

mestule **Highly Voted** 2 years, 4 months ago

Selected Answer: BE

I think it's B (directly attach) and E (attach via OU).

upvoted 6 times

Dz1990 **Most Recent** 1 year ago

Selected Answer: BE

🔗 Approach 1: Direct Account Attachment (Option B)

SCP → Directly attached to each nonproduction account

Pros: Simple, direct, immediate

Cons: Must manage policy on multiple accounts

🔗 Approach 2: Organizational Unit Structure (Option E)

SCP → Nonproduction OU → All nonproduction accounts inherit

Pros: Cleaner organization, easier to manage, scalable

Cons: Requires OU restructuring

upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: BE

B - Attach the SPC to the three accounts

E - Creates an OU > moves the member accounts to OU > attach the SCP to OU

"If you apply an authorization policy (for example, a service control policy (SCP)), to the root, it applies to all organizational units (OUs) and member accounts in the organization."

"A" would also affect the one production account, which we clearly don't want.

You can "attach an SCP to a root, OU, or account"

upvoted 2 times

MatAlves 1 year, 8 months ago

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps.html

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_getting-started_concepts.html

upvoted 2 times

sandordini 2 years, 1 month ago

Selected Answer: BE

Only the non-prods need to be limited.

upvoted 4 times

67a3f49 2 years, 3 months ago

According to GPT-4 it's AE:

A. Attach the SCP to the root OU for the organization. This approach will apply the SCP to all accounts under the organization, including both nonproduction and production accounts. However, without additional context or actions, this does not meet the requirement to exclude the production account from the restrictions.

E. Create an OU for the required accounts. Attach the SCP to the OU. Move the nonproduction member accounts into the new OU. This is the correct approach as it directly addresses the requirement. By creating a separate OU for nonproduction accounts and attaching the SCP to this OU, you can specifically target the policy to only those accounts, effectively exempting the production account from the restrictions.

upvoted 1 times

1Alpha1 2 years, 3 months ago

Selected Answer: AC

AC - same answer

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_getting-started_concepts.html

upvoted 1 times

MatAlves 1 year, 8 months ago

The link you provided says:

"If you apply an authorization policy (for example, a service control policy (SCP)), to the root, it applies to all organizational units (OUs) and member accounts in the organization."

"A" would also affect the one production account, which we clearly don't want.

You can "attach an SCP to a root, OU, or account"

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps.html

upvoted 2 times

Cali182 2 years, 4 months ago

Selected Answer: AD

From Chat

A. Attach the SCP to the root OU for the organization: Attaching the SCP to the root OU ensures that it applies to all member accounts within the organization, including both nonproduction and production accounts.

D. Create an OU for the production account. Attach the SCP to the OU. Move the production member account into the new OU: By creating a separate OU for the production account and attaching the SCP to that OU, you can ensure that the SCP only affects the nonproduction accounts while allowing the production account to operate without restrictions.

upvoted 2 times

Andy_09 2 years, 4 months ago

CE should be the correct answer

upvoted 1 times

A company's website hosted on Amazon EC2 instances processes classified data stored in Amazon S3. Due to security concerns, the company requires a private and secure connection between its EC2 resources and Amazon S3.

Which solution meets these requirements?

- A. Set up S3 bucket policies to allow access from a VPC endpoint. **Most Voted**
- B. Set up an IAM policy to grant read-write access to the S3 bucket.
- C. Set up a NAT gateway to access resources outside the private subnet.
- D. Set up an access key ID and a secret access key to access the S3 bucket.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Ashy1313 **Highly Voted** 1 year, 10 months ago

Selected Answer: A

A VPC endpoint enables customers to privately connect to supported AWS services .
upvoted 7 times

sandordini **Highly Voted** 1 year, 7 months ago

Selected Answer: A

I think this question asks about the connection not about authorization, and for a secure S3 connection (e.g. without internet exposure, etc.) should be a VPC endpoint.
upvoted 7 times

Naveena_Devanga **Most Recent** 1 year, 9 months ago

D is the correct answer.
upvoted 1 times

Darshan07 1 year, 10 months ago

Selected Answer: A

A is the correct answer
upvoted 3 times

An ecommerce company runs its application on AWS. The application uses an Amazon Aurora PostgreSQL cluster in Multi-AZ mode for the underlying database. During a recent promotional campaign, the application experienced heavy read load and write load. Users experienced timeout issues when they attempted to access the application.

A solutions architect needs to make the application architecture more scalable and highly available.

Which solution will meet these requirements with the LEAST downtime?

A. Create an Amazon EventBridge rule that has the Aurora cluster as a source. Create an AWS Lambda function to log the state change events of the Aurora cluster. Add the Lambda function as a target for the EventBridge rule. Add additional reader nodes to fail over to.

B. Modify the Aurora cluster and activate the zero-downtime restart (ZDR) feature. Use Database Activity Streams on the cluster to track the cluster status.

C. Add additional reader instances to the Aurora cluster. Create an Amazon RDS Proxy target group for the Aurora cluster.

Most Voted

D. Create an Amazon ElastiCache for Redis cache. Replicate data from the Aurora cluster to Redis by using AWS Database Migration Service (AWS DMS) with a write-around approach.

Correct Answer: C

Community vote distribution

C (100%)

Comments

Marunio **Highly Voted** 2 years, 3 months ago

Selected Answer: C

Only C is real viable option - Adding Reader replica for handling Read load and RDS Proxy for connections.
upvoted 7 times

Andy_09 **Highly Voted** 2 years, 4 months ago

I would go for option C
upvoted 5 times

sk1974 **Most Recent** 1 year, 3 months ago

Selected Answer: C

C is the most applicable among the 4 available options.
upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: C

C - Explanation below:
upvoted 2 times

MatAlves 1 year, 8 months ago

A - "Lambda function to log state changes" - doesn't help with read/write load.

B - ZDR applies to restarts that Aurora performs automatically to resolve error conditions: doesn't help with read/write load.

D - Write-around approach: data is always written to the database and the data that is read goes to the cache. Doesn't help with read/write load.

C - CORRECT. Even though it doesn't address "write operations", Aurora Replicas to offload read workloads from the primary DB instance. Amazon RDS Proxy allows applications to pool and share connections established with the database, improving database efficiency and application scalability

upvoted 4 times

Scheldon 1 year, 11 months ago

Selected Answer: C

AnswerC.

Proxy should help with the problem

upvoted 2 times

alawada 2 years, 2 months ago

Selected Answer: C

RDX proxy to handle timeout issue

upvoted 2 times

xBUGx 2 years, 2 months ago

Selected Answer: C

I go with C bc there is no better option

upvoted 4 times

jaswantn 2 years, 4 months ago

RDX proxy to handle timeout issue. option C

upvoted 2 times

A company is designing a web application on AWS. The application will use a VPN connection between the company's existing data centers and the company's VPCs.

The company uses Amazon Route 53 as its DNS service. The application must use private DNS records to communicate with the on-premises services from a VPC.

Which solution will meet these requirements in the MOST secure manner?

- A. Create a Route 53 Resolver outbound endpoint. Create a resolver rule. Associate the resolver rule with the VPC. **Most Voted**
- B. Create a Route 53 Resolver inbound endpoint. Create a resolver rule. Associate the resolver rule with the VPC.
- C. Create a Route 53 private hosted zone. Associate the private hosted zone with the VPC.
- D. Create a Route 53 public hosted zone. Create a record for each service to allow service communication

Correct Answer: A

Community vote distribution

A (88%)

12%

Comments

haci **Highly Voted** 2 years, 3 months ago

Selected Answer: A

If you have workloads that leverage both VPCs and on-premises resources, you also need to resolve DNS records hosted on-premises. Similarly, these on-premises resources may need to resolve names hosted on AWS. Through Resolver endpoints and conditional forwarding rules, you can resolve DNS queries between your on-premises resources and VPCs to create a hybrid cloud setup over VPN or Direct Connect (DX). Specifically:

Inbound Resolver endpoints allow DNS queries to your VPC from your on-premises network or another VPC.

Outbound Resolver endpoints allow DNS queries from your VPC to your on-premises network or another VPC.

Reference: <https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/resolver.html>

upvoted 18 times

MatAlves 1 year, 8 months ago

Right to the point!

upvoted 2 times

JCVDB23 **Highly Voted** 2 years, 2 months ago

Selected Answer: A

Amazon Route 53 Resolver provides DNS resolution for VPCs and on-premises networks over a Direct Connect or VPN connection. An outbound resolver endpoint forwards DNS queries from your VPC to your on-premises DNS service. A resolver rule specifies the domain names for the DNS queries that you want to forward (such as example.com), and the IP addresses of the DNS resolvers in your on-premises network. Option C is not suitable because private hosted zones are used to route traffic within a VPC

<https://aws.amazon.com/blogs/architecture/using-route-53-private-hosted-zones-for-cross-account-multi-region-architectures/>

upvoted 6 times

Joshua721 **Most Recent** 1 year, 4 months ago

Selected Answer: A

Inbound Resolver endpoints allow DNS queries to your VPC from your on-premises network or another VPC.

Outbound Resolver endpoints allow DNS queries from your VPC to your on-premises network or another VPC.

Resolver rules enable you to create one forwarding rule for each domain name and specify the name of the domain for which you want to forward DNS queries from your VPC to an on-premises DNS resolver and from your on-premises to your VPC. Rules are applied directly to your VPC and can be shared across multiple accounts.

upvoted 2 times

Duddle88 1 year, 5 months ago

Selected Answer: C

"The company uses Amazon Route 53 as its DNS service" THERE IS NO DNS ON-PREMISES

Private DNS Resolution: It allows you to create DNS records that are only accessible within your VPC, ensuring secure and private communication between your AWS resources and on-premises network.

Forwarding DNS Queries: This is useful if you need to forward DNS queries from your VPC to your on-premises DNS servers. It's typically used when you have existing DNS infrastructure on-premises that you want to continue using.

upvoted 2 times

MatAlves 1 year, 8 months ago

Selected Answer: A

AWS <-> On-premises = Route 53 Resolver

- Outbound Resolver = From your VPC (AWS) to On-premises or another VPC

- Inbound Resolver = From on-premises network or another VPC TO your VPC.

upvoted 3 times

Jacky_S 1 year, 11 months ago

Selected Answer: C

The reason why i vote on C, because the question mentioned that "The company uses Amazon Route53 as it's DNS service" and did not mention that is using multiple accounts, so it should be the most secure way to just add the record in it's private host zone of it's own account due to dns poisoning concern.

Of cause, i totally agree on A if the dns zone owner is in on-premises dns server which reduce the operation efforts.

upvoted 3 times

cjace 1 year, 12 months ago

C. Create a Route 53 private hosted zone. Associate the private hosted zone with the VPC.

This setup allows the application within the VPC to resolve DNS queries using private DNS records, ensuring that the communication remains within the AWS network and is not exposed to the public internet. Associating the private hosted zone with the VPC ensures that only the resources within the VPC can resolve the DNS queries, maintaining a secure environment for application and on-premises service communication.

The outbound resolver endpoint and rule would be more relevant if the requirement was for resources within the VPC to resolve DNS queries for domain names that are located in the on-premises network. In that case, the outbound resolver would forward queries from the VPC to the on-premises DNS server for resolution. However, for private DNS communication from the VPC to on-premises services, the private hosted zone is the most secure method.

upvoted 3 times

alawada 2 years, 2 months ago

Selected Answer: A

Amazon Route 53 Resolver provides DNS resolution for VPCs and on-premises networks

upvoted 2 times

anikolov 2 years, 4 months ago

Selected Answer: A

Should be A "Create a Route 53 Resolver outbound endpoint."

upvoted 5 times

Andy_09 2 years, 4 months ago

Looks correct

upvoted 2 times

A company is running a photo hosting service in the us-east-1 Region. The service enables users across multiple countries to upload and view photos. Some photos are heavily viewed for months, and others are viewed for less than a week. The application allows uploads of up to 20 MB for each photo. The service uses the photo metadata to determine which photos to display to each user.

Which solution provides the appropriate user access MOST cost-effectively?

- A. Store the photos in Amazon DynamoDB. Turn on DynamoDB Accelerator (DAX) to cache frequently viewed items.
- B. Store the photos in the Amazon S3 Intelligent-Tiering storage class. Store the photo metadata and its S3 location in DynamoDB. **Most Voted**
- C. Store the photos in the Amazon S3 Standard storage class. Set up an S3 Lifecycle policy to move photos older than 30 days to the S3 Standard-Infrequent Access (S3 Standard-IA) storage class. Use the object tags to keep track of metadata.
- D. Store the photos in the Amazon S3 Glacier storage class. Set up an S3 Lifecycle policy to move photos older than 30 days to the S3 Glacier Deep Archive storage class. Store the photo metadata and its S3 location in Amazon OpenSearch Service.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago

B is the correct option

upvoted 10 times

Typewriter101 **Highly Voted** 2 years, 3 months ago

Selected Answer: B

The Intelligent-Tiering storage class automatically moves objects between two access tiers (frequent access and infrequent access) based on their access patterns, which aligns well with the varying view frequencies of the photos. Storing metadata in DynamoDB allows for efficient querying and retrieval of photo metadata.

upvoted 10 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: B

AnswerB

Seem to be the best solution from provided

upvoted 2 times

alawada 2 years, 2 months ago

Selected Answer: B

Store the photos in the Amazon S3 Intelligent-Tiering = Unpredictable scenario

upvoted 3 times

Indrasis 2 years, 3 months ago

Correct option: B

upvoted 2 times

A company runs a highly available web application on Amazon EC2 instances behind an Application Load Balancer. The company uses Amazon CloudWatch metrics.

As the traffic to the web application increases, some EC2 instances become overloaded with many outstanding requests. The CloudWatch metrics show that the number of requests processed and the time to receive the responses from some EC2 instances are both higher compared to other EC2 instances. The company does not want new requests to be forwarded to the EC2 instances that are already overloaded.

Which solution will meet these requirements?

- A. Use the round robin routing algorithm based on the RequestCountPerTarget and ActiveConnectionCount CloudWatch metrics.
- B. Use the least outstanding requests algorithm based on the RequestCountPerTarget and ActiveConnectionCount CloudWatch metrics. **Most Voted**
- C. Use the round robin routing algorithm based on the RequestCount and TargetResponseTime CloudWatch metrics.
- D. Use the least outstanding requests algorithm based on the RequestCount and TargetResponseTime CloudWatch metrics.

Correct Answer: B

Community vote distribution

B (80%)

D (20%)

Comments

h0ng97_spare_002 **Highly Voted** 2 years, 2 months ago

Selected Answer: B

Option B is correct because can use "RequestCountPerTarget" to identify the amount of requests for each EC2 instance. Then use "least outstanding requests algorithm" to route to targets with the lowest number of in progress requests.

Option D is wrong because "RequestCount" cannot identify the amount of requests for each EC2 instance. "RequestCount" is for the whole ALB.
upvoted 11 times

Tatai2015 2 years ago

<https://aws.amazon.com/about-aws/whats-new/2019/11/application-load-balancer-now-supports-least-outstanding-requests-algorithm-for-load-balancing-requests/>
upvoted 3 times

Andy_09 **Highly Voted** 2 years, 4 months ago

Option B would be the correct choice
upvoted 8 times

chest_jd **Most Recent** 1 year, 1 month ago

Selected Answer: D

D is correct.
<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-cloudwatch-metrics.html>
ActiveConnectionCount - The total number of concurrent TCP connections active from clients to the load balancer
RequestCountPerTarget - The average request count per target, in a target group
Both metrics do not show stage of every target.

RequestCount - The number of requests processed over IPv4 and IPv6. This metric is only incremented for requests where the load balancer node was able to choose a target.

We can say how many requests go to targets.

TargetResponseTime - The time elapsed, in seconds, after the request leaves the load balancer until the target starts to send the response headers.
We have metric to get statistics for every target/instance.

In fact Item D is correct

upvoted 1 times

sk1974 1 year, 3 months ago

Selected Answer: B

Actually ,I now vote for B and not D

upvoted 1 times

sk1974 1 year, 3 months ago

Selected Answer: D

I will go with D since the question asks for the target response.

upvoted 1 times

FlyingHawk 1 year, 5 months ago

Selected Answer: B

D will be correct if change ReqeustCount to RequestCountPerTarget. RequestCount applies to the ALB rather than individual targets, making it unsuitable for identifying overloaded instances.

ActiveConnectionCount:

This metric indicates the current number of active TCP connections. It complements RequestCountPerTarget by helping identify instances that might be overwhelmed with connections, even if their request count seems moderate.

upvoted 1 times

Scheldon 1 year, 11 months ago

Selected Answer: B

AnswerB

upvoted 2 times

TruthWS 2 years, 2 months ago

Option B

upvoted 3 times

dkw2342 2 years, 2 months ago

IMO the correct answer is option D:

This is from an earlier version of the AWS documentation on ALB target groups - for some reason they removed this information in the current revision:

"Consider using least outstanding requests when the requests for your application vary in complexity or your targets vary in processing capability. Round robin is a good choice when the requests and targets are similar, or if you need to distribute requests equally among targets. You can compare the effect of round robin versus least outstanding requests using the following CloudWatch metrics: RequestCount, TargetConnectionErrorCount, and TargetResponseTime."

<https://web.archive.org/web/20200426172626/https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-target-groups.html#modify-routing-algorithm>

upvoted 2 times

h0ng97_spare_002 2 years, 2 months ago

The link is just saying that you can view "RequestCount, TargetConnectionErrorCount, and TargetResponseTime" to understand the difference in effect between round robin vs least outstanding requests. It is not the direct answer to this question.

upvoted 3 times

Salilgen 1 year, 5 months ago

I agree.

"The CloudWatch metrics show that the number of requests processed and the time to receive the responses from some EC2 instances are both higher compared to other EC2 instances". Then we must monitor time and number of requests metrics

upvoted 1 times

xBUGx 2 years, 2 months ago

Selected Answer: D

I think TargetResponseTime is the best indicator for telling is a server is overloaded or not

upvoted 1 times

alawada 2 years, 2 months ago

Selected Answer: B

distribute the number of requests among instances

upvoted 2 times

Kezuko 2 years, 2 months ago

Selected Answer: B

<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-cloudwatch-metrics.html>

To understand the types
upvoted 4 times

haci 2 years, 3 months ago

Selected Answer: B

The question is not asking for better performance in response time. It is just simply asking to distribute the number of requests among instances. So B seems more logical.

upvoted 3 times

osmk 2 years, 3 months ago

Selected Answer: D

The least outstanding requests routing algorithm routes requests to the targets with the lowest number of in progress requests >
<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-target-groups.html>

upvoted 3 times

osmk 2 years, 3 months ago

D>>> The least outstanding requests routing algorithm routes requests to the targets with the lowest number of in progress requests >
<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-target-groups.html>

upvoted 1 times

Moon239 2 years, 4 months ago

Why not D?

upvoted 2 times

jaswantn 2 years, 4 months ago

With Least outstanding requests algorithm, new request will send it to the "target" with least number of outstanding requests. Targets processing long-standing requests or having lower processing capabilities are not burdened with more requests. That's why option B is correct and not option D.

upvoted 3 times

A company uses Amazon EC2, AWS Fargate, and AWS Lambda to run multiple workloads in the company's AWS account. The company wants to fully make use of its Compute Savings Plans. The company wants to receive notification when coverage of the Compute Savings Plans drops.

Which solution will meet these requirements with the MOST operational efficiency?

A. Create a daily budget for the Savings Plans by using AWS Budgets. Configure the budget with a coverage threshold to send notifications to the appropriate email message recipients. **Most Voted**

B. Create a Lambda function that runs a coverage report against the Savings Plans. Use Amazon Simple Email Service (Amazon SES) to email the report to the appropriate email message recipients.

C. Create an AWS Budgets report for the Savings Plans budget. Set the frequency to daily.

D. Create a Savings Plans alert subscription. Enable all notification options. Enter an email address to receive notifications.

Correct Answer: A

Community vote distribution

A (65%)

D (35%)

Comments

anikolov **Highly Voted** 2 years, 4 months ago

Selected Answer: A

My vote is for A : <https://docs.aws.amazon.com/savingsplans/latest/userguide/sp-usingBudgets.html>

upvoted 10 times

lenotc **Highly Voted** 2 years, 2 months ago

Selected Answer: D

D:

<https://aws.amazon.com/about-aws/whats-new/2020/11/savings-plans-alerts-now-available-in-aws-cost-management/>

upvoted 6 times

sk1974 **Most Recent** 1 year, 3 months ago

Selected Answer: D

Question asks when 'The company wants to receive notification when coverage of the Compute Savings Plans drops.' . So . I will say answer id D

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: A

D is alert for expiration. A is for coverage drop.

upvoted 2 times

Mistwalker 1 year, 4 months ago

Selected Answer: D

It sure seems to be this:

<https://aws.amazon.com/about-aws/whats-new/2020/11/savings-plans-alerts-now-available-in-aws-cost-management/>

upvoted 1 times

rosanna 1 year, 5 months ago

Selected Answer: D

AWS budget is mostly used to avoid exceeding the threshold, not under utilization alert.

upvoted 2 times

MatAlves 1 year, 8 months ago

Selected Answer: A

A - describes exactly what is said in this link:

"You can use AWS Budgets to enable simple-to-complex cost and usage tracking. Some examples include:
(...)

Setting a daily utilization or coverage budget to track your RI or Savings Plans. You can choose to be notified through email and Amazon SNS topics when your utilization drops below 80 percent for a given day."

<https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html>

upvoted 4 times

MatAlves 1 year, 8 months ago

<https://docs.aws.amazon.com/savingsplans/latest/userguide/sp-alert.html>

upvoted 2 times

MatAlves 1 year, 8 months ago

D - clearly not what the question is asking:

"You can track your Savings Plans expirations and upcoming queued Savings Plans in Cost Explorer. You can use Savings Plans alerts to receive advance email alerts 1, 7, 30, or 60 days before your Savings Plan expiration date, or in when a commitment is queued for purchase. These notifications also alert you on the expiration date, and can be sent to up to 10 email recipients."

"Savings Plans Expiration Alerts notify you as your existing Savings Plans approach expiration. These alerts can help you easily renew your Savings Plans."

upvoted 3 times

Scheldon 1 year, 11 months ago

Selected Answer: A

AnswerA

We can set Savings Plan in AWS Budgets which will notify us if utilization, coverage and costs will be not in set range.

<https://docs.aws.amazon.com/savingsplans/latest/userguide/sp-usingBudgets.html>

upvoted 3 times

Kezuko 2 years, 2 months ago

Selected Answer: D

<https://aws.amazon.com/blogs/aws-cloud-financial-management/launch-savings-plans-expiration-and-queued-alerts-now-available-in-aws-cost-management/>

upvoted 4 times

YGHUIWRHF1234 2 years, 2 months ago

Selected Answer: A

Correct answer is A

upvoted 2 times

xBUGx 2 years, 2 months ago

Selected Answer: A

A is precisely targeted

upvoted 3 times

ManishGup 2 years, 3 months ago

Ny vote going to D.

<https://aws.amazon.com/blogs/aws-cloud-financial-management/launch-savings-plans-expiration-and-queued-alerts-now-available-in-aws-cost-management/>

upvoted 1 times

Indrasis 2 years, 3 months ago

Selected Answer: A

A is correct

upvoted 2 times

jaswantn 2 years, 4 months ago

Option D...In the Savings Plans Overview page indicate how many days in advance you would like to receive Savings Plans Alerts for Plan's expiration and upcoming queued purchase notifications.

upvoted 3 times

Andy_09 2 years, 4 months ago

Option D

upvoted 3 times

hajra313 2 years, 4 months ago

alert subscription will notify u before ending saving plan

upvoted 1 times

A company runs a real-time data ingestion solution on AWS. The solution consists of the most recent version of Amazon Managed Streaming for Apache Kafka (Amazon MSK). The solution is deployed in a VPC in private subnets across three Availability Zones.

A solutions architect needs to redesign the data ingestion solution to be publicly available over the internet. The data in transit must also be encrypted.

Which solution will meet these requirements with the MOST operational efficiency?

A. Configure public subnets in the existing VPC. Deploy an MSK cluster in the public subnets. Update the MSK cluster security settings to enable mutual TLS authentication. **Most Voted**

B. Create a new VPC that has public subnets. Deploy an MSK cluster in the public subnets. Update the MSK cluster security settings to enable mutual TLS authentication.

C. Deploy an Application Load Balancer (ALB) that uses private subnets. Configure an ALB security group inbound rule to allow inbound traffic from the VPC CIDR block for HTTPS protocol.

D. Deploy a Network Load Balancer (NLB) that uses private subnets. Configure an NLB listener for HTTPS communication over the internet.

Correct Answer: A

Community vote distribution

A (96%)

4%

Comments

haci **Highly Voted** 2 years, 3 months ago

Selected Answer: A

Since we are talking about real-time data (UDP packets) ALB is not a viable solution. You don't need to listen HTTPS, so D is eliminated. If you create a new VPC, you must create link between the old one and this is not mentioned in B. So It is A for me.

upvoted 11 times

Dantecito **Most Recent** 1 year, 3 months ago

Selected Answer: D

D. NLB forward traffic from the internet to private subnet where the cluster is. MSK use layer 4.

A. B. use public subnets exposing the cluster

C. Similar to D but MSK requires a Layer 4.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: A

For MSK to get public access, you must associate it with a public subnet, so rule out C and D, B is to create a new VPC, which is more complicated and more operations, so A is correct.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

To turn on public access to an MSK Provisioned cluster, first ensure that the cluster meets all of the following conditions:

The subnets that are associated with the cluster must be public. Each public subnet has a public IPv4 address associated with it and public IPv4 addresses are priced as shown in Amazon VPC pricing page. This means that the subnets must have an associated route table with an internet gateway attached. For information about how to create and attach an internet gateway, see Enable VPC internet access using internet gateways in the Amazon VPC User Guide.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Unauthenticated access control must be off and at least one of the following access-control methods must be on: SASL/IAM, SASL/SCRAM, mTLS. For information about how to update the access-control method of a cluster, see Update security settings of a Amazon MSK cluster.

upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: A

"You can turn on public access to an MSK cluster at no additional cost...

To turn on public access to a cluster, first ensure that the cluster meets all of the following conditions:

- The subnets that are associated with the cluster must be public.
- Unauthenticated access control must be off and at least one of the following access-control methods must be on: SASL/IAM, SASL/SCRAM, mTLS.
- ..."

<https://docs.aws.amazon.com/msk/latest/developerguide/public-access.html>

upvoted 4 times

MatAlves 1 year, 8 months ago

There is no reference to a NEW VPC being required in the documentation. We can simply configure subnets in the existing VPC.

upvoted 2 times

Scheldon 1 year, 11 months ago

Selected Answer: A

AnswerA

I need to agree that answer will probably be Option A.

upvoted 1 times

Indrasis 2 years, 3 months ago

Selected Answer: A

A is correct

upvoted 2 times

Marunio 2 years, 3 months ago

Selected Answer: A

A, since Kafka is loadbalancing itself. - <https://dattell.com/data-architecture-blog/load-balancing-with-kafka/#:~:text=Load%20balancing%20with%20Kafka%20is,partitions%20while%20preserving%20message%20ordering.>

B - why create new VPC?

C / D - Kafka is loadbalancing itself, also NLB can't handle HTTPS.

upvoted 3 times

Andy_09 2 years, 4 months ago

Option A

upvoted 4 times

A company wants to migrate an on-premises legacy application to AWS. The application ingests customer order files from an on-premises enterprise resource planning (ERP) system. The application then uploads the files to an SFTP server. The application uses a scheduled job that checks for order files every hour.

The company already has an AWS account that has connectivity to the on-premises network. The new application on AWS must support integration with the existing ERP system. The new application must be secure and resilient and must use the SFTP protocol to process orders from the ERP system immediately.

Which solution will meet these requirements?

- A. Create an AWS Transfer Family SFTP internet-facing server in two Availability Zones. Use Amazon S3 storage. Create an AWS Lambda function to process order files. Use S3 Event Notifications to send s3:ObjectCreated:* events to the Lambda function.
- B. Create an AWS Transfer Family SFTP internet-facing server in one Availability Zone. Use Amazon Elastic File System (Amazon EFS) storage. Create an AWS Lambda function to process order files. Use a Transfer Family managed workflow to invoke the Lambda function.
- C. Create an AWS Transfer Family SFTP internal server in two Availability Zones. Use Amazon Elastic File System (Amazon EFS) storage. Create an AWS Step Functions state machine to process order files. Use Amazon EventBridge Scheduler to invoke the state machine to periodically check Amazon EFS for order files.
- D. Create an AWS Transfer Family SFTP internal server in two Availability Zones. Use Amazon S3 storage. Create an AWS Lambda function to process order files. Use a Transfer Family managed workflow to invoke the Lambda function. **Most Voted**

Correct Answer: D

Community vote distribution

D (84%)

A (16%)

Comments

anikolov **Highly Voted** 2 years, 4 months ago

Selected Answer: D

D looks more secure over existing on-prem to AWS connection
 -Transfer Family SFTP internal server in two Availability Zones.
 -Use Amazon S3 storage.
 -Use a Transfer Family managed workflow to invoke the Lambda function"
 upvoted 11 times

hajra313 2 years, 4 months ago

If the legacy application needs to ingest customer order files from an on-premises ERP system and upload them to an SFTP server, an internet-facing AWS Transfer Family SFTP server would be the appropriate choice.

In this scenario, the SFTP server needs to be accessible from the internet to facilitate the file transfer between the on-premises system and AWS. Therefore, an internet-facing server is required to securely receive the files.

upvoted 2 times

Mr_Marcus 2 years ago

"The company already has an AWS account that has connectivity to the on-premises network." Internal Server.

upvoted 2 times

bora4motion **Most Recent** 1 year ago

Selected Answer: D

A is wrong, the company already has a connection between aws and on prem, this means a dx or s2s vpn, therefore you dont have to expose the sftp to internet and transfer the data over the internet.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: D

A and D are similar, but A internet-facing server is less secure than D, also no need to use S3 Event Notifications to send s3:ObjectCreated:* events to the Lambda function because Transfer Family managed workflows can directly invoke Lambda..

<https://docs.aws.amazon.com/transfer/latest/userguide/transfer-workflows.html>

upvoted 1 times

FlyingHawk 1 year, 4 months ago

AWS Transfer Family supports managed workflows for file processing. With managed workflows, you can kick off a workflow after a file has been transferred over SFTP, FTPS, or FTP. Using this feature, you can securely and cost effectively meet your compliance requirements for business-to-business (B2B) file exchanges by coordinating all the necessary steps required for file processing. In addition, you benefit from end-to-end auditing and visibility.

<https://aws.amazon.com/blogs/storage/customize-file-delivery-notifications-using-aws-transfer-family-managed-workflows/>

upvoted 1 times

ckhemani 1 year, 6 months ago

Selected Answer: D

A and B is out since it is public facing and we don't need public facing SFTP since Company already have Connection to AWS thro ON-PREMISES.

Initially i thought it will C since it was doing scheduled as they were doing scheduled job every hour on on-premises. Then the read thro question and saw they are saying with new solution, they want to process the file immediately after it is uploaded so D is correct since it will trigger the Lambda file as soon as file appears on S3.

upvoted 3 times

Scheldon 1 year, 11 months ago

Selected Answer: D

Answer D

upvoted 2 times

sandordini 2 years, 1 month ago

Selected Answer: D

"order files from an on-premises enterprise resource planning (ERP)" - Therefore Internal Endpoint is enough, no need for Internet-facing, although Internet-facing also handles on-prem connections as well, but "most secure". Even tho we are talking about SecureFTP.... Very bad wording of the question... :(

Definitely S3 against EFS, so D should be the answer...

upvoted 3 times

sandordini 2 years, 1 month ago

Also: With managed workflows, you can kick off a workflow after a file has been transferred over SFTP

upvoted 2 times

Hung23 2 years, 2 months ago

Selected Answer: A

Correct answer is A because must support integration with existing erp system we need to choose sftp internal-facing

upvoted 3 times

buzzinmumbai 2 years, 2 months ago

Answer is D . Both A&D are right but the question says it must support integration with existing erp system. I believe you can use transfer family for the existing job onprem as well to check for files.

upvoted 1 times

alawada 2 years, 2 months ago

Selected Answer: D

has an AWS account that has connectivity to the on-premises network.

upvoted 2 times

xBUGx 2 years, 3 months ago

Selected Answer: D

The company already has an AWS account that has connectivity to the on-premises network. So no need internet.

upvoted 3 times

67a3f49 2 years, 3 months ago

I would go in D as it's internal network.

upvoted 2 times

NayeraB 2 years, 3 months ago

Selected Answer: A

I think A makes more sense

upvoted 2 times

Andy_09 2 years, 4 months ago

A is the correct option

upvoted 3 times

A company's applications use Apache Hadoop and Apache Spark to process data on premises. The existing infrastructure is not scalable and is complex to manage.

A solutions architect must design a scalable solution that reduces operational complexity. The solution must keep the data processing on premises.

Which solution will meet these requirements?

- A. Use AWS Site-to-Site VPN to access the on-premises Hadoop Distributed File System (HDFS) data and application. Use an Amazon EMR cluster to process the data.
- B. Use AWS DataSync to connect to the on-premises Hadoop Distributed File System (HDFS) cluster. Create an Amazon EMR cluster to process the data.
- C. Migrate the Apache Hadoop application and the Apache Spark application to Amazon EMR clusters on AWS Outposts. Use the EMR clusters to process the data. **Most Voted**
- D. Use an AWS Snowball device to migrate the data to an Amazon S3 bucket. Create an Amazon EMR cluster to process the data.

Correct Answer: C

Community vote distribution

C (86%)

14%

Comments

anikolov **Highly Voted** 2 years, 4 months ago

Selected Answer: C

C cover requirement: The solution must keep the data processing on premises
upvoted 14 times

Andy_09 **Highly Voted** 2 years, 4 months ago

I would go for option C, as data processing has to be done on premise.
upvoted 9 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: C

Answer C
upvoted 2 times

sandordini 2 years, 1 month ago

Selected Answer: C

Only solution to keep the processing on-prem.
upvoted 2 times

Hung23 2 years, 2 months ago

Selected Answer: B

Create an Amazon EMR Cluster: With the data now available in Amazon S3, the company can create an Amazon EMR cluster for data processing. EMR provides scalable Hadoop and Spark clusters that can process data stored in S3, enabling the company to leverage cloud-based processing resources while still keeping the data processing on premises.
upvoted 3 times

example_ 1 year, 11 months ago

Selected response : C

upvoted 2 times

A company is migrating a large amount of data from on-premises storage to AWS. Windows, Mac, and Linux based Amazon EC2 instances in the same AWS Region will access the data by using SMB and NFS storage protocols. The company will access a portion of the data routinely. The company will access the remaining data infrequently.

The company needs to design a solution to host the data.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create an Amazon Elastic File System (Amazon EFS) volume that uses EFS Intelligent-Tiering. Use AWS DataSync to migrate the data to the EFS volume.
- B. Create an Amazon FSx for ONTAP instance. Create an FSx for ONTAP file system with a root volume that uses the auto tiering policy. Migrate the data to the FSx for ONTAP volume. **Most Voted**
- C. Create an Amazon S3 bucket that uses S3 Intelligent-Tiering. Migrate the data to the S3 bucket by using an AWS Storage Gateway Amazon S3 File Gateway.
- D. Create an Amazon FSx for OpenZFS file system. Migrate the data to the new volume.

Correct Answer: B

Community vote distribution

B (80%)

C (20%)

Comments

ogerber **Highly Voted** 2 years, 3 months ago

Selected Answer: B

Amazon FSx for NetApp ONTAP feature: Multi-protocol access to data using the Network File System (NFS), Server Message Block (SMB), and Internet Small Computer Systems Interface (iSCSI) protocols

upvoted 18 times

jaswantn **Highly Voted** 2 years, 4 months ago

option C SMB and NFS storage protocols -> S3 file gateway

upvoted 7 times

FlyingHawk 1 year, 4 months ago

Initially I thought C is correct, but Windows, Mac, and Linux based Amazon EC2 instances in the same AWS Region will access the data by using SMB and NFS storage protocols, S3 does not provide the smb and NFS access to these ec2 instances, so C is incorrect, B is correct.

upvoted 3 times

MatAlves 1 year, 8 months ago

"S3 File Gateway is used for on-premises data intensive applications that need file protocol access to objects in S3. "

<https://aws.amazon.com/storagegateway/file/s3/>

upvoted 2 times

MatAlves 1 year, 8 months ago

The company is MIGRATING data from on-premises to AWS.

Amazon FSx for NetApp ONTAP offers high-performance file storage that's broadly accessible from Linux, Windows, and macOS compute instances via the industry-standard NFS, SMB, iSCSI, and NVMe-over-TCP protocols.

<https://aws.amazon.com/fsx/netapp-ontap/features/>

upvoted 3 times

MatAlves **Most Recent** 1 year, 8 months ago

Selected Answer: B

Amazon FSx for NetApp ONTAP offers high-performance file storage that's broadly accessible from Linux, Windows, and macOS compute instances via the industry-standard NFS, SMB, iSCSI, and NVMe-over-TCP protocols.

<https://aws.amazon.com/fsx/netapp-ontap/features/>

"S3 File Gateway is used for on-premises data intensive applications that need file protocol access to objects in S3. "

<https://aws.amazon.com/storagegateway/file/s3/>

upvoted 4 times

Scheldon 1 year, 11 months ago

Selected Answer: B

Answer B

upvoted 2 times

rjjkc 2 years ago

Selected Answer: C

I think it is "C"

Not "B" - The question never indicates the company is using "NetApp ONTAP file systems", so I am not sure what it means by "Migrate the data to the FSx for ONTAP volume". Please correct if misunderstood.

"C" clearly indicates how to migrate the data to S3, the S3 Intelligent-Tiering addressed the access pattern in the question and you can SMB/NFS mount S3 bucket

<https://docs.aws.amazon.com/filegateway/latest/files3/using-smb-fileshare.html>

<https://docs.aws.amazon.com/filegateway/latest/files3/GettingStartedAccessFileShare.html>

upvoted 1 times

TwinSpark 2 years ago

Selected Answer: B

FSx for ONTAP support NFS and SMB Protocol, even AWS Storage Gateway Amazon S3 File Gateway support them but it is used to connect on premises devices to file on s3, not to connect ec2 instances in the same aws region

upvoted 3 times

Linuslin 2 years ago

Selected Answer: B

Amazon FSx for NetApp ONTAP provides fully managed shared storage in the AWS Cloud with the popular data access and management capabilities of ONTAP.

Move workloads running on NetApp or other NFS/SMB/iSCSI servers to AWS without modifying application code or how you manage data.

And FSx for NetAPP ONTAP support "Reducing storage costs with automatic and intelligent storage tiering."

<https://aws.amazon.com/tw/fsx/netapp-ontap/faqs/#product-faqs#netapp-ontap-faq#reducing-storage-costs-with-automatic-and-intelligent-storage-tiering>

upvoted 4 times

camp5 2 years, 2 months ago

it's C

upvoted 1 times

TruthWS 2 years, 2 months ago

B - FSx for ONTAP support SMB and NFS

upvoted 2 times

alawada 2 years, 2 months ago

Selected Answer: B

Amazon FSx for NetAPP ONTAP feature: Multi-protocol access to data using the Network File System (NFS), Server Message Block (SMB), and Internet Small Computer Systems Interface (iSCSI) protocols

Option C: make no sense I see it as a distractor

upvoted 2 times

Kezuko 2 years, 2 months ago

Selected Answer: C

Both B and C works, but it seems like C has a least operational overhead

upvoted 2 times

Kezuko 2 years, 2 months ago

<https://www.amazonaws.cn/en/storagegateway/faqs/#:~:text=The%20Amazon%20S3%20File%20Gateway,be%20directly%20accessed%20in%20S3.>

upvoted 1 times

dkw2342 2 years, 2 months ago

It's B, option C makes no sense.

1. "Migrate the data to the S3 bucket using an AWS Storage Gateway Amazon S3 File Gateway." -> Nothing about running the gateway to access the files via SMB and NFS afterwards.

2. Even if you ignore this, the S3 File Gateway requires a virtual appliance to be deployed (on EC2 in this case), which contradicts the "LEAST operational overhead" requirement.

upvoted 2 times

rondelldell 2 years, 2 months ago

The company will access the remaining data infrequently."

upvoted 1 times

Indrasis 2 years, 3 months ago

Selected Answer: C

Option C looks correct.

"The company will access a portion of the data routinely. The company will access the remaining data infrequently."

upvoted 3 times

Appon 2 years, 3 months ago

Selected Answer: B

option B

upvoted 2 times

MattBJ 2 years, 3 months ago

Selected Answer: C

C is correct

upvoted 3 times

hajra313 2 years, 4 months ago

Option A and D do not support SMB and NFS file system . Option b looks correvt

upvoted 5 times

Andy_09 2 years, 4 months ago

Option with S3 usage looks corrcet

upvoted 1 times

A manufacturing company runs its report generation application on AWS. The application generates each report in about 20 minutes. The application is built as a monolith that runs on a single Amazon EC2 instance. The application requires frequent updates to its tightly coupled modules. The application becomes complex to maintain as the company adds new features.

Each time the company patches a software module, the application experiences downtime. Report generation must restart from the beginning after any interruptions. The company wants to redesign the application so that the application can be flexible, scalable, and gradually improved. The company wants to minimize application downtime.

Which solution will meet these requirements?

- A. Run the application on AWS Lambda as a single function with maximum provisioned concurrency.
- B. Run the application on Amazon EC2 Spot Instances as microservices with a Spot Fleet default allocation strategy.
- C. Run the application on Amazon Elastic Container Service (Amazon ECS) as microservices with service auto scaling.
- D. Run the application on AWS Elastic Beanstalk as a single application environment with an all-at-once deployment strategy.

Most Voted

Correct Answer: C

Community vote distribution

C (100%)

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago
Microservices using ECS
upvoted 10 times

MatAlves **Most Recent** 1 year, 8 months ago
Selected Answer: C
Monolith -> Microservices = ECS.
upvoted 1 times

Hung23 2 years, 2 months ago
Selected Answer: C
Sure 100%
upvoted 3 times

asdfcdsxdxc 2 years, 3 months ago
Selected Answer: C
Microservices using Elastic Container Service is correct
upvoted 1 times

Indrasis 2 years, 3 months ago
Selected Answer: C
C is correct
upvoted 1 times

Typewriter101 2 years, 3 months ago
Selected Answer: C
B will not help
spot instances provide cost savings but using it for a stateful task isn't right cause spot instances can be interrupted
upvoted 1 times

sandordini 2 years, 1 month ago

Correct answer but incorrect reasoning. Spot fleet includes on-demand AND spot instances to provide the desired capacity.
upvoted 1 times

A company wants to rearchitect a large-scale web application to a serverless microservices architecture. The application uses Amazon EC2 instances and is written in Python.

The company selected one component of the web application to test as a microservice. The component supports hundreds of requests each second. The company wants to create and test the microservice on an AWS solution that supports Python. The solution must also scale automatically and require minimal infrastructure and minimal operational support.

Which solution will meet these requirements?

- A. Use a Spot Fleet with auto scaling of EC2 instances that run the most recent Amazon Linux operating system.
- B. Use an AWS Elastic Beanstalk web server environment that has high availability configured.
- C. Use Amazon Elastic Kubernetes Service (Amazon EKS). Launch Auto Scaling groups of self-managed EC2 instances.
- D. Use an AWS Lambda function that runs custom developed code. **Most Voted**

Correct Answer: D

Community vote distribution

D (82%)

Other

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago
Lambda looks like a better option
upvoted 11 times

Typewriter101 **Highly Voted** 2 years, 3 months ago
Selected Answer: D
Lambda
serverless, scalable, minimal infrastructure, handling hundreds of requests per second
upvoted 8 times

In2718 **Most Recent** 9 months, 3 weeks ago
Selected Answer: B
The question does not consider if the transaction would take less than 15 minutes in which case D is a failure. Answer B would be the ideal choice without this information, though it would require slightly more effort
upvoted 1 times

asosa_adorati 11 months, 3 weeks ago
Selected Answer: D
D is the only solution...it is requiring a serverless solution...AWS Lambda is
upvoted 1 times

sandordini 2 years, 1 month ago
Selected Answer: D
A: auto-scaling of EC2 instances - Lot of overhead + Infra
B: The company selected one component of the web application to test as a microservice. The component supports hundreds of requests each second. > Elastic Beanstalk is a bad choice if you need worker processes. The whole point of a worker process is to perform a task in the background without slowing down your main web app. But Elastic Beanstalk doesn't support this option in a scalable way.
Also, they want to test just 1 selected microservice and I think it's a bit of overkill to do it using Elastic Beanstalk. Happy to be challenged though!
C: self-managed EC2 instances > infra + operational overhead
D: Lambda supports Python, microservice should be quicker than 15 mins, worst case scenario the test will fail.. (that's the purpose tests are conducted for anyway..)
I'd go for D
upvoted 5 times

gsgdga 2 years, 2 months ago

Selected Answer: C

microservice => EKS, ECS

upvoted 1 times

LuongTo 1 year, 7 months ago

containerized application then go with EKS, ECS is absolutely yes, but there would be more solution for microservices e.g. lambda

upvoted 2 times

alawada 2 years, 2 months ago

Selected Answer: C

C is the correct answer. The best way to deploy microservice is to use container-based service

upvoted 1 times

MatAlves 1 year, 8 months ago

"Maintain nodes yourself with self-managed nodes"

<https://docs.aws.amazon.com/eks/latest/userguide/worker.html>

upvoted 1 times

dkw2342 2 years, 2 months ago

Microservices doesn't automatically mean ECS or EKS. Read the question again: "Serverless" clearly contradicts "self-managed EC2 instances".

D is the only option that fits the criteria.

upvoted 2 times

rubiteb 2 years, 3 months ago

Best answer is C.

The application is a large-scale web app as mentioned in the question.

upvoted 1 times

MatAlves 1 year, 8 months ago

You cannot have 'minimal infrastructure and minimal operational support' with 'Auto Scaling groups of self-managed EC2 instances.'

upvoted 2 times

rubiteb 2 years, 3 months ago

I mean B for Elastic Beans Stalk not C. EBS is the best solution for running large-scale application.

upvoted 1 times

Umuntu 2 years, 4 months ago

C is the correct answer. The best way to deploy microservice is to use container-based service such as EKS or ECS. So C is great

upvoted 3 times

Typewriter101 2 years, 3 months ago

Using ECS or EKS involves managing cluster and ec2 which will increase the infrastructure and operational overhead compared to lambda which is serverless.

upvoted 2 times

Andy_09 2 years, 4 months ago

EBS for minimal infra maintenance

upvoted 1 times

A company has an AWS Direct Connect connection from its on-premises location to an AWS account. The AWS account has 30 different VPCs in the same AWS Region. The VPCs use private virtual interfaces (VIFs). Each VPC has a CIDR block that does not overlap with other networks under the company's control.

The company wants to centrally manage the networking architecture while still allowing each VPC to communicate with all other VPCs and on-premises networks.

Which solution will meet these requirements with the LEAST amount of operational overhead?

- A. Create a transit gateway, and associate the Direct Connect connection with a new transit VIF. Turn on the transit gateway's route propagation feature. **Most Voted**
- B. Create a Direct Connect gateway. Recreate the private VIFs to use the new gateway. Associate each VPC by creating new virtual private gateways.
- C. Create a transit VPC. Connect the Direct Connect connection to the transit VPC. Create a peering connection between all other VPCs in the Region. Update the route tables.
- D. Create AWS Site-to-Site VPN connections from on premises to each VPC. Ensure that both VPN tunnels are UP for each connection. Turn on the route propagation feature.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago
Option A
upvoted 7 times

Umuntu **Highly Voted** 2 years, 4 months ago
A is the best solution
upvoted 6 times

Besisco **Most Recent** 1 year, 4 months ago

Selected Answer: A

A, but... why there is not mentioned transit gateway attachments?
if no attachments, how VPCs are going to communicate onprem through the transit gateway?
upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: A

"You can use AWS Direct Connect gateway to connect your Direct Connect connection over a transit virtual interface to the VPCs or VPNs that are attached to your transit gateway. You associate a Direct Connect gateway with the transit gateway. Then, create a transit virtual interface for your AWS Direct Connect connection to the Direct Connect gateway."
<https://docs.aws.amazon.com/directconnect/latest/UserGuide/direct-connect-transit-gateways.html>
upvoted 4 times

alawada 2 years, 2 months ago

Selected Answer: A

Turn on the transit gateway's route propagation feature.
upvoted 2 times

cedser8 2 years, 3 months ago

<https://docs.aws.amazon.com/directconnect/latest/UserGuide/direct-connect-gateways-intro.html>

upvoted 2 times

Typewriter101 2 years, 3 months ago

Selected Answer: A

transit gateway -> hub and spoke

upvoted 6 times

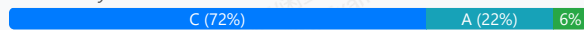
A company has applications that run on Amazon EC2 instances. The EC2 instances connect to Amazon RDS databases by using an IAM role that has associated policies. The company wants to use AWS Systems Manager to patch the EC2 instances without disrupting the running applications.

Which solution will meet these requirements?

- A. Create a new IAM role. Attach the AmazonSSMManagedInstanceCore policy to the new IAM role. Attach the new IAM role to the EC2 instances and the existing IAM role.
- B. Create an IAM user. Attach the AmazonSSMManagedInstanceCore policy to the IAM user. Configure Systems Manager to use the IAM user to manage the EC2 instances.
- C. Enable Default Host Configuration Management in Systems Manager to manage the EC2 instances. **Most Voted**
- D. Remove the existing policies from the existing IAM role. Add the AmazonSSMManagedInstanceCore policy to the existing IAM role.

Correct Answer: C

Community vote distribution



Comments

jaswantn **Highly Voted** 2 years, 4 months ago

option C....Default Host Management Configuration creates and applies a default IAM role to ensure that Systems Manager has permissions to manage all instances in the Region and perform automated patch scans using Patch Manager.

upvoted 13 times

Pics00094 **Highly Voted** 2 years, 3 months ago

Selected Answer: C

C is the answer

upvoted 6 times

In2718 **Most Recent** 9 months, 3 weeks ago

Selected Answer: D

Chat AI -

C. Enable Default Host Configuration Management...

This is an optional feature of Systems Manager, but not related to providing the EC2 instances the right IAM permissions. Doesn't solve the problem.

D. Remove the existing policies from the existing IAM role. Add the AmazonSSMManagedInstanceCore policy to the existing IAM role.

Correct. You don't replace the role (so applications using it for RDS keep working).

You simply add the AmazonSSMManagedInstanceCore policy to the existing IAM role.

This gives Systems Manager the access it needs, without disrupting existing functionality.

upvoted 1 times

Resilient_k 7 months ago

<https://docs.aws.amazon.com/systems-manager/latest/userguide/fleet-manager-default-host-management-configuration.html>

upvoted 1 times

In2718 10 months, 2 weeks ago

Selected Answer: C

This is a terrible question because both A and C require a new role and cannot have more than one role for an EC2 - A is eliminated because it says two roles, C is eliminated because it will require two roles.... but it doesn't kill itself by saying such. So the least wrong answer to this question is C.

upvoted 1 times

lalaraj 10 months, 3 weeks ago

Selected Answer: A

Answer: A. Create a new IAM role. Attach the AmazonSSMManagedInstanceCore policy to the new IAM role. Attach the new IAM role to the EC2 instances and the existing IAM role.

upvoted 1 times

FlyingHawk 1 year, 5 months ago

Selected Answer: C

<https://docs.aws.amazon.com/systems-manager/latest/userguide/setup-instance-permissions.html>, DHMC eliminates the need to manually create and attach a new IAM role for Systems Manager, reducing operational overhead.

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: C

Well.. Actually I don't know what to choose. A or C ??

A - EC2 instances can have only one IAM role attached at a time. Whether attaching multiple roles is possible or not depends on whether AWS allow: such configuration in your environment.

B - Systems Manager manages EC2 instances through the instance's attached IAM role, not an IAM user.

C - It doesn't address the core requirement of maintaining the existing RDS connectivity. It still requires the EC2 instances to have the necessary IAM permissions to be managed by SSM. The IAM role associated with the EC2 instance must still have the AmazonSSMManagedInstanceCore policy attached to enable Systems Manager capabilities.

D - Removing the existing policies would break the application's ability to connect to the RDS database. This directly contradicts the requirement of not disrupting running applications.

upvoted 1 times

FlyingHawk 1 year, 5 months ago

A is the way before having C, details see this blog:

<https://aws.amazon.com/blogs/mt/enable-management-of-your-amazon-ec2-instances-in-aws-systems-manager-using-default-host-management-configuration/>

upvoted 2 times

rosanna 1 year, 5 months ago

Selected Answer: A

I'd vote for A because the request here is not to disrupt the existing workload, meaning the existing IAM role must be intact with the addition to the new permission set that gives SSM patch manager patching capabilities.

upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: C

"The Default Host Management Configuration setting allows AWS Systems Manager to manage your Amazon EC2 instances automatically as managed instances.

Default Host Management Configuration makes it possible to manage EC2 instances without your having to manually create an AWS Identity and Access Management (IAM) instance profile. Instead, Default Host Management Configuration creates and applies a default IAM role to ensure that Systems Manager has permissions to manage all instances in the AWS account and AWS Region where it's activated."

upvoted 3 times

MatAlves 1 year, 8 months ago

<https://docs.aws.amazon.com/systems-manager/latest/userguide/fleet-manager-default-host-management-configuration.html>

upvoted 2 times

88f8032 2 years, 1 month ago

Selected Answer: A

i think A

upvoted 2 times

NayeraB 2 years, 3 months ago

So is C same as A, but automated?

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

No, A is impossible because EC2 instances can have only one IAM role attached at a time.

upvoted 1 times

osmk 2 years, 3 months ago

C is fine

upvoted 2 times

Andy_09 2 years, 4 months ago

C is a better option
upvoted 3 times

Andy_09 2 years, 4 months ago
Correct answer A
upvoted 3 times

arunkpskpm 2 years, 3 months ago

"Attach the new IAM role to the EC2 instances and the existing IAM role" - You can't attach multiple policies to an EC2 instance. So A is wrong.
upvoted 6 times

A company runs container applications by using Amazon Elastic Kubernetes Service (Amazon EKS) and the Kubernetes Horizontal Pod Autoscaler. The workload is not consistent throughout the day. A solutions architect notices that the number of nodes does not automatically scale out when the existing nodes have reached maximum capacity in the cluster, which causes performance issues.

Which solution will resolve this issue with the LEAST administrative overhead?

- A. Scale out the nodes by tracking the memory usage.
- B. Use the Kubernetes Cluster Autoscaler to manage the number of nodes in the cluster. **Most Voted**
- C. Use an AWS Lambda function to resize the EKS cluster automatically.
- D. Use an Amazon EC2 Auto Scaling group to distribute the workload.

Correct Answer: B

Community vote distribution

B (100%)

Comments

FlyingHawk 1 year, 5 months ago

I think the better solution now is using Karpenter or turn on EKS Auto mode

<https://docs.aws.amazon.com/eks/latest/userguide/autoscaling.html>

upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: B

Refer to

<https://www.examttopics.com/discussions/amazon/view/109702-exam-aws-certified-solutions-architect-associate-saa-c03/>

upvoted 2 times

Scheldon 1 year, 11 months ago

Selected Answer: B

AnswerB

Using of Kubernetes Cluster Autoscaler seems to be the best solution here

upvoted 2 times

alawada 2 years, 2 months ago

Selected Answer: B

When the workload increases and existing nodes reach maximum capacity, the Cluster Autoscaler detects the need for additional nodes and request them from the underlying AWS infrastructure.

upvoted 2 times

osmk 2 years, 3 months ago

Selected Answer: B

Bcorrect

upvoted 2 times

Naveena_Devanga 2 years, 3 months ago

B is the correct answer. The Kubernetes Cluster Autoscaler automatically adjusts the number of nodes in your cluster when pods fail or are rescheduled onto other nodes. The Cluster Autoscaler uses Auto Scaling groups

upvoted 4 times

jaswantn 2 years, 4 months ago

option B.

upvoted 2 times

Andy_09 2 years, 4 months ago

Kubernetes Cluster Autoscaler looks correct

upvoted 4 times

A company maintains about 300 TB in Amazon S3 Standard storage month after month. The S3 objects are each typically around 50 GB in size and are frequently replaced with multipart uploads by their global application. The number and size of S3 objects remain constant, but the company's S3 storage costs are increasing each month.

How should a solutions architect reduce costs in this situation?

- A. Switch from multipart uploads to Amazon S3 Transfer Acceleration.
- B. Enable an S3 Lifecycle policy that deletes incomplete multipart uploads. **Most Voted**
- C. Configure S3 inventory to prevent objects from being archived too quickly.
- D. Configure Amazon CloudFront to reduce the number of objects stored in Amazon S3.

Correct Answer: B

Community vote distribution

B (100%)

Comments

MatAlves 1 year, 8 months ago

Selected Answer: B

B - No brainer.

upvoted 1 times

MatAlves 1 year, 8 months ago

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/mpu-abort-incomplete-mpu-lifecycle-config.html>

upvoted 1 times

alawada 2 years, 2 months ago

Selected Answer: B

Optimize multipart uploads to reduce costs associated with storing incomplete multipart upload parts. Ensure that multipart uploads are completed and the parts are assembled into complete objects in a timely manner to avoid unnecessary storage costs.

upvoted 4 times

Typewriter101 2 years, 3 months ago

Selected Answer: B

when primary concern is cost and the data transfer multipart upload may be the more cost-effective than S3 transfer acceleration. So switching to s3 TA is won't be reasonable.

upvoted 4 times

Umntu 2 years, 4 months ago

Option B is correct

upvoted 2 times

Andy_09 2 years, 4 months ago

Option B

upvoted 3 times

A company has deployed a multiplayer game for mobile devices. The game requires live location tracking of players based on latitude and longitude. The data store for the game must support rapid updates and retrieval of locations.

The game uses an Amazon RDS for PostgreSQL DB instance with read replicas to store the location data. During peak usage periods, the database is unable to maintain the performance that is needed for reading and writing updates. The game's user base is increasing rapidly.

What should a solutions architect do to improve the performance of the data tier?

- A. Take a snapshot of the existing DB instance. Restore the snapshot with Multi-AZ enabled.
- B. Migrate from Amazon RDS to Amazon OpenSearch Service with OpenSearch Dashboards.
- C. Deploy Amazon DynamoDB Accelerator (DAX) in front of the existing DB instance. Modify the game to use DAX.
- D. Deploy an Amazon ElastiCache for Redis cluster in front of the existing DB instance. Modify the game to use Redis.

Most Voted

Correct Answer: D

Community vote distribution

D (76%)

C (24%)

Comments

Rhydian25 **Highly Voted** 1 year, 11 months ago

Selected Answer: D

Why are people voting for C? PostgreSQL is a relational DB. DynamoDB is NoSQL.

It makes no sense
upvoted 9 times

1e22522 1 year, 10 months ago
its bezos with his alt accounts.
upvoted 6 times

Dantecito **Most Recent** 1 year, 3 months ago

Selected Answer: D

To add more to all the answer Redis has built-in support for geospatial data, making it ideal for updates and retrieval of locations.
upvoted 1 times

MatAlves 1 year, 8 months ago

Selected Answer: D

Refer to <https://www.examtopycs.com/discussions/amazon/view/53854-exam-aws-certified-solutions-architect-associate-saa-c02/>
upvoted 1 times

MatAlves 1 year, 8 months ago

Specific to Redis, ElastiCache lets you "scale in" or "scale out" both reads and writes. Cluster mode offers added shard support, enabling write scaling.

<https://aws.amazon.com/blogs/database/building-a-real-time-gaming-leaderboard-with-amazon-elasticache-for-redis/>
upvoted 2 times

Lin878 1 year, 11 months ago

Selected Answer: D

I confuse, Is DAX working with RDS?

upvoted 1 times

Jacky_S 1 year, 11 months ago

Selected Answer: D

The answer C is not making any sense with "Deploy Amazon DynamoDB Accelerator (DAX) in front of the existing DB instance.", because AWS DynamoDB is a DBaaS.

upvoted 2 times

Scheldon 1 year, 11 months ago

Selected Answer: D

AnswerD

ElastiCache is a fully managed, in-memory caching service that provides microsecond read and write latencies that support flexible, real-time use cases.

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/creating-elasticache-cluster-with-RDS-settings.html>

upvoted 2 times

Scheldon 1 year, 11 months ago

Beside DAX is for DynamoDB and I think it will not work with RDS

upvoted 1 times

ug56c 1 year, 11 months ago

Selected Answer: D

Amazon ElastiCache for Redis for RDS

upvoted 1 times

sheilawu 2 years ago

Selected Answer: C

" writing updates" so it should be DAX.

upvoted 2 times

f07ed8f 2 years ago

Selected Answer: C

Vote for C (DAX) as ElastiCache for Redis cluster only helps on read operation but not write.

upvoted 4 times

f07ed8f 2 years ago

Vote for C (DAX) as ElastiCache for Redis cluster only helps on read operation but not write operation.

upvoted 1 times

sirajtr47 2 years, 1 month ago

Performance >> Amazon ElastiCache for Redis cluster

upvoted 4 times

Rylz 2 years ago

Yeah, but this is an online game that need to update everything instantly so i don't think that caching is the right method here think about it, you are making movements in the online game its like loading new data all the time, so each time you move you load the cache for next time usage. what if you dont need to use it again?

i dont think elasticache is the right method here so it leave us with DAX

what do you think?

upvoted 1 times

Jacky_S 1 year, 11 months ago

i agree with DAX, but how it can be deploying in front of DB instance? Since that is a DBaaS.

upvoted 1 times

FZA24 2 years, 3 months ago

Selected Answer: D

D looks correct

upvoted 2 times

Umntu 2 years, 4 months ago

D looks correct

upvoted 4 times

Andy_09 2 years, 4 months ago

Looks correct

upvoted 2 times

A company stores critical data in Amazon DynamoDB tables in the company's AWS account. An IT administrator accidentally deleted a DynamoDB table. The deletion caused a significant loss of data and disrupted the company's operations. The company wants to prevent this type of disruption in the future.

Which solution will meet this requirement with the LEAST operational overhead?

- A. Configure a trail in AWS CloudTrail. Create an Amazon EventBridge rule for delete actions. Create an AWS Lambda function to automatically restore deleted DynamoDB tables.
- B. Create a backup and restore plan for the DynamoDB tables. Recover the DynamoDB tables manually.
- C. Configure deletion protection on the DynamoDB tables. **Most Voted**
- D. Enable point-in-time recovery on the DynamoDB tables.

Correct Answer: C

Community vote distribution

C (100%)

Comments

BillaRanga **Highly Voted** 1 year, 9 months ago

Selected Answer: C

<https://aws.amazon.com/about-aws/whats-new/2023/03/amazon-dynamodb-table-deletion-protection/>

Deletion protection is now available for Amazon DynamoDB tables in all AWS Regions. DynamoDB now makes it possible for you to protect your tables from accidental deletion when performing regular table management operations. When creating new tables or managing existing tables, authorized administrators can set the deletion protection property for each table, which will govern whether a table can be deleted.

upvoted 15 times

BillaRanga 1 year, 9 months ago

Option B and D talks about recovering but not preventing. A is toooooo much work

upvoted 3 times

Andy_09 **Highly Voted** 1 year, 10 months ago

Option C

upvoted 5 times

Typewriter101 **Most Recent** 1 year, 10 months ago

Selected Answer: C

B involves more operations.

upvoted 3 times

A company has an on-premises data center that is running out of storage capacity. The company wants to migrate its storage infrastructure to AWS while minimizing bandwidth costs. The solution must allow for immediate retrieval of data at no additional cost.

How can these requirements be met?

- A. Deploy Amazon S3 Glacier Vault and enable expedited retrieval. Enable provisioned retrieval capacity for the workload.
- B. Deploy AWS Storage Gateway using cached volumes. Use Storage Gateway to store data in Amazon S3 while retaining copies of frequently accessed data subsets locally. **Most Voted**
- C. Deploy AWS Storage Gateway using stored volumes to store data locally. Use Storage Gateway to asynchronously back up point-in-time snapshots of the data to Amazon S3.
- D. Deploy AWS Direct Connect to connect with the on-premises data center. Configure AWS Storage Gateway to store data locally. Use Storage Gateway to asynchronously back up point-in-time snapshots of the data to Amazon S3.

Correct Answer: B

Community vote distribution

B (58%)

C (42%)

Comments

67a3f49 **Highly Voted** 2 years, 3 months ago

B is the correct one because:

"A company has an on-premises data center that is running out of storage capacity".

So when they keep data on-prem and do the backup to S3 they'll run out of data and this is not their purpose.

upvoted 10 times

Sergantus 1 year, 7 months ago

Storage Gateway in Stored mode DOS does NOT improve capacity – the main copy of data is stored on the gateway. It's B for that same reason

upvoted 1 times

JCVDB23 **Highly Voted** 2 years, 2 months ago

Selected Answer: B

B. Deploy AWS Storage Gateway using cached volumes. Use Storage Gateway to store data in Amazon S3 while retaining copies of frequently accessed data subsets locally.

AWS Storage Gateway's cached volumes let you use Amazon S3 as your primary data storage while retaining frequently accessed data locally in your storage gateway. Cached volumes minimize the need to scale your on-premises storage infrastructure, while still providing your applications with low-latency access to their frequently accessed data. All data transferred between your gateway and AWS storage is encrypted for security. You can also save on data transfer costs as AWS Storage Gateway compresses all data transferred between the gateway and AWS, allowing you to store more data in AWS while reducing your data transfer costs.

upvoted 7 times

In2718 **Most Recent** 10 months, 2 weeks ago

Selected Answer: C

I'm going with answer C for all the reasons mentioned. The stored volumes is better in this case for the parameters specified including cost.

upvoted 1 times

TEC65 1 year, 1 month ago

Selected Answer: B

they want to limit onsite storage

upvoted 1 times

tch 1 year, 4 months ago

Selected Answer: B

In the cached Volume Gateway mode, your primary data is stored in Amazon S3, while retaining your frequently accessed data locally in the cache for low latency access. In the stored Volume Gateway mode, your primary data is stored locally and your entire dataset is available for low latency access on premises while also asynchronously getting backed up to Amazon S3

upvoted 2 times

LeonSauveterre 1 year, 5 months ago

Selected Answer: B

Another headache one. Well, I'll go for B for now.

A - Really cost-effective, but immediate retrievals incurs additional costs.

B - Less optimal for datasets where most data is accessed frequently and must be immediately available locally.

C - The on-prem data center "is running out of storage capacity". Then why store everything locally? Why even migrate storage infrastructure to AWS...

D - Like C, and also, bandwidth costs are not acceptable.

upvoted 1 times

Lin878 1 year, 11 months ago

Selected Answer: C

I vote "C" because Question doesn't mention to access frequent data. If they want access frequent data, I will vote "B" with cached volume.

upvoted 1 times

1e22522 1 year, 10 months ago

you don't fool me bezos boy

upvoted 3 times

Scheldon 1 year, 11 months ago

Selected Answer: B

AnswerB

I think B answer is the best option here. We will store only data which are frequently accessed and all other we will sent to the cloud. So we will have access to all data but hence most frequently accessed data will be stored in On-Premises Cache we will not pay a lot of additionally \$\$ for data transfer if any.

upvoted 1 times

ike001 1 year, 12 months ago

C is the correct. answer

upvoted 1 times

mohammadthainat 2 years, 2 months ago

Selected Answer: B

1- "The company wants to migrate its storage infrastructure to AWS" ->> B as data will be migrated to AWS.

2- "The solution must allow for immediate retrieval of data at no additional cost." ->> B as data will be stored in S3 Standard storage class which provide immediate data retrieval

upvoted 3 times

gsgdga 2 years, 2 months ago

Selected Answer: C

immediate retrieval of data → should have full data set on-premises => stored volumes AWS Storage Gateway

upvoted 4 times

rubiteb 2 years, 3 months ago

B - as the company is migrating their data to AWS so data has to be stored in the cloud.

upvoted 2 times

osmk 2 years, 3 months ago

C>>>

Cached Mode: In this mode, your primary data resides in Amazon S3, while frequently accessed data is cached locally for low-latency access.

Stored Mode: Here, your entire dataset is stored locally, allowing low-latency access on premises. Simultaneously, the data is asynchronously backed up to Amazon S3.

upvoted 4 times

MatAlves 1 year, 8 months ago

How does that address the fact the company is "running out of storage capacity"?

upvoted 1 times

BillaRanga 2 years, 3 months ago

Selected Answer: C

D -> It takes One month to set up AWS Direct Connect setup

A -> No sense as it talks nothing about On-Prem

B -> Cached volume only stores frequently access data On-Prem, But requirement tells "Data" so we assume it tells All data

C -> Correct, as Stored volumes stores everything in Storage Gateway On-Prem while asynchronously backing up to the cloud
upvoted 5 times

xBUGx 2 years, 2 months ago

i was voting for C, but C doesnt solve on prem out of capacity issue.

upvoted 1 times

sandordini 2 years, 1 month ago

D: It never said one month would be a problem.. Question doesn't state a matter of urgency, but it still stores the data on-prem, and synchronizes to AWS.

C: The same issue as D. Stores data locally, but our on-prem storage is full. Thats why the company wants cloud.

A: Has retrieval costs.

upvoted 1 times

jaswantn 2 years, 4 months ago

option C... data being accessible through stored volume reduces bandwidth cost and provides immediate retrieval of data.

upvoted 2 times

Andy_09 2 years, 4 months ago

Option C, as it makes all the data available for low-latency access.

upvoted 2 times

A company runs a three-tier web application in a VPC across multiple Availability Zones. Amazon EC2 instances run in an Auto Scaling group for the application tier.

The company needs to make an automated scaling plan that will analyze each resource's daily and weekly historical workload trends. The configuration must scale resources appropriately according to both the forecast and live changes in utilization.

Which scaling strategy should a solutions architect recommend to meet these requirements?

- A. Implement dynamic scaling with step scaling based on average CPU utilization from the EC2 instances.
- B. Enable predictive scaling to forecast and scale. Configure dynamic scaling with target tracking **Most Voted**
- C. Create an automated scheduled scaling action based on the traffic patterns of the web application.
- D. Set up a simple scaling policy. Increase the cooldown period based on the EC2 instance startup time.

Correct Answer: B

Community vote distribution

B (100%)

Comments

sandordini 1 year, 7 months ago

Selected Answer: B

Not A: Only handles Dynamic scaling, not pattern-based/predictive scaling.

B: Both Predictive and dynamic

Not C: Manual version of predictive, lacks live circumstances..

Not D: The question doesn't talk about cool down period...

upvoted 3 times

alawada 1 year, 8 months ago

<https://aws.amazon.com/blogs/aws/new-predictive-scaling-for-ec2-powered-by-machine-learning/>

upvoted 2 times

BillaRanga 1 year, 9 months ago

Selected Answer: B

By configuring dynamic scaling with target tracking, the company can automatically adjust resources based on the forecasted demand while also responding to live changes in utilization

upvoted 4 times

Andy_09 1 year, 10 months ago

Option B

upvoted 4 times

A package delivery company has an application that uses Amazon EC2 instances and an Amazon Aurora MySQL DB cluster. As the application becomes more popular, EC2 instance usage increases only slightly. DB cluster usage increases at a much faster rate.

The company adds a read replica, which reduces the DB cluster usage for a short period of time. However, the load continues to increase. The operations that cause the increase in DB cluster usage are all repeated read statements that are related to delivery details. The company needs to alleviate the effect of repeated reads on the DB cluster.

Which solution will meet these requirements MOST cost-effectively?

- A. Implement an Amazon ElastiCache for Redis cluster between the application and the DB cluster. **Most Voted**
- B. Add an additional read replica to the DB cluster.
- C. Configure Aurora Auto Scaling for the Aurora read replicas.
- D. Modify the DB cluster to have multiple writer instances.

Correct Answer: A

Community vote distribution

A (93%)

7%

Comments

Andy_09 **Highly Voted** 2 years, 4 months ago

Option A

upvoted 7 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: C

AnswerC

After topic reconsideration I will go with Aurora autoscaling

To meet your connectivity and workload requirements, Aurora Auto Scaling dynamically adjusts the number of Aurora Replicas (reader DB instances) provisioned for an Aurora DB cluster. Aurora Auto Scaling is available for both Aurora MySQL and Aurora PostgreSQL. Aurora Auto Scaling enables your Aurora DB cluster to handle sudden increases in connectivity or workload. When the connectivity or workload decreases, Aurora Auto Scaling removes unnecessary Aurora Replicas so that you don't pay for unused provisioned DB instances.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Integrating.AutoScaling.html>

upvoted 1 times

MatAlves 1 year, 8 months ago

repeated reads = perfect scenario for CACHING.

How adding more reader instances will "alleviate the effect of repeated reads on the DB cluster"?

upvoted 2 times

Scheldon 1 year, 11 months ago

Selected Answer: A

AnswerA

upvoted 3 times

sandordini 2 years, 1 month ago

Selected Answer: A

A. Although Redis is not typically cheap, the question statement clearly shouts for a cached solution, which is Redis... Also, that's the only long-term solution, as we don't know anything about the volumes, scale of trends, etc...

upvoted 4 times

Kezuko 2 years, 2 months ago

Selected Answer: A

"repeated read statements" -> Cache layer
upvoted 3 times

BillaRanga 2 years, 3 months ago

Selected Answer: A

The question says, "The operations that cause the increase in DB cluster usage are all ****repeated read statements**** that are related to delivery details." - Read statements mean we can cache the results - hence, we need No read-replicas; we need only a cache layer to improve the performance.. Also, Adding read replicas costs money. The requirement is to meet them **MOST** cost-effectively

upvoted 4 times

A company has an application that uses an Amazon DynamoDB table for storage. A solutions architect discovers that many requests to the table are not returning the latest data. The company's users have not reported any other issues with database performance. Latency is in an acceptable range.

Which design change should the solutions architect recommend?

- A. Add read replicas to the table.
- B. Use a global secondary index (GSI).
- C. Request strongly consistent reads for the table. **Most Voted**
- D. Request eventually consistent reads for the table.

Correct Answer: C

Community vote distribution

C (100%)

Comments

alawada **Highly Voted** 1 year, 8 months ago

Selected Answer: C

DynamoDB by default provides eventual consistency for read operations, which means that a query may not reflect the most recent data changes immediately after an update. Instead, it may take some time for the data to propagate across all replicas in the DynamoDB global table.

To ensure that read operations return the latest data and address the issue of stale data being returned to users, the solutions architect should recommend switching the read consistency level from eventually consistent reads to strongly consistent reads.

upvoted 7 times

BillaRanga **Most Recent** 1 year, 9 months ago

Selected Answer: C

Both tables and LSIs provide two read consistency options: eventually consistent (default) and strongly consistent reads.

1) Eventually Consistent Reads

Eventually consistent is the default read consistent model for all read operations. When issuing eventually consistent reads to a DynamoDB table or an index, the responses may not reflect the results of a recently completed write operation. If you repeat your read request after a short time, the response should eventually return the more recent item.

upvoted 4 times

BillaRanga 1 year, 9 months ago

2) Strongly Consistent Reads

Read operations such as GetItem, Query, and Scan provide an optional ConsistentRead parameter. If you set ConsistentRead to true, DynamoDB returns a response with the most up-to-date data, reflecting the updates from all prior write operations that were successful.

Hence it is C

A) Read-replicas are Async again, Which will persist the same problem.

B) Indexing will further cause latency, this has nothing to do with the question

upvoted 5 times

Andy_09 1 year, 10 months ago

Option C

upvoted 4 times

A company has deployed its application on Amazon EC2 instances with an Amazon RDS database. The company used the principle of least privilege to configure the database access credentials. The company's security team wants to protect the application and the database from SQL injection and other web-based attacks.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use security groups and network ACLs to secure the database and application servers.
- B. Use AWS WAF to protect the application. Use RDS parameter groups to configure the security settings. **Most Voted**
- C. Use AWS Network Firewall to protect the application and the database.
- D. Use different database accounts in the application code for different functions. Avoid granting excessive privileges to the database users.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Andy_09 **Highly Voted** 1 year, 10 months ago

Option B

upvoted 8 times

BillaRanga **Highly Voted** 1 year, 9 months ago

Selected Answer: B

protect the application and the database from SQL injection and other web-based attacks. -> WAF

upvoted 5 times

03beafc **Most Recent** 1 year, 7 months ago

It's probably still B, but waf can't be attached directly to ec2's

upvoted 2 times

Typewriter101 1 year, 10 months ago

Selected Answer: B

SQL injection -> WAF

upvoted 3 times

An ecommerce company runs applications in AWS accounts that are part of an organization in AWS Organizations. The applications run on Amazon Aurora PostgreSQL databases across all the accounts. The company needs to prevent malicious activity and must identify abnormal failed and incomplete login attempts to the databases.

Which solution will meet these requirements in the MOST operationally efficient way?

- A. Attach service control policies (SCPs) to the root of the organization to identify the failed login attempts.
- B. Enable the Amazon RDS Protection feature in Amazon GuardDuty for the member accounts of the organization. **Most Voted**
- C. Publish the Aurora general logs to a log group in Amazon CloudWatch Logs. Export the log data to a central Amazon S3 bucket.
- D. Publish all the Aurora PostgreSQL database events in AWS CloudTrail to a central Amazon S3 bucket.

Correct Answer: B

Community vote distribution

B (100%)

Comments

BillaRanga **Highly Voted** 2 years, 3 months ago

Selected Answer: B

A -> SCPs are not for monitoring or logging

B-> correct

After you enable the RDS Protection feature, GuardDuty immediately starts monitoring RDS login activity from Aurora databases in your account. GuardDuty continuously monitors and profiles RDS login activity for suspicious activity, for example, unauthorized access to Aurora database in your account, from a previously unseen external actor.

upvoted 9 times

zinabu **Highly Voted** 2 years, 2 months ago

malicious activity=gurd duty

upvoted 6 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: B

AnswerB

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/CHAP_BestPractices.Security.html

Amazon GuardDuty is a threat detection service that helps protect your accounts, containers, workloads, and the data within your AWS environment. Using machine learning (ML) models, and anomaly and threat detection capabilities, GuardDuty continuously monitors different log sources and runtime activity to identify and prioritize potential security risks and malicious activities in your environment.

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/guard-duty-rds-protection.html>

upvoted 2 times

Naveena_Devanga 2 years, 3 months ago

B is the correct answer.

Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior to protect your Amazon Web Services accounts, workloads, and data stored in Amazon S3.

upvoted 4 times

Andy_09 2 years, 4 months ago

Option B

upvoted 4 times

A company has an AWS Direct Connect connection from its corporate data center to its VPC in the us-east-1 Region. The company recently acquired a corporation that has several VPCs and a Direct Connect connection between its on-premises data center and the eu-west-2 Region. The CIDR blocks for the VPCs of the company and the corporation do not overlap. The company requires connectivity between two Regions and the data centers. The company needs a solution that is scalable while reducing operational overhead.

What should a solutions architect do to meet these requirements?

- A. Set up inter-Region VPC peering between the VPC in us-east-1 and the VPCs in eu-west-2.
- B. Create private virtual interfaces from the Direct Connect connection in us-east-1 to the VPCs in eu-west-2.
- C. Establish VPN appliances in a fully meshed VPN network hosted by Amazon EC2. Use AWS VPN CloudHub to send and receive data between the data centers and each VPC.
- D. Connect the existing Direct Connect connection to a Direct Connect gateway. Route traffic from the virtual private gateways of the VPCs in each Region to the Direct Connect gateway. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

BillaRanga **Highly Voted** 2 years, 3 months ago

Selected Answer: D

"If you want to set up a Direct Connect to one or more VPC in many different regions (same account), you must use a Direct Connect Gateway." upvoted 5 times

BillaRanga 2 years, 3 months ago

CloudHub is a VPN (encrypted) connection, so it goes over the public Internet., Whereas DirectConnect is Private (but not encrypted). So CloudHub is not suited for this useCase

upvoted 3 times

JA2018 **Most Recent** 1 year, 6 months ago

Selected Answer: D

Connect the existing Direct Connect connection to a Direct Connect gateway. Route traffic from the virtual private gateways of the VPCs in each Region to the Direct Connect gateway.

Explanation:

Scalability and Reduced Overhead:

Using a Direct Connect gateway allows for easy scaling by adding more virtual private gateways from different regions to the same Direct Connect gateway, eliminating the need to manage separate connections for each region and minimizing operational overhead.

Inter-Region Connectivity:

A Direct Connect gateway acts as a central hub for routing traffic between VPCs in different regions through the Direct Connect connection, providing the required connectivity between the US-East-1 and EU-West-2 regions

upvoted 1 times

jaswantn 2 years, 4 months ago

option D

upvoted 2 times

Andy_09 2 years, 4 months ago

Changing to Option D for simpler implementation.

upvoted 3 times

1e22522 1 year, 10 months ago

sure "Andy"

upvoted 3 times

Andy_09 2 years, 4 months ago

Option A

upvoted 2 times

A company is developing a mobile game that streams score updates to a backend processor and then posts results on a leaderboard. A solutions architect needs to design a solution that can handle large traffic spikes, process the mobile game updates in order of receipt, and store the processed updates in a highly available database. The company also wants to minimize the management overhead required to maintain the solution.

What should the solutions architect do to meet these requirements?

A. Push score updates to Amazon Kinesis Data Streams. Process the updates in Kinesis Data Streams with AWS Lambda. Store the processed updates in Amazon DynamoDB. **Most Voted**

B. Push score updates to Amazon Kinesis Data Streams. Process the updates with a fleet of Amazon EC2 instances set up for Auto Scaling. Store the processed updates in Amazon Redshift.

C. Push score updates to an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe an AWS Lambda function to the SNS topic to process the updates. Store the processed updates in a SQL database running on Amazon EC2.

D. Push score updates to an Amazon Simple Queue Service (Amazon SQS) queue. Use a fleet of Amazon EC2 instances with Auto Scaling to process the updates in the SQS queue. Store the processed updates in an Amazon RDS Multi-AZ DB instance.

Correct Answer: A

Community vote distribution

A (100%)

Comments

BillaRanga **Highly Voted** 2 years, 3 months ago

Selected Answer: A

requirement -1: "Stream + process in order + Minimum Overhead" = Kinesis Data Stream + Lambda
requirement-2: "Highly available database + Min Management overhead" = DynamoDB

Setting Up Ec2 instance or MultiAZ DB = overhead

upvoted 12 times

Andy_09 **Highly Voted** 2 years, 4 months ago

Option A

upvoted 8 times

Scheldon **Most Recent** 1 year, 11 months ago

Selected Answer: A

AnswerA

upvoted 2 times

sandordini 2 years, 1 month ago

Selected Answer: A

Even though it looks like SQS, but EC2 and Multi-AZ DB fail when it comes to minimal operational overhead.

upvoted 4 times

mohammadthainat 2 years, 2 months ago

Selected Answer: A

easy one: mobile game ->> DynamoDB

upvoted 2 times